Chiral Koszul Duality for Ordered Quantum Fields

Noncommutative Factorisation D-Modules, Defects,
Bar–Cobar Duality, and Holomorphic–Topological Field Theory

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The local geometry of a protected field theory on a complex curve times an oriented real line has two operations. Holomorphic collision produces a diagonal singularity; ordered motion along the line produces associative fusion. This volume constructs the common algebraic object, proves its bar–cobar and Morita theorems, derives its exchange and sewing structures, and develops a sequence of exact noncommutative examples from free tensor algebras to Yangians, Hall algebras, and mixed holomorphic–topological brane observables.

Abstract

Let X be a smooth complex algebraic curve. A protected field theory on $X^{\mathrm{an}} \times \mathbb{R}$ has two elementary local motions. Points collide in the holomorphic curve and produce a singular operator product. Intervals compose in the oriented real line and produce an ordered associative product. The common algebraic object is an augmented E_1 algebra in factorisation right D-modules:

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}\big(\mathrm{FactD}(X), \otimes^!\big).$$

We call A an *ordered chiral algebra*. Its factorisation coefficient retains the filtered distribution along every collision diagonal. Its internal E_1 multiplication retains transverse order.

The volume constructs this object from first principles. A finite-set model of $D(\mathrm{Ran}\, X)$ fixes the variance and completion conventions. In the linear species model, Hadamard tensor and Cauchy convolution admit canonical maps between common and independent decompositions; their composite is the projector onto aligned decompositions. For sheaf-valued factorisation objects, locality is the direct family of factorisation squares on every disjointness locus. This exact formulation uses precisely the interchange structure available in the general coefficient category.

The ordered bar is the interval amplitude

$$\overline{B}^{\mathrm{ord}}_X(A) \simeq \Bbbk \otimes^{\mathbb{L}}_A \Bbbk \simeq \int_{([0,1],\partial[0,1])} A.$$

Its normalized complex is the conilpotent tensor coalgebra on $s\bar{A}$; its differential fuses adjacent transverse letters while every term remains a factorisation coefficient on X. Bar and cobar form an equivalence on nilcomplete augmented algebras and conilcomplete coalgebras. The augmentation boundary has endomorphism algebra $A^! = \mathrm{RHom}_A(\Bbbk, \Bbbk)$, and the double centralizer is the derived augmentation completion. The two-sided Koszul transform carries the diagonal bimodule to the diagonal bicomodule and identifies Hochschild with coHochschild theory.

The chiral coefficient is developed through principal parts, normal cones, partition lattices, and Fulton–MacPherson screens. A well-pointed, Σ-cofibrant two-coloured Boardman–Vogt resolution has transverse and chiral edges. The product orientation makes the edge-contraction differentials anticommute. Stasheff identities, chiral Jacobi identities, and their interchange become the three codimension-two faces of one cellular boundary equation.

Exchange begins with a second transverse direction or a flat meromorphic connection. Hochschild cochains are the universal E_2 endomorphism algebra of the identity bimodule. A rigid meromorphic tensor category with a strong monoidal fibre functor reconstructs a coquasitriangular Hopf algebra. KZ transport and the Drinfeld–Kohno theorem identify the standard affine braid category with its quantum-group realization. Rational difference transport, RTT, PBW, and evaluation modules give the Yangian model. BRST operators act factorwise on the ordered bar, and the resulting bicomplex compares bar construction with quantum Hamiltonian reduction.

Two sewing theories complete the geometry. A Calabi–Yau trace closes the ordered interval and gives a ribbon-graph open TCFT. A chiral modular action plumbs algebraic curves and gives conformal-block sewing.

Their compatible combination is a bicoloured graph transform with two loop operators and two genus gradings. A factorising perturbative BV quantization supplies the graph weights; its quantum master equation is the Stokes identity on the compactified graph configurations.

The final parts compute increasingly noncommutative examples: free and square-zero algebras, quiver path algebras, the quantum plane, Weyl and Clifford PBW deformations, Zamolodchikov–Faddeev exchange algebras, the RTT Yangian, the A_2 Ringel–Hall algebra, and the mixed holomorphic–topological matrix-brane trace algebra. The last gives the exact stable Poisson bracket

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1,b+d-1}.$$

The automorphic application identifies the bar Euler character of a positive Borcherds algebra with its denominator. For the Igusa algebra,

$$e^{-\rho}\,\chi_\Gamma \overline{B}^{\mathrm{ord}}\big(U(\mathfrak{n}_{\Delta_5,+})\big) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

A polarized Gram transform then preserves additive physical charge while encoding the three quadratic Fourier degrees. Ordered fusion, chiral collision, exchange, sewing, and automorphic character thus arise as typed constructions from one local object.

Contents

Overture: five motions of one protected experiment

Place two protected observables on a complex curve and on an oriented real line. Their relative position admits five elementary motions.

geometric carrier	motion	algebraic structure
complex diagonal	holomorphic collision	chiral distribution
ordered interval	transverse fusion	E_1 multiplication
transverse plane	exchange	braid transport and R-matrix
circle	closure	Hochschild trace
stable curve	plumbing	modular sewing

The diagonal and the interval form the primitive pair. The diagonal carries the singular coefficient of the operator product. The interval carries the order of composition. A second transverse direction supplies exchange. A trace closes the interval. A modular action varies the complex curve and contracts the states at a node.

On the curve the binary chiral operation has the invariant form

$$\mu_{\mathrm{ch}} : j_* j^! (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}, \qquad j : X^2 \setminus \Delta(X) \hookrightarrow X^2.$$

On the oriented line the binary fusion is

$$m_\perp : A \otimes^! A \longrightarrow A.$$

The ordered chiral algebra is the object carrying both operations together with the factorisation compatibility that identifies their common rectangular experiment.

Every relation in the volume has the same geometric origin. A compactified parameter space has codimension-one faces labelled by elementary compositions. A codimension-two corner can be reached by two ordered face contractions. The induced orientations differ by a sign. Coefficient maps therefore turn the cellular equation $\partial^2 = 0$ into an algebraic master equation.

Theorem 0.0.1 (Stokes-to-master principle)**.** *Let C be an oriented manifold with corners, let E be a cochain complex, and assign to every oriented codimension-one face F a degree-one operator ρ_F on E. Assume that the assignment respects restriction to corners and that each ρ_F is a cochain map of odd degree. Put*

$$D = d_E + \sum_F \rho_F .$$

For every codimension-two stratum S, the component of D^2 supported on S is the oriented sum of the composites $\rho_{F_2}\rho_{F_1}$ over the flags $S \subset F_2 \subset C$. Consequently, a coefficient system extends to a chain map from the cellular boundary complex of C precisely when the corresponding corner identities hold.

Proof. Expand D^2. The term d_E^2 vanishes. The mixed terms vanish because the face operators are cochain maps of odd degree. The remaining terms are ordered pairs of face contractions. Group them by their common codimension-two stratum. The two induced boundary orientations of each corner are opposite, so the grouped sum is the cellular coefficient of ∂^2. This proves the assertion. □

For three ordered intervals, Theorem 0.0.1 gives associativity. For three chiral points, it gives the Jacobi–Borcherds identity. For three exchange lines, it gives the Yang–Baxter equation. For renormalised graph weights, it gives the BV master equation. For stable curves, it gives the modular sewing equation. Geometry fixes the signs; the theory supplies the operators.

Part I

The ordered chiral object

Chapter 1

Two local directions

1.1 The local experiment

Let X be a smooth complex algebraic curve. A rectangular open set in $X^{\mathrm{an}} \times \mathbb{R}$ has the form $U \times I$, with U a holomorphic disc and I an oriented interval. A holomorphic–topological theory assigns a complex of observables $\mathrm{Obs}(U \times I)$ whose dependence on U is holomorphic and whose dependence on I is locally constant.

Choose disjoint discs $U_1, U_2 \subset U$ and ordered intervals $I_1 < I_2 \subset I$. Factorisation gives

$$\mathrm{Obs}(U_1 \times I_1) \otimes \mathrm{Obs}(U_2 \times I_2) \longrightarrow \mathrm{Obs}(U \times I).$$

Moving U_1 toward U_2 probes the diagonal singularity. Moving the two intervals until they become adjacent probes ordered composition. The two limits commute because they arise from one inclusion of disjoint rectangles.

Principle 1.1.1 (Rectangular locality). The holomorphic collision and the transverse product are two restrictions of the same factorisation map on rectangles. Their compatibility is therefore a functorial identity before any coordinate expansion is chosen.

1.2 Order, exchange, and symmetry

The ordered chamber

$$\mathrm{Conf}_n^{<}(\mathbb{R}) = \{(t_1, \ldots, t_n) \in \mathbb{R}^n : t_1 < \cdots < t_n\}$$

is contractible. It retains the order of the labels and supplies coherent associative fusion. The configuration space $\mathrm{Conf}_n(\mathbb{R}^2)$ has braid-group fundamental group and supplies exchange. Passing from one transverse direction to two therefore adds the geometric operation of exchange.

At the algebraic level the distinction is:

$$\begin{aligned} \mathsf{E}_1 &: \text{ ordered multiplication,} \\ \mathsf{E}_2 &: \text{ ordered multiplication with coherent exchange,} \\ \mathsf{E}_\infty &: \text{ coherently symmetric multiplication.} \end{aligned}$$

A quantum group algebra is ordinarily an E_1 algebra. Its representation category becomes braided through a coproduct and an R-matrix. Its derived Hochschild centre is an E_2 algebra. These three structures occupy different categorical levels and will remain distinct throughout the volume.

1.3 The invariant definition

Definition 1.3.1 (Ordered chiral algebra)**.** An *ordered chiral algebra* on X is an augmented E_1-algebra

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}\big(\mathrm{FactD}(X), \otimes^!\big).$$

Its multiplication and augmentation are denoted

$$m : A \otimes^! A \to A, \qquad \varepsilon : A \to \Bbbk.$$

The augmentation ideal is $\bar{A} = \mathrm{fib}(\varepsilon)$.

The adjective *chiral* names the coefficient category. The adjective *ordered* names the internal operadic multiplication. The first carries support and singularity on the curve; the second carries fusion along the oriented line.

1.4 Boundary origin

Let C be a factorisation category on X and let b be a boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_C(b, b)$$

composes in a specified order. Its insertion points vary factorisably on X. This is the native source of ordered chiral algebras.

Theorem 1.4.1 (Factorisation Morita chart)**.** *Let C be a presentable* $\mathrm{FactD}(X)$*-linear factorisation category and let b be a factorisation-compatible compact dualisable generator: its restrictions to a factorising basis are compact generators and its disjoint restriction maps identify $b_{U_1 \sqcup U_2}$ with $b_{U_1} \boxtimes b_{U_2}$. Composition makes $A_b = \mathrm{RHom}_C(b, b)$ an E_1-algebra in* $\mathrm{FactD}(X)$*. If the local compact-generator equivalences commute with restriction and Weiss descent, then*

$$\mathrm{RHom}_C(b, -) : C \xrightarrow{\sim} \mathrm{Mod}_{A_b}\big(\mathrm{FactD}(X)\big)$$

is an equivalence of factorisation categories.

Proof. The enrichment supplies the coefficient object A_b and the composition map. Associativity and the unit are the enriched categorical axioms. The factorisation maps of b identify endomorphisms on disjoint discs with the external tensor product of local endomorphisms. The ordinary compact-generator Morita theorem applies on each member of a factorising basis. Compatibility with restriction identifies these equivalences on intersections, and descent glues them to the displayed equivalence. □

A choice of b gives a chart. The factorisation category is the Morita invariant primitive object; the algebra A_b is its concrete presentation.

1.5 The bar direction

The reduced bar is generated by ordered words

$$[a_1 | \cdots | a_r], \qquad a_i \in s\bar{A}.$$

Its multiplication differential joins adjacent letters. Word order records the order of intervals in $\mathbb{R}$. Every letter remains an object of $\mathrm{FactD}(X)$, so the complete chiral coefficient survives at every bar length.

Proposition 1.5.1 (Arity-three bar identity)**.** *For homogeneous $a, b, c \in \bar{A}$, the multiplication component of the bar differential satisfies*

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc]$$

under the suspension convention of Chapter A. Its square on a three-letter word is the suspended associator.

Proof. There are two codimension-one contractions of the interval configuration of three ordered subintervals. The induced orientations of their common codimension-two endpoint are opposite. The coefficient maps are the two parenthesisations $(ab)c$ and $a(bc)$. Their equality is associativity. □

Chapter 2

Factorisation coefficients on a curve

2.1 The finite-set model of the Ran category

Let $\mathsf{Fin}^{\mathrm{surj}}_{\neq\varnothing}$ be the category of nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines the partial diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{\alpha(i)})_{i\in I}.$$

For every finite set I, write $\mathcal{D}_I = \mathrm{DMod}(X^I)$ for the presentable stable category of right D_{X^I}-modules. Extraordinary pullback along the partial diagonals defines the diagram

$$I \longmapsto \mathcal{D}_I, \qquad \alpha \longmapsto \Delta_\alpha^!.$$

We use the standard limit presentation

$$\mathrm{DMod}(\mathrm{Ran}\, X) := \varprojlim_{I\in(\mathsf{Fin}^{\mathrm{surj}}_{\neq\varnothing})^{\mathrm{op}}} \mathcal{D}_I.$$

Thus a Ran D-module is a family $F_I \in \mathcal{D}_I$ together with coherent identifications $\Delta_\alpha^! F_I \simeq F_J$. Symmetric-group covariance is part of this coherence. The unital completion adjoins the empty set and sets $X^\varnothing = \mathrm{pt}$; every augmented construction below is formed in this completion. This is the finite-set realization of the Ran formalism of Beilinson–Drinfeld and Francis–Gaitsgory [BD04, FG12].

For a partition $\pi = \{B_1, \dots, B_r\}$ of I, let $U_\pi \subset X^I$ be the open locus on which points belonging to distinct blocks are pairwise distinct. Points in a single block may coincide. Write $j_\pi : U_\pi \hookrightarrow X^I$.

Definition 2.1.1 (Factorisation right D-module)**.** A *factorisation right D-module* on X is a Ran D-module F together with coherent equivalences

$$j_\pi^! F_I \xrightarrow{\sim} j_\pi^! \Big(\boxtimes_{B\in\pi} F_B\Big)$$

for every partition π. Coherence means compatibility with permutation, refinement, and every partial-diagonal identification. The resulting stable category is denoted $\mathrm{FactD}(X)$.

The restriction to U_π gives the value on separated supports. The extension of F_I across partial diagonals gives the collision coefficient. The factorisation object contains both pieces as one invariant datum.

2.2 The pointwise coefficient product

For right D-modules on a smooth variety Y, the coefficient product is

$$M \otimes^! N := \Delta_Y^!(M \boxtimes N).$$

It is symmetric monoidal and preserves colimits separately in each variable. Define it componentwise on the Ran limit:

$$(F \otimes^! G)_I = F_I \otimes^! G_I.$$

Proposition 2.2.1 (Coefficient monoidal category)**.** *The componentwise* $!$*-tensor product preserves factorisation objects and makes* $\mathrm{FactD}(X)$ *a presentable stable symmetric monoidal category. Its geometric realizations and finite tensor powers are computed componentwise.*

Proof. The category of Ran families is a limit in the category of presentable stable categories and colimit-preserving functors, hence is presentable and stable. A factorisation structure is a compatible system of equivalences in a small diagram of open-restriction functors. The category of such systems is the corresponding accessible limit; equivalently, it is the inverse image of the subspace of equivalences in the diagram category. Thus $\mathrm{FactD}(X)$ is a presentable stable category and its colimits are computed componentwise.

On U_π, apply the factorisation equivalences of F and G and the external-product identity

$$\left(\boxtimes_{B\in\pi} F_B\right) \otimes^! \left(\boxtimes_{B\in\pi} G_B\right) \simeq \boxtimes_{B\in\pi}(F_B \otimes^! G_B).$$

The refinement diagrams commute by functoriality of external product and extraordinary pullback. Componentwise !-tensor therefore restricts to $\mathrm{FactD}(X)$, preserves colimits separately, and supplies the asserted symmetric monoidal structure. ◻

This is the coefficient category used by ordered fusion. An internal multiplication is a morphism in $\mathrm{FactD}(X)$ and therefore intertwines every disjointness equivalence and every diagonal filtration.

2.3 Chiral convolution

For an ordered decomposition $d = (I_1, I_2)$ of a finite set I, let $j_d : U_d \hookrightarrow X^I$ be the locus on which the two labelled blocks are disjoint. Chiral convolution is the finite direct sum

$$(F \star_{\mathrm{ch}} G)_I = \bigoplus_{I=I_1\sqcup I_2} (j_d)_* j_d^!(F_{I_1} \boxtimes G_{I_2}).$$

The empty block supplies the unit in the unital completion. Associativity is the canonical identification of both iterated products with the sum indexed by ordered decompositions $I = I_1 \sqcup I_2 \sqcup I_3$.

Let $\mathcal{D}(\mathrm{Ran}\, X)_{\geq r}$ denote objects whose components below cardinality r vanish.

Theorem 2.3.1 (Cardinality pro-nilpotence)**.** *Chiral convolution satisfies*

$$\mathcal{D}(\mathrm{Ran}\, X)_{\geq r} \star_{\mathrm{ch}} \mathcal{D}(\mathrm{Ran}\, X)_{\geq s} \subseteq \mathcal{D}(\mathrm{Ran}\, X)_{\geq r+s}.$$

Hence the quotient by cardinality $> N$ *is nilpotent on reduced objects, and the inverse limit over* N *is pro-nilpotent for chiral convolution.*

Proof. Every summand is indexed by a disjoint union of the two supporting finite sets, so cardinalities add. An $(N+1)$-fold convolution of reduced objects has cardinality at least $N+1$ and vanishes in the Nth quotient. The inverse system of these nilpotent quotients is the cardinality completion. ◻

This support-growth mechanism is the geometric source of chiral Koszul duality [FG12]. Chiral convolution assembles disjoint finite subsets. The ordered bar uses the pointwise coefficient product. A comparison between the two constructions includes an explicit monoidal or filtered comparison map.

2.4 The complete diagonal coefficient

Let $D = \Delta(X) \subset X^2$. The meromorphic extension $\mathcal{O}_{X^2}(*D)$ carries the pole filtration

$$P_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}((m+1)D), \qquad m \geq 0.$$

For an effective Cartier divisor,

$$\mathrm{gr}^P_m \mathcal{O}_{X^2}(*D) \simeq \mathcal{O}_D((m+1)D) \simeq N^{\otimes(m+1)}_{D/X^2}.$$

Since $N_{\Delta/X^2} \simeq T_X$, the coefficient of $(z-w)^{-m-1}$ is the coordinate expression of the $(m+1)$st normal symbol. A coordinate change acts upper triangularly on this filtered object. The filtration is intrinsic; the mode coordinates form local splittings of it.

Principle 2.4.1 (Complete coefficient). A chiral operation consists of a factorisation object, its diagonal extension, and its structure morphism. Principal parts are the normal symbols of the filtered morphism. The regular product is its regular quotient together with the unit and translation structure.

2.5 Regular-holonomic realization

On the regular-holonomic subcategory, the de Rham Riemann–Hilbert functor carries each F_I to a constructible complex on $X^{I,\mathrm{an}}$. External products and restrictions to disjointness loci carry the algebraic factorisation equivalences to constructible factorisation equivalences [Del70].

Proposition 2.5.1 (Regular-holonomic realization)**.** *Riemann–Hilbert is symmetric monoidal for the pointwise coefficient product on the regular-holonomic factorisation subcategory. It carries ordered chiral algebras, modules, and finite bar stages to constructible factorisation objects. It carries a completed bar whenever the chosen completion is preserved by the realization functor.*

Proof. The statement is componentwise on X^I. Regular-holonomic external product, open restriction, and !-tensor commute with the de Rham realization. Finite simplicial realizations follow from exactness. The completed statement is the continuity assertion in the final hypothesis. □

Chapter 3

Aligned decompositions and the exact locality law

3.1 Hadamard and Cauchy products of linear species

Let $\mathcal{V}$ be a stable symmetric monoidal category with finite direct sums. A linear species is a functor

$$F : \mathsf{Fin}^{\simeq} \longrightarrow \mathcal{V}.$$

Species carry two products. The Hadamard product is componentwise,

$$(F \boxtimes_{\mathrm{H}} G)(I) = F(I) \otimes G(I),$$

and the Cauchy product is Day convolution for disjoint union,

$$(F \boxtimes_{\mathrm{C}} G)(I) = \bigoplus_{I = I_1 \sqcup I_2} F(I_1) \otimes G(I_2).$$

These products isolate the finite-set combinatorics underlying pointwise coefficient tensor and chiral convolution; compare the general Day and duoidal formalisms in [Gla16, Tor25].

For four species F_1, F_2, G_1, G_2, the arity-I component of

$$(F_1 \boxtimes_{\mathrm{C}} F_2) \boxtimes_{\mathrm{H}} (G_1 \boxtimes_{\mathrm{C}} G_2)$$

is the direct sum indexed by two ordered decompositions (d, e) of I. The arity-I component of

$$(F_1 \boxtimes_{\mathrm{H}} G_1) \boxtimes_{\mathrm{C}} (F_2 \boxtimes_{\mathrm{H}} G_2)$$

is the direct sum indexed by one ordered decomposition d. It is the common-decomposition sector (d, d) of the first sum.

Theorem 3.1.1 (Aligned-decomposition splitting)**.** *There are canonical natural transformations*

$$\iota : (F_1 \boxtimes_{\mathrm{H}} G_1) \boxtimes_{\mathrm{C}} (F_2 \boxtimes_{\mathrm{H}} G_2) \longrightarrow (F_1 \boxtimes_{\mathrm{C}} F_2) \boxtimes_{\mathrm{H}} (G_1 \boxtimes_{\mathrm{C}} G_2),$$
$$\pi : (F_1 \boxtimes_{\mathrm{C}} F_2) \boxtimes_{\mathrm{H}} (G_1 \boxtimes_{\mathrm{C}} G_2) \longrightarrow (F_1 \boxtimes_{\mathrm{H}} G_1) \boxtimes_{\mathrm{C}} (F_2 \boxtimes_{\mathrm{H}} G_2).$$

They satisfy

$$\pi\iota = \mathrm{id}, \qquad e := \iota\pi, \qquad e^2 = e.$$

The image of e is the common-decomposition summand. The transformations are compatible with associativity, units, and the symmetry of the Hadamard product.

Proof. At a finite set I, the source and target are finite direct sums. The map ι includes the summands indexed by

$$d \longmapsto (d, d),$$

and π is the canonical projection onto those summands. Their two composites are the identity and the corresponding direct-summand projector. For three Cauchy factors, both coherence composites include, respectively project onto, triples carrying one common ordered decomposition. The unit is the empty species, and symmetry exchanges the two blocks. These descriptions prove every coherence identity componentwise. □

The theorem is an exact statement in the additive category of species. It identifies the combinatorial projector selected by one common laboratory decomposition.

3.2 The sheaf-level locality square

Return to D-modules on the Ran space. Fix an ordered decomposition $d = (I_1, I_2)$ and its disjointness locus $j_d : U_d \hookrightarrow X^I$. Factorisation identifies

$$j_d^! A_I \simeq j_d^! (A_{I_1} \boxtimes A_{I_2}).$$

An internal multiplication $m : A \otimes^! A \to A$ belongs to $\mathrm{FactD}(X)$ exactly when the diagram

$$\begin{tikzcd} j_d^!(A_I \otimes^! A_I) \arrow[r, "j_d^! m"] \arrow[d, "\Delta_d \otimes^! \Delta_d"'] & j_d^! A_I \arrow[d, "\Delta_d"] \\ j_d^!\big((A_{I_1} \otimes^! A_{I_1}) \boxtimes (A_{I_2} \otimes^! A_{I_2})\big) \arrow[r, "m \boxtimes m"'] & j_d^!(A_{I_1} \boxtimes A_{I_2}) \end{tikzcd}$$

commutes for every d. Here Δ_d denotes the inverse factorisation equivalence.

Theorem 3.2.1 (Locality criterion)**.** *Let A be a factorisation right D-module and let $m : A \otimes^! A \to A$ be an E_1 multiplication in the ambient Ran category. The multiplication defines an E_1 algebra in* $\mathrm{FactD}(X)$ *precisely when all disjoint-locus squares above commute. Under the finite-set indexing functor, these squares occupy the common-decomposition sector of Theorem 3.1.1.*

Proof. A morphism in $\mathrm{FactD}(X)$ is a Ran morphism intertwining every factorisation equivalence. Applying this definition to m gives the displayed family of squares. Each square uses the same decomposition d before and after multiplication, which is exactly the common-decomposition index selected by the species projector. □

This family of squares is the complete sheaf-level bialgebra law for ordered locality. Extension across partial diagonals remains part of the factorisation D-module; the indexing projector records the separated-support combinatorics.

3.3 Aligned coefficient windows

Definition 3.3.1 (Aligned species window)**.** A full replete subcategory $\mathcal{S}$ of linear species is *aligned* when it contains the two units, is closed under the Hadamard and Cauchy products, and the idempotent e of Theorem 3.1.1 acts as the identity on every iterated mixed product of objects of $\mathcal{S}$. The inclusions and projections are required to remain morphisms in $\mathcal{S}$.

Corollary 3.3.2. *On an aligned species window, ι and π are inverse interchange isomorphisms. Their componentwise coherence identities restrict to the window, so the Hadamard and Cauchy products form a strong duoidal pair there.*

Proof. The equality $e = \mathrm{id}$ gives $\iota\pi = \mathrm{id}$, while $\pi\iota = \mathrm{id}$ holds by Theorem 3.1.1. □

Finite support conditions and fixed decomposition types supply elementary aligned windows. The general factorisation theory uses the locality squares of Theorem 3.2.1.

3.4 Bar compatibility

The simplicial ordered bar has level

$$B_r^{\mathrm{ord}}(A) = \Bbbk \otimes^! A^{\otimes^! r} \otimes^! \Bbbk.$$

Every face is built from m or the augmentation, and every degeneracy is built from the unit.

Proposition 3.4.1 (Factorising ordered bar)**.** *The simplicial ordered bar is a simplicial object in* $\mathrm{FactD}(X)$. *Its geometric realization inherits the chiral factorisation structure, and the deconcatenation coproduct is a morphism of factorisation objects.*

Proof. The locality criterion applies to every simplicial operator. Geometric realization is computed componentwise. Deconcatenation cuts a transverse word, while a factorisation map separates the chiral support letter by letter. The resulting maps commute on every disjointness locus, which is the morphism condition in $\mathrm{FactD}(X)$. □

Chapter 4

Rectangular factorisation and block-Ran recognition

4.1 The rectangular basis

Let Disk_X be the category of sufficiently small analytic discs in X^{an}, with disjoint embeddings as multimorphisms. Let $\mathsf{Disk}_1^{\mathrm{ord}}$ be the little oriented intervals operad. Their product has objects $U \times I$ and multimorphisms given by pairwise disjoint rectangular embeddings that preserve the interval order.

Definition 4.1.1 (Rectangular holomorphic–topological factorisation algebra)**.** A rectangular factorisation algebra on $X^{\mathrm{an}} \times \mathbb{R}$ is a factorisation algebra on the product-disk basis satisfying:

1. holomorphic algebraicity in the X variable, represented by a factorisation right D-module;
2. local constancy under isotopy in the interval variable;
3. Weiss descent for covers by product rectangles.

The resulting category is denoted $\mathrm{Fact}_{\mathrm{hol,lc}}(X \times \mathbb{R})$.

4.2 Exponential law

The little intervals operad acts on the category of holomorphic factorisation coefficients. Conversely, an internal E_1 algebra assigns operations to ordered families of intervals while retaining the factorisation coefficient in X.

Theorem 4.2.1 (Rectangular recognition)**.** *There is a natural equivalence*

$$\mathrm{Fact}_{\mathrm{hol,lc}}(X \times \mathbb{R}) \simeq \mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X), \otimes^!\big).$$

Under this equivalence, restriction to $X \times I$ gives the underlying factorisation coefficient, and fusion of ordered subintervals gives the internal E_1 multiplication.

Proof. A product-disk algebra is an algebra over the Boardman–Vogt tensor product $\mathsf{Disk}_X \otimes \mathsf{Disk}_1^{\mathrm{ord}}$: the tensor-product relation identifies operations performed in the two coordinate directions. The exponential law for operadic algebras identifies such structures with E_1-algebras in Disk_X-algebras. Holomorphic algebraisation identifies the latter with E_1-algebras in $\mathrm{FactD}(X)$. Explicitly, choose a standard interval I. Local constancy identifies every interval with I up to a contractible space of choices. Ordered embeddings $I_1 < \cdots < I_r \subset I$ produce the r-ary E_1 operation. The factorisation maps in the disc variable make these operations morphisms of factorisation coefficients. Conversely, an internal E_1 algebra assigns to a family of rectangles the external chiral coefficient followed by the ordered E_1 operation. The two constructions are inverse by the Segal and factorisation axioms and by the universal interchange relation of the operadic tensor product. □

This theorem gives a geometric realization of Theorem 1.3.1. It also fixes the physical meaning of its multiplication: the operation is fusion of boundary operators or parallel line defects along the topological coordinate.

4.3 Block-Ran presentation

An ordered block profile is a finite ordered list

$$\mathbf{I} = (I_1, \dots, I_r)$$

of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral refinements occur inside one block. Transverse fusion joins adjacent blocks. Units insert an empty transverse slot in the unital model.

Let BRan_X be the Grothendieck construction of the simplicial E_1 indexing category acting on the finite-set Ran diagram. A section over BRan_X is *Segal* in the ordered direction when its value on r blocks is the pointwise tensor product of its one-block values. It is *factorising* in the chiral direction when it satisfies Theorem 2.1.1 inside every block.

Theorem 4.3.1 (Block-Ran recognition)**.** *The category of ordered chiral algebras is equivalent to the category of coCartesian sections over* BRan_X *that are Segal in the block direction and factorising in every chiral block.*

Proof. The simplicial nerve of an E_1 algebra is a Segal object with rth term $A^{\otimes^! r}$. Applying this construction in the category $\mathrm{FactD}(X)$ yields a section on the block-Ran Grothendieck construction. Conversely, the Segal maps recover the multiplication from the active map $[2] \to [1]$, the unit from $[0] \to [1]$, and all higher coherence from the simplicial identities. The factorisation condition ensures that these maps belong to $\mathrm{FactD}(X)$. Straightening and unstraightening give inverse functors. □

The theorem constructs the ordered chiral operator category directly. A symbol such as $E_1 \otimes \mathsf{Ch}_X$ may be used after this construction to denote its operadic envelope; the block-Ran category supplies the actual source of that notation.

4.4 Modules and interfaces

A marked block at the left or right endpoint gives the standard coloured operad for a module. A marked block at both ends gives a bimodule. The relative tensor product is gluing of marked intervals.

Corollary 4.4.1 (Marked block recognition)**.** *Right, left, and bi-modules over an ordered chiral algebra are equivalent to marked Segal sections over the corresponding coloured block-Ran categories. For an* (A, B)*-bimodule* M *and a* (B, C)*-bimodule* N*, interval gluing gives*

$$M \otimes_B^{\mathbb{L}} N \simeq \int_{I \cup \partial I} (M, B, N).$$

4.5 Opposite order

Reflection $t \mapsto -t$ reverses the interval order and preserves the complex curve. It gives

$$\mathrm{rev} : A \longmapsto A^{\mathrm{op}}$$

with the same factorisation coefficient. Exchange-path reversal belongs to a later E_2 or meromorphic structure; Verdier duality belongs to the D-module coefficient. These three involutions act on distinct data.

Chapter 5

The two-direction cofibrant resolution

5.1 Three resolutions and three roles

The simplex indexes the monadic two-sided bar. The associahedron resolves homotopy-coherent ordered multiplication. A collision compactification resolves nested holomorphic collisions and retains the relative screen of each cluster. These constructions meet through comparison maps while retaining their separate universal properties.

5.2 Tensoring two cofibrant operadic resolutions

Fix a cellular, well-pointed, Σ-cofibrant resolution

$$P_\perp \xrightarrow{\sim} \mathsf{E}_1$$

of the associative operad and a cellular, well-pointed, Σ-cofibrant resolution

$$Q_X \xrightarrow{\sim} \mathsf{Ch}_X$$

of the selected chiral coefficient operad. Work in a monoidal model category of dg collections satisfying the following exact properties:

1. the two operads and their tensor product are admissible;
2. $P_\perp \otimes Q_X$ is cofibrant;
3. tensoring the two displayed weak equivalences produces a weak equivalence

$$P_\perp \otimes Q_X \xrightarrow{\sim} \mathsf{E}_1 \otimes \mathsf{Ch}_X;$$

4. change of operad along this map is a Quillen equivalence.

The Boardman–Vogt and rectification theorems provide these properties for the standard well-pointed Σ-cofibrant models used here [BM06, BM07].

A cellular element of $P_\perp \otimes Q_X$ can be represented by a levelled interleaving of a planar transverse tree and a chiral collision tree. Equivalently, it is represented by a two-direction forest whose internal edges record which resolution contributes the corresponding contraction. For such a forest T, write $E_\perp(T)$ and $E_{\mathrm{ch}}(T)$ for the two edge sets and let $\mathrm{Dec}_{P,Q}(T)$ be the tensor product of its operadic cell decorations. The orientation line is

$$\mathrm{or}(T) = \det\bigl(\Bbbk^{E_\perp(T)}\bigr) \otimes \det\bigl(\Bbbk^{E_{\mathrm{ch}}(T)}\bigr).$$

Definition 5.2.1 (Abstract mixed forest complex)**.** For a finite label set I, set

$$\mathsf{W}^{\mathrm{ord}}_{P,Q}(I) = \bigoplus_{T} \mathrm{Dec}_{P,Q}(T) \otimes \mathrm{or}(T).$$

Its differential is

$$d = d_{\mathrm{dec}} + d_{\perp} + d_{\mathrm{ch}},$$

where d_{dec} is the cellular differential of the decorations and the last two terms contract one internal edge in the indicated resolution.

Theorem 5.2.2 (Mixed boundary identity)**.** *With the product orientation,*

$$d_{\perp}^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_{\perp}d_{\mathrm{ch}} + d_{\mathrm{ch}}d_{\perp} = 0,$$

and d_{dec} supercommutes with both contraction differentials. Hence $d^2 = 0$.

Proof. A codimension-two cell contains two contracted edges. For two edges from the same resolution, the two contraction orders induce the opposite boundary orientations of that cellular operad. For one edge of each type, both orders produce the same decorated forest and exchange the two one-dimensional factors in the determinant line, producing a minus sign. Naturality of the cellular boundary with respect to every face map gives the two remaining anticommutators. These four cancellations are the cellular identity $\partial^2 = 0$ for the tensor product resolution. □

When endomorphism operations decorate the vertices, $d_{\perp}^2 = 0$ becomes the Stasheff relation, $d_{\mathrm{ch}}^2 = 0$ becomes the chiral coherence equation, and the mixed anticommutator becomes the interchange law saying that ordered fusion is a morphism of chiral coefficients.

Theorem 5.2.3 (Cofibrant comparison)**.** *Under the model-categorical hypotheses above,*

$$P_{\perp} \otimes Q_X \longrightarrow \mathsf{E}_1 \otimes \mathsf{Ch}_X$$

is a cofibrant resolution. Its algebras are homotopy-coherent ordered chiral algebras, and its localized algebra category is equivalent to

$$\mathrm{Alg}_{E_1}\big(\mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V})\big).$$

For the selected D-module realization of Ch_X, this is $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. Cofibrancy and weak equivalence are hypotheses 2 and 3. Admissibility and hypothesis 4 turn this weak equivalence into a Quillen equivalence of algebra categories. The universal property of the operadic tensor product identifies an algebra over $\mathsf{E}_1 \otimes \mathsf{Ch}_X$ with an E_1 algebra internal to Ch_X-algebras. □

5.3 Geometric collision screens

For points on the real surface underlying a complex curve, the real oriented blowup replaces a collision cluster by its relative real configuration modulo translation and positive dilation. The algebraic Fulton–MacPherson compactification is the complex-algebraic companion [FM94]. Their strata are indexed by nested sets or rooted collision forests. The incidence category records the nested clusters. The screen records their relative positions and carries genuine cohomology.

For three points on the affine complex line, the complex-algebraic screen modulo complex translation and complex dilation is

$$\mathbb{P}^1 \setminus \{0, 1, \infty\},$$

whose first cohomology has rank two. The corresponding incidence fibre has a terminal corolla and hence contractible nerve. The screen coefficient therefore contains information absent from the incidence poset.

Assume now that a chosen collision compactification has a finite normal-crossing boundary filtration and that its screen-chain coefficient system presents the chiral resolution Q_X. Define the geometric mixed complex by

$$\mathsf{W}^{\mathrm{ord}}_{X,\mathrm{geom}}(I) = \bigoplus_T C_*\big(\mathrm{Screen}_X(T)\big) \otimes C_*\big(K_\perp(T)\big) \otimes \mathrm{or}(T),$$

where $K_\perp(T)$ is the associahedral cell attached to the transverse planar tree.

Comparison Theorem 5.3.1 (Geometric forest comparison)**.** *If the compactification face maps agree with forest contraction and the screen chains define Q_X, there is a filtered quasi-isomorphism*

$$\mathsf{W}^{\mathrm{ord}}_{X,\mathrm{geom}} \xrightarrow{\simeq} \mathsf{W}^{\mathrm{ord}}_{P,Q}.$$

Proof. Filter both complexes by the total number of internal edges. The associated graded of the geometric boundary complex is the direct sum over strata of screen chains, associahedral cells, and the determinant of the normal-edge set. This is the associated graded of the abstract tensor-product resolution because the screen coefficient presents Q_X and the associahedral coefficient presents $P_\perp$. The comparison is an isomorphism on the first page. The filtration is finite in each arity, so the spectral sequences converge and the comparison is a quasi-isomorphism. □

5.4 Homotopy transfer and Feynman weights

Let

$$(H, d_H) \underset{p}{\overset{i}{\rightleftarrows}} (V, d_V), \qquad ip - \mathrm{id}_V = d_V h + h d_V$$

be a deformation retract. Evaluation of a decorated forest places the original operations at vertices and h on internal edges. The mixed boundary identity proves the Maurer–Cartan equation for the transferred operations.

In perturbative field theory, the same evaluation places interaction vertices at vertices and propagators on edges. Renormalisation extends each weight to the compactification, and restriction to a boundary face contracts the corresponding divergent subgraph.

Theorem 5.4.1 (Feynman weight morphism)**.** *Let a perturbatively renormalised holomorphic–topological BV theory have weights extending to the geometric mixed compactifications. Assume that restriction to every boundary stratum factorises by graph contraction and that the renormalised quantum master equation holds. Then the graph weights define a dg-operad morphism*

$$\mathsf{W}^{\mathrm{ord}}_{X,\mathrm{geom}} \longrightarrow \mathrm{End}_{\mathrm{Obs}_\partial}.$$

Composing with a homotopy inverse of Theorem 5.3.1 gives the corresponding action of the abstract cofibrant resolution.

Proof. Boundary factorisation is the operadic composition identity. Stokes' theorem for the extended weights, including BV and counterterm faces, is the chain-map identity. The quantum master equation supplies the cancellation of the total codimension-one boundary. These are exactly the axioms of a dg-operad morphism. □

Part II

Bar, cobar, Morita, and centre

Chapter 6

The ordered bar construction

6.1 The simplicial object

Let A be an augmented ordered chiral algebra. The two-sided simplicial bar in $\mathrm{FactD}(X)$ is

$$B_r^{\mathrm{ord}}(A) = \Bbbk \otimes^! A^{\otimes^! r} \otimes^! \Bbbk, \qquad r \geq 0.$$

The interior face d_i multiplies the ith and $(i+1)$st factors. The two outer faces use the augmentation. A degeneracy inserts the unit. Its geometric realization is

$$\overline{B}_X^{\mathrm{ord}}(A) = \left|B_\bullet^{\mathrm{ord}}(A)\right|.$$

All tensor products and colimits are taken in the coefficient category, so $\overline{B}_X^{\mathrm{ord}}(A)$ remains a factorisation right D-module.

6.2 Normalized chains

Put $\bar{A} = \mathrm{fib}(A \to \Bbbk)$. Under the cohomological suspension convention, normalization identifies the underlying graded object with

$$T^c(s\bar{A}) = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes^! r}.$$

For a strict dg algebra the differential is

$$d_B = d_{\mathrm{int}} + b_m,$$

where d_{int} differentiates one letter and b_m multiplies two adjacent letters. For an A_∞ algebra it becomes

$$d_B = d_{\mathrm{int}} + \sum_{k \geq 2} b_{m_k},$$

with b_{m_k} replacing k consecutive letters by m_k.

Theorem 6.2.1 (Ordered bar coalgebra)**.** *The deconcatenation formula*

$$\Delta[a_1|\cdots|a_r] = \sum_{p=0}^{r} [a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r]$$

makes $\overline{B}_X^{\mathrm{ord}}(A)$ a conilpotent coassociative dg coalgebra object in $\mathrm{FactD}(X)$*. The bar differential is a coderivation. Its coradical filtration is the transverse word-length filtration.*

Proof. The two ways of cutting a word twice give the same sum over two cut positions, which proves coassociativity. Each term of the bar differential is local on a consecutive interval of letters. Comparing a cut before and after this local contraction gives the coderivation identity. Iterating the reduced coproduct of a word of length r more than r times gives zero, which proves conilpotence. □

Under the Francis–Gaitsgory realization of a factorisation object as a commutative coalgebra for chiral convolution, Chapter 3 equips the bar with the corresponding chiral factorisation coproduct. The two coproducts record two independent cuts: one cuts the ordered interval, the other separates the holomorphic support.

6.3 Interval factorisation homology

Let the two endpoints of $[0,1]$ carry the augmentation module $\Bbbk$. Excision identifies the interval amplitude with the derived relative tensor product.

Theorem 6.3.1 (Bar as an interval amplitude)**.** *There are natural equivalences in* $\mathrm{FactD}(X)$

$$\overline{B}_X^{\mathrm{ord}}(A) \simeq \Bbbk \otimes_A^{\mathbb{L}} \Bbbk \simeq \int_{([0,1],\partial[0,1])} A.$$

More generally, for a right A-module R and a left A-module L,

$$\mathrm{Bar}(R, A, L) \simeq R \otimes_A^{\mathbb{L}} L \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Cover the interval by an ordered chain of small intervals and endpoint collars. The factorisation Čech diagram is the two-sided simplicial bar. Its colimit computes both factorisation homology and the derived relative tensor product. Refinement of the cover is cofinal, so the equivalence is canonical. □

The theorem gives the bar its physical meaning. A bar word is an ordered sequence of intermediate boundary operations. The differential composes adjacent operations. The endpoints select the boundary states.

6.4 The associahedral model

A strict multiplication gives the simplicial bar. A homotopy-coherent multiplication is resolved by the Stasheff associahedra K_r. Their cellular chains form a cofibrant model of the associative operad.

Theorem 6.4.1 (Associahedral comparison)**.** *Work in the admissible monoidal dg model fixed in Chapter 5, and let A be cofibrant as an algebra over the Boardman–Vogt resolution of* Ass*. The operadic bar admits a cellular filtration with terms*

$$\bigoplus_{r\geq 0} C_*(K_r) \otimes (s\bar{A})^{\otimes^! r}.$$

After strictification, there is a natural zigzag of quasi-isomorphisms

$$\mathrm{Bar}_{A_\infty}(A) \simeq \overline{B}_X^{\mathrm{ord}}(A^{\mathrm{str}}).$$

Proof. The cellular boundary of K_r is the signed sum over graftings of two planar trees. Evaluating a cell by the corresponding A_∞ operation produces the Stasheff bar differential. The Boardman–Vogt augmentation $C_*(K_\bullet) \to \mathsf{Ass}$ is a cofibrant resolution in this model category. The rectification Quillen equivalence and homotopy invariance of derived relative tensor products give the zigzag. □

The simplex and the associahedron now have exact roles: the simplex presents the derived tensor product; the associahedron resolves the coherent multiplication used to evaluate it.

6.5 Completion

The completed bar is

$$\widehat{\overline{B}^{\mathrm{ord}}}_X(A) = \prod_{r \geq 0} (s\bar{A})^{\otimes^! r}.$$

A weight grading with strictly positive augmentation ideal makes the product complete and separated. More generally, let F^N be a decreasing complete filtration for which every m_k raises total filtration by at least $k - 1$.

Proposition 6.5.1 (Convergence of the completed bar)**.** *Assume that $A/F^N A$ has finite bar length in each total weight and that the tower satisfies the Mittag–Leffler condition on cohomology. Then*

$$H^*(\widehat{\overline{B}^{\mathrm{ord}}}_X(A)) \cong \varprojlim_N H^*(\overline{B}^{\mathrm{ord}}_X(A/F^N A)).$$

The completion is separated, and the bar differential is continuous.

Proof. The finite quotients form a tower of complete complexes. The Mittag–Leffler condition kills the derived inverse-limit term $\varprojlim^1$. Continuity follows because every summand of the bar differential raises the filtration by a controlled amount. □

6.6 Fubini

Let $p : X \times \mathbb{R} \to X$ and $q : X \times \mathbb{R} \to \mathbb{R}$ be the projections. When the relevant factorisation homologies exist and preserve the geometric realizations above, the two integrations commute.

Theorem 6.6.1 (Ordered chiral Fubini)**.** *Suppose $\int_X$ preserves the tensor powers and geometric realizations of the ordered bar. Then*

$$\int_X \overline{B}^{\mathrm{ord}}_X(A) \simeq \mathrm{Bar}\left(\int_X A\right).$$

If the cyclic realization is preserved as well, then

$$\int_X \mathrm{HH}_*(A) \simeq \mathrm{HH}_*\left(\int_X A\right).$$

Proof. Both sides of the first equivalence are the colimit of the bisimplicial diagram obtained from chiral disks in X and the two-sided bar in the interval. Colimits commute. The second statement applies the same argument to the cyclic bar. □

Chapter 7

Bar–cobar reconstruction and Koszul duality

7.1 Cobar

Let C be a coaugmented conilpotent coassociative coalgebra in $\mathrm{FactD}(X)$ and write $\bar{C}$ for its coaugmentation coideal. Its cobar is

$$\mathrm{Cobar}_X(C) = \left(T(s^{-1}\bar{C}), d_C + d_\Delta\right),$$

where d_Δ is the derivation induced by the reduced coproduct. The universal twisting morphism

$$\tau_A : \overline{B}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is the projection to one-letter words followed by desuspension.

7.2 Nilcompletion and conilcompletion

For an augmented algebra A, the operadic lower central tower produces its nilcompletion $A^\wedge_{\mathrm{nil}}$. For a coaugmented coalgebra C, the coradical tower produces its conilcompletion $C^\wedge_{\mathrm{conil}}$. In a positively weighted algebra these agree with ordinary completion in the augmentation and coradical filtrations.

The precise general scope of operadic Koszul duality is the nilcomplete and conilcomplete locus. This is the maximal equivalence theorem of Heuts [Heu24].

Theorem 7.2.1 (Internal associative Koszul duality)**.** *Let $\mathcal{V}$ be a stable presentable symmetric monoidal ∞-category whose tensor product preserves colimits separately. Bar and cobar restrict to inverse equivalences*

$$\mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathcal{V})_{\mathrm{nilcomp}} \simeq \mathrm{Coalg}^{\mathrm{coaug}}_{E_1}(\mathcal{V})_{\mathrm{conilcomp}}.$$

Applied to $\mathcal{V} = \mathrm{FactD}(X)$, this gives ordered chiral bar–cobar duality.

Proof. The associative operad is reduced and Koszul self-dual up to suspension. The general bar–cobar adjunction therefore identifies the unit and counit after nilcompletion and conilcompletion. Heuts' maximality theorem proves that these are the largest full subcategories on which both unit and counit are equivalences. The hypotheses on $\mathrm{FactD}(X)$ follow from Theorem 2.2.1. □

7.3 Positive-weight proof

The abstract theorem has a direct proof in the connected positive-weight case. Let

$$A = \Bbbk \oplus \bigoplus_{w\geq 1} A_w, \qquad A_u \otimes^! A_v \longrightarrow A_{u+v}.$$

Filter $\mathrm{Cobar}\,\overline{B}^{\mathrm{ord}}(A)$ by the number of bars that have survived the cobar multiplication. The associated graded is the tensor algebra on a contractible simplicial resolution of each positive-weight word.

Theorem 7.3.1 (Positive reconstruction)**.** *Let $A = \Bbbk \oplus \bigoplus_{w>0} A_w$ be connected and complete, and assume that each weight has finitely many decompositions into positive weights. Then the counit*

$$\mathrm{Cobar}_X\,\overline{B}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is a filtered quasi-isomorphism. For a conilpotent weight-complete coalgebra with the same finite-decomposition property, the unit

$$C \longrightarrow \overline{B}_X^{\mathrm{ord}}\,\mathrm{Cobar}_X(C)$$

is a filtered quasi-isomorphism.

Proof. Fix total weight w. Finitely many bar words contribute. The bar of a free word has the augmented simplex as its indexing complex, and the extra degeneracy contracts every subdivided word onto the single unsplit word. Thus the associated graded counit is a quasi-isomorphism in every weight. Completeness and exhaustiveness promote the associated-graded equivalence to the filtered equivalence. The coalgebra statement is dual, using the finite coradical filtration in each weight. □

7.4 Boundary Koszul dual

Definition 7.4.1 (Augmentation-boundary algebra)**.** The Koszul dual of A at its augmentation is

$$A^! := \mathrm{RHom}_A(\Bbbk, \Bbbk).$$

Composition of derived endomorphisms gives an E_1 algebra in $\mathrm{FactD}(X)$. The bar resolution gives

$$A^! \simeq \mathrm{RHom}_{\mathrm{FactD}(X)}(\overline{B}_X^{\mathrm{ord}}(A), \Bbbk)$$

whenever the bar terms are dualisable or the continuous internal Hom is used.

Theorem 7.4.2 (Double centralizer)**.** *Let A be a connective augmented algebra for which the derived augmentation-completion functor is computed by the cobar totalization of the Amitsur complex of $A \to \Bbbk$, and assume that $\Bbbk$ is compact in the completed module category. Then*

$$A^{!!} = \mathrm{RHom}_{A^!}(\Bbbk, \Bbbk) \simeq A^{\wedge}_{\varepsilon},$$

the derived augmentation completion of A.

Proof. The bar resolution identifies the thick subcategory generated by $\Bbbk$ with perfect $A^!$-modules. The endomorphisms of the corresponding Yoneda generator are the double centralizer. Completion along the kernel of the augmentation is the universal algebra acting identically on this augmentation-generated subcategory. The cobar totalization is the derived augmentation completion, which gives the displayed equivalence. On the nilcomplete locus this completion agrees with A itself. □

7.5 Quadratic Koszul algebras

Let V be a dualisable coefficient object and

$$A = T(V)/(R), \qquad R \hookrightarrow V \otimes^! V.$$

The quadratic dual algebra is

$$A^{q!} = T(V^*[-1])/(R^\perp).$$

The quadratic dual coalgebra embeds in the bar and generates the Koszul complex.

Theorem 7.5.1 (Quadratic recognition)**.** *Assume that the weight components are dualisable and that the diagonal Koszul complex is exact in positive homological degree. Then the inclusion of the quadratic dual coalgebra into $\overline{B}_X^{\mathrm{ord}}(A)$ is a quasi-isomorphism, and*

$$A^! \simeq A^{q!}.$$

Proof. Filter the bar by tensor weight. Its linear strand is the quadratic Koszul complex. Exactness in positive degree collapses the spectral sequence onto the quadratic dual coalgebra. Continuous duality converts the latter into the stated quadratic dual algebra. □

7.6 PBW and curvature

A filtered relation may contain a linear term and a constant term:

$$r - \alpha(r) - \beta(r), \qquad r \in R, \qquad \alpha : R \to V, \quad \beta : R \to \Bbbk.$$

The linear term becomes a differential on the quadratic dual. The constant term becomes a curvature element.

Theorem 7.6.1 (Curved PBW duality)**.** *Suppose the Braverman–Gaitsgory PBW conditions hold for the filtered algebra $A_{\alpha,\beta}$. Then its associated graded is $T(V)/(R)$, and its Koszul dual is the curved dg algebra*

$$\big(A^{q!}, d_\alpha, h_\beta\big), \qquad d_\alpha^2 = [h_\beta, -].$$

Conversely, this curved identity is equivalent to the overlap conditions in filtration degree three.

Proof. The degree-three overlap ambiguity lies in $(R \otimes V) \cap (V \otimes R)$. Its filtration-two part gives the derivation d_α; its filtration-one and filtration-zero parts give the equation $d_\alpha^2 = [h_\beta, -]$ and $d_\alpha(h_\beta) = 0$. These are exactly the PBW conditions of [BG96]. The Rees spectral sequence then identifies the associated graded with the quadratic algebra. □

Chapter 8

The Koszul–Morita square

8.1 The positive-complete module categories

Let
$$A = \Bbbk \oplus \prod_{w \geq 1} A_w, \qquad A_u \otimes^! A_v \longrightarrow A_{u+v},$$
be a connected positively weighted augmented algebra in $\mathrm{FactD}(X)$, complete for the positive-weight filtration. Assume that every weight has finitely many decompositions into positive weights. Put
$$C = \overline{B}_X^{\mathrm{ord}}(A) = \Bbbk \oplus \prod_{w \geq 1} C_w$$
and let $\tau : C \to A$ be the universal twisting cochain.

A *positive-complete left A-module* is a module
$$M = \prod_{w \geq w_0} M_w$$
for some integer w_0, with $A_u \otimes^! M_v \to M_{u+v}$, complete for its weight filtration. A *conilcomplete left C-comodule* is a weight-complete comodule
$$N = \prod_{w \geq w_0} N_w$$
whose reduced coaction raises the coradical length and whose weight-w component meets only finitely many iterated coaction summands. Write these categories as
$$\mathrm{Mod}_A^{+\wedge} \quad \text{and} \quad \mathrm{CoMod}_C^{+\wedge}.$$

8.2 One-sided twisting

For $M \in \mathrm{Mod}_A^{+\wedge}$ define
$$K_A(M) = C \otimes^! M$$
with differential
$$d_K = d_C + d_M + (\mathrm{id}_C \otimes \mu_M)(\mathrm{id}_C \otimes \tau \otimes \mathrm{id}_M)(\Delta_C \otimes \mathrm{id}_M).$$
The coproduct on the first factor makes $K_A(M)$ a left C-comodule. For $N \in \mathrm{CoMod}_C^{+\wedge}$ define
$$L_C(N) = A \otimes^! N$$
with the dual twisting differential induced by the C-coaction, τ, and the multiplication of A.

Theorem 8.2.1 (Positive-complete module–comodule duality)**.** *The twisting functors are inverse equivalences*

$$K_A : \mathrm{Mod}_A^{+\wedge} \simeq \mathrm{CoMod}_C^{+\wedge} : L_C.$$

The unit and counit are morphisms in $\mathrm{FactD}(X)$*; hence the equivalence retains every chiral factorisation map and every diagonal filtration carried by the coefficients.*

Proof. Filter $L_C K_A(M)$ by the total number of bar cuts. In fixed total weight only finitely many cut patterns occur. The associated graded is the augmented simplicial subdivision complex of each ordered positive-weight word, tensored with its coefficient in M. The extra degeneracy that joins the first cut contracts this complex onto the unsplit word. Thus the associated-graded unit

$$M \longrightarrow L_C K_A(M)$$

is an equivalence in every weight. Completeness of M and finite decomposition of weights identify the total object with the inverse limit of its finite weight windows, so the unit is an equivalence.

Filter $K_A L_C(N)$ by coradical length. In each fixed weight, conilpotence makes this filtration finite. The associated graded is the same augmented subdivision complex, now contracted onto the original comodule cogenerator. The counit

$$K_A L_C(N) \longrightarrow N$$

is therefore an equivalence in every finite window, and conilcompletion gives the full equivalence. The twisting terms use only multiplication, coaction, and τ, all of which are morphisms in $\mathrm{FactD}(X)$. □

8.3 Two-sided twisting

Let A and B be positive-complete augmented algebras with bars C_A and C_B. For a complete (A, B)-bimodule M define

$$K^{\mathrm{bi}}_{A,B}(M) = C_A \otimes^! M \otimes^! C_B$$

with the left and right twisting terms. The two terms anticommute because the left and right module actions commute. The result is a (C_A, C_B)-bicomodule. The reverse construction is

$$L^{\mathrm{bi}}_{C_A,C_B}(N) = A \otimes^! N \otimes^! B$$

with the two coaction-twisting terms.

Theorem 8.3.1 (Bimodule–bicomodule duality)**.** *The two-sided twisting functors give inverse equivalences between positive-complete (A, B)-bimodules and conilcomplete (C_A, C_B)-bicomodules. For $A = B$, the regular diagonal bimodule A is sent to the regular diagonal bicomodule C_A.*

Proof. Apply Theorem 8.2.1 first to the left action and then to the right action. The two weight filtrations form a bounded-below bifiltration in every finite total-weight window, so the two contractions commute. For the regular bimodule, the resulting two-sided twisted object is the standard two-sided bar resolution. Multiplication contracts its middle A-factor, leaving C_A with its left and right regular coactions. □

8.4 Tensor becomes cotensor

Let M be a positive-complete (A, B)-bimodule and let N be a positive-complete (B, D)-bimodule. Their composition is the derived relative tensor product $M \otimes^{\mathbb{L}}_B N$. On the coalgebra side, composition is the derived cotensor product over C_B.

Theorem 8.4.1 (Tensor–cotensor transport)**.** *There is a natural equivalence of (C_A, C_D)-bicomodules*

$$K^{\mathrm{bi}}_{A,D}\big(M \otimes^{\mathbb{L}}_{B} N\big) \simeq K^{\mathrm{bi}}_{A,B}(M)\square^{\mathbb{R}}_{C_B} K^{\mathrm{bi}}_{B,D}(N).$$

The equivalence is associative under composition of three bimodules.

Proof. Compute $M \otimes^{\mathbb{L}}_{B} N$ by the two-sided simplicial bar

$$\big|[r] \longmapsto M \otimes^{!} B^{\otimes^{!} r} \otimes^{!} N\big|.$$

Apply the twisting functor in each simplicial degree. The outer A- and D-twisting factors remain unchanged. The middle B-bar becomes the cosimplicial cobar object that computes the homotopy equalizer of the two C_B-coactions. Fixed-weight finiteness permits interchange of realization and totalization; completeness then passes the equivalence to the inverse limit. Both parenthesizations for three bimodules arise from the same trisimplicial bar, which proves associativity. $\square$

8.5 Hochschild and coHochschild theories

For a positive-complete A-bimodule M, write

$$C^{\bullet}(A, M) = \mathrm{RHom}_{A^e}(A, M), \qquad C_{\bullet}(A, M) = M \otimes^{\mathbb{L}}_{A^e} A, \qquad A^e = A \otimes^{!} A^{\mathrm{op}}.$$

For a C-bicomodule P, define

$$\mathrm{coCH}^{\bullet}(C, P) = \mathrm{RHom}_{C\text{-bicomod}}(C, P), \qquad \mathrm{coCH}_{\bullet}(C, P) = P \square^{\mathbb{R}}_{C^e} C.$$

Theorem 8.5.1 (Hochschild–coHochschild comparison)**.** *For every positive-complete A-bimodule M,*

$$C^{\bullet}(A, M) \simeq \mathrm{coCH}^{\bullet}\big(C, K^{\mathrm{bi}}_{A,A}(M)\big),$$

and

$$C_{\bullet}(A, M) \simeq \mathrm{coCH}_{\bullet}\big(C, K^{\mathrm{bi}}_{A,A}(M)\big).$$

These equivalences intertwine composition of Hochschild cochains with composition of bicomodule endomorphisms.

Proof. The equivalence of Theorem 8.3.1 sends the regular bimodule A to the regular bicomodule C and preserves derived mapping objects. This gives the cochain statement. The chain statement is Theorem 8.4.1 applied to the two regular module actions. Since the equivalence is enriched, composition of maps is preserved. $\square$

8.6 The Koszul–Morita square

The augmentation object $\Bbbk$ belongs to both sides of Theorem 8.2.1. Its derived endomorphisms therefore agree:

$$A^{!} = \mathrm{RHom}_A(\Bbbk, \Bbbk) \simeq \mathrm{RHom}_{C\text{-comod}}(\Bbbk, \Bbbk).$$

Together with $C = \Bbbk \otimes^{\mathbb{L}}_{A} \Bbbk$, this gives the commutative square

$$\begin{tikzcd}
A \arrow[r, "\overline{B}^{\mathrm{ord}}_{X}"] \arrow[d, "\Bbbk\text{-module}"'] & C \arrow[d, "\Bbbk\text{-comodule}"] \\
A^{!} = \mathrm{RHom}_A(\Bbbk, \Bbbk) \arrow[r, "\simeq"'] & \mathrm{RHom}_{C\text{-comod}}(\Bbbk, \Bbbk).
\end{tikzcd}$$

The interval evaluates the upper arrow. The augmentation boundary evaluates the lower arrow. The annulus and the derived centre arise from the corresponding two-sided transform.

8.7 Calabi–Yau trace transport

Suppose the positive-complete perfect A-module category carries a d-Calabi–Yau trace

$$\mathrm{Tr}_A : \mathrm{HH}_d(A) \longrightarrow \Bbbk.$$

Under Theorem 8.5.1, this becomes a cyclic cotrace on the regular bar bicomodule. Evaluation against the transported pairing contracts every ribbon edge. This is the algebraic input for the surface sewing of Chapter 18.

Chapter 9

The derived centre and the bulk action

9.1 Internal Hochschild cochains

For an ordered chiral algebra A, define its derived centre by

$$Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A) := \mathrm{RHom}_{A\otimes^! A^{\mathrm{op}}}(A, A) \in \mathrm{FactD}(X).$$

A bar model is

$$Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A) \simeq \mathrm{RHom}\big(\overline{B}^{\mathrm{ord}}(A), A\big),$$

with the Hochschild differential and brace operations.

Theorem 9.1.1 (Internal higher Deligne theorem)**.** *The object $Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A)$ carries a canonical E_2-algebra structure internal to* $\mathrm{FactD}(X)$*. Its restriction along $E_1 \hookrightarrow E_2$ is the cup product, and its shifted Lie bracket is the Gerstenhaber bracket.*

Proof. The brace operad acts on the Hochschild cochains of every E_1 algebra in a suitable symmetric monoidal category. The brace operad is a chain model of E_2 [Tam02, Lur17]. All brace insertions are built from the multiplication and tensor product in $\mathrm{FactD}(X)$, so the action remains internal to the coefficient category. □

The E_2 structure belongs to the centre. It is distinct from an E_2 structure on the original ordered algebra and from a braiding on its module category.

9.2 Universal central action

A central action of an E_2 algebra Z on A is an E_2 map from Z to the endomorphisms of the identity A-bimodule. The derived centre represents these actions.

Theorem 9.2.1 (Universal bulk algebra)**.** *For every E_2 algebra Z in* $\mathrm{FactD}(X)$*, maps*

$$Z \longrightarrow Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A)$$

are naturally equivalent to coherent central actions of Z on A. Thus $Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A)$ is the universal algebraic bulk acting on the ordered boundary algebra.

Proof. A central action is a map into the E_2 endomorphism object of the diagonal bimodule. By definition this endomorphism object is $\mathrm{RHom}_{A^e}(A, A)$. The brace structure records the compatibility of multiple bulk insertions and their action on boundary words. □

9.3 Categorical centre and the identity bimodule

Let Bimod(FactD(X)) be the Morita 2-category of algebra objects, bimodules, and bimodule maps. The identity 1-morphism of A is the diagonal bimodule ${}_AA_A$. Its derived endomorphisms form an E_2 algebra by the two compositions available in a 2-category.

Proposition 9.3.1 (Identity-bimodule endomorphisms)**.** *There are canonical equivalences*

$$\mathrm{End}_{\mathrm{Bimod}(\mathrm{FactD}(X))}({}_AA_A) \simeq \mathrm{End}_{\mathrm{Fun}^L(\mathrm{Mod}_A,\mathrm{Mod}_A)}(\mathrm{id}_{\mathrm{Mod}_A}) \simeq Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A).$$

The two categorical compositions correspond to the cup product and the brace E_2 structure.

Proof. Enriched Morita theory identifies colimit-preserving endofunctors of Mod_A with A-bimodules and sends the identity functor to the diagonal bimodule. Derived natural endomorphisms of the identity therefore equal $\mathrm{RHom}_{A^e}(A, A)$. Horizontal and vertical composition give the interchange structure whose chain model is the brace action. □

When a coproduct equips Mod_A with a monoidal product, its Drinfeld centre is the category of modules with half-braidings. The proposition identifies the Hochschild bulk at the Morita 2-categorical level; a model-specific functor can then compare it with objects and morphisms in that Drinfeld centre.

9.4 The physical bulk-to-centre map

Let a holomorphic–topological theory on a half-space have bulk observables $\mathrm{Obs}_{\mathrm{bulk}}$ and boundary algebra A. Bringing a bulk observable to the boundary produces a coherent central action, hence a canonical map

$$\mathfrak{oc} : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A).$$

Recognition Theorem 9.4.1 (Open–closed completeness)**.** *Assume:*

1. *the chosen boundary condition is a factorisation-compatible compact generator of the protected boundary category;*
2. *the action map detects bulk cohomology, so $H^*(\mathfrak{oc})$ is injective;*
3. *every cohomology class of a coherent central natural transformation of the identity boundary functor is represented by a protected bulk observable, so $H^*(\mathfrak{oc})$ is surjective.*

Then

$$\mathfrak{oc} : \mathrm{Obs}_{\mathrm{bulk}} \xrightarrow{\simeq} Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A)$$

is a quasi-isomorphism of internal E_2 algebras.

Proof. The first hypothesis identifies the boundary category with perfect A-modules and identifies coherent central natural transformations with $\mathrm{RHom}_{A^e}(A, A)$. The second and third hypotheses make the cohomology map of $\mathfrak{oc}$ injective and surjective. Hence $\mathfrak{oc}$ is a quasi-isomorphism. Its construction by bulk insertion respects the configuration-space operations, so the equivalence preserves the internal E_2 structures. □

This recognition theorem expresses the exact content of bulk reconstruction. The map exists for every boundary condition; the equivalence records the additional completeness property.

9.5 Morita invariance

Theorem 9.5.1 (Morita invariance of the centre)**.** *An invertible factorisation* (A, B)*-bimodule induces an equivalence*

$$Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A) \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{ord,ch}}(B)$$

of internal E_2 *algebras.*

Proof. Tensoring with the invertible bimodule gives an equivalence of bimodule categories and carries the diagonal A-bimodule to the diagonal B-bimodule. Derived endomorphisms and their brace compositions are preserved. □

9.6 Bar and bulk

The bar $\overline{B}^{\mathrm{ord}}(A)$ is the interval resolution of the augmentation. The bulk $Z^{\mathrm{der}}(A)$ is the endomorphism algebra of the diagonal bimodule. The bar computes the centre through

$$Z^{\mathrm{der}}(A) \simeq \mathrm{RHom}(\overline{B}^{\mathrm{ord}}_{A^e}(A), A),$$

and remains the coalgebraic resolution. The centre is the displayed derived endomorphism object. This separation is the structural hinge connecting Koszul duality and open–closed field theory.

Part III

The geometry of the chiral coefficient

Chapter 10

The diagonal and the complete operator product

10.1 Chiral operations

Let $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$ and $\Delta : X \hookrightarrow X^2$. For right D_X-modules M, N, P, a binary chiral operation is a morphism

$$\mu : j_* j^! (M \boxtimes N) \longrightarrow \Delta_! P.$$

The source permits meromorphic singularities along the diagonal; the target is a distribution supported on the diagonal. Composition is defined on partial diagonals in X^n. The chiral operad is the system of these multi-input operations.

A chiral Lie algebra is a right D_X-module $\mathfrak{g}$ with an antisymmetric binary chiral operation satisfying the Jacobi identity on X^3. A unital chiral algebra in the sense of Beilinson–Drinfeld is equivalently a factorisation algebra with the corresponding diagonal operations [BD04, FBZ04].

10.2 Formal principal parts

Choose a formal coordinate z at $x \in X$ and write $t = z_1 - z_2$. A local section of $j_* j^! (M \boxtimes N)$ has a finite principal part

$$\sum_{r=0}^{N-1} t^{-r-1} c_r(z_2)$$

after choosing local trivialisations. The coefficient c_r is the local representative of the $(r+1)$st normal symbol.

For the diagonal divisor D, the exact sequence

$$0 \longrightarrow \mathcal{O}(mD) \longrightarrow \mathcal{O}((m+1)D) \longrightarrow N_{D/X^2}^{\otimes(m+1)} \longrightarrow 0$$

shows that the full principal part is a filtered geometric object. The residue is the first quotient. Higher poles are the higher quotients. The regular product is the image in the quotient by the entire polar submodule.

Theorem 10.2.1 (Principal-part reconstruction)**.** *Let K be an object equipped with a finite pole filtration*

$$P_{-1}K \subset P_0 K \subset \cdots \subset P_{N-1}K = K$$

in the stable category of right D-modules, and put $Q_r = \mathrm{cofib}(P_{r-1}K \to P_rK)$. Then K is reconstructed from $P_{-1}K$, the graded pieces Q_r, and actual connecting morphisms

$$\kappa_r : Q_r \longrightarrow P_{r-1}K[1], \qquad 0 \le r < N.$$

The homotopy class $[\kappa_r] \in \mathrm{Ext}^1(Q_r, P_{r-1}K)$ determines the equivalence class of the rth extension. For the standard meromorphic coefficient $K = E \otimes \mathcal{O}_{X^2}(ND)$ with E locally free near D, one has

$$Q_r \simeq E|_D \otimes N_{D/X^2}^{\otimes(r+1)}.$$

A filtered chiral operation is equivalently a compatible morphism of these iterated extension diagrams.

Proof. Every filtration step sits in a distinguished triangle

$$P_{r-1}K \longrightarrow P_rK \longrightarrow Q_r \xrightarrow{\kappa_r} P_{r-1}K[1].$$

The middle term is the shifted fibre of the actual connecting morphism,

$$P_rK \simeq \mathrm{fib}(\kappa_r)[-1],$$

and this fibre is functorial up to a contractible choice. Replacing κ_r by a homotopic map gives an equivalent extension. Induction reconstructs the full filtered object. For the standard meromorphic coefficient, the divisor exact sequence identifies the displayed graded quotient. A filtered morphism is exactly a morphism of the resulting tower of distinguished triangles. □

This theorem gives a precise meaning to the complete OPE. A list of residues determines the first symbol. The filtered extension determines the complete singularity and its coordinate covariance.

10.3 Delta derivatives

On the formal disc, the distribution

$$\frac{1}{(z-w)^{r+1}}$$

corresponds under residue pairing to

$$\frac{1}{r!}\partial_w^r \delta(z-w).$$

Thus a singular OPE

$$a(z)b(w) \sim \sum_{r=0}^{N-1} \frac{a_{(r)}b(w)}{(z-w)^{r+1}}$$

is the coordinate form of the diagonal distribution

$$\sum_{r=0}^{N-1} \frac{1}{r!}(a_{(r)}b)(w)\, \partial_w^r \delta(z-w).$$

The D-module packages the derivatives and their coordinate-change law.

10.4 Coordinate changes

Let $u = f(z)$ with $f'(z) \neq 0$. Expanding $f(z_1) - f(z_2)$ in powers of $t = z_1 - z_2$ expresses $(u_1 - u_2)^{-r-1}$ as

$$(f'(z_2))^{-r-1} t^{-r-1} + \sum_{s<r} a_{rs}(f) t^{-s-1} + \text{regular}.$$

The transformation is upper triangular in pole order. The highest normal symbol is tensorial; lower symbols receive derivatives of the coordinate change. A conformal structure provides the corresponding splitting.

For the Virasoro field,

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

Its coordinate transformation is

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12}\{f, z\}, \qquad \{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2.$$

The Schwarzian term is the extension class coupling the fourth normal symbol to the regular projective connection.

Calculation 10.4.1 (Schwarzian cocycle)**.** *For composable coordinates $u = f(z)$ and $v = g(u)$,*

$$\{g \circ f, z\} = f'(z)^2 \{g, f(z)\} + \{f, z\}.$$

Hence the Virasoro transformation law is functorial.

Proof. Differentiate $g(f(z))$ three times, divide by the first derivative, and collect the quadratic second-derivative terms. The cross terms cancel into the displayed identity. □

10.5 Nested diagonals

For I a finite set, partial diagonals are indexed by partitions of I. Nested collisions are indexed by chains of partitions or rooted forests. The multi-normal cone records one normal direction for each internal forest edge. The associated graded of the simultaneous pole filtration is a tensor product of the corresponding normal bundles. This is the geometric source of the edge orientation line in Chapter 5.

10.6 Collision Rees object

Let $P_\bullet$ be the pole filtration. Its Rees object is

$$\mathcal{R}_P(K) = \bigoplus_{m \geq -1} P_m K \, \hbar^{m+1}.$$

At $\hbar = 1$ it recovers the chiral coefficient. At $\hbar = 0$ it recovers the normal-symbol algebra. Multiplication of filtered chiral operations induces multiplication of Rees objects.

Theorem 10.6.1 (Chiral-to-symbol degeneration)**.** *A chiral algebra with an exhaustive separated good pole filtration whose Rees object is $\hbar$-torsion-free has a flat Rees family. Its special fibre carries the corresponding classical symbol operation. On a translation-equivariant formal disc this special fibre is a Poisson vertex algebra whenever the commutator lowers the filtration by one.*

Proof. Filtration preservation defines the Rees operation. The good-filtration hypothesis makes the Rees object coherent, and $\hbar$-torsion-freeness gives flatness over the one-dimensional regular base. If the commutator lowers degree, its leading symbol defines a biderivation of degree -1. Associativity and the Jacobi identity pass to the associated graded and give the Poisson vertex axioms. □

Chapter 11

Partition collisions and Fulton–MacPherson screens

11.1 The partition complex

Let I be a finite label set and let $\Pi(I)$ be its partition lattice. For a partition π, write $|\pi|$ for the number of blocks and $\Delta_\pi : X^\pi \to X^I$ for the corresponding partial diagonal.

Let $\mathfrak{g}$ be a chiral Lie algebra. Define the labelled collision complex

$$C_I^{\mathrm{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \Pi(I)} (\Delta_\pi)_! \left(\boxtimes_{B\in\pi} \mathfrak{g}_B\right) [|\pi| - |I|] \otimes \mathrm{or}(I/\pi).$$

The orientation line is the determinant of the set of mergers required to reach π from the discrete partition. The differential merges two blocks and applies the chiral bracket to their labels.

Theorem 11.1.1 (Square of the collision differential)**.** *The merger rule descends from ordered pairs of blocks to the partition complex precisely when the binary chiral operation is antisymmetric. Once it descends, the resulting collision differential satisfies $d_{\mathrm{coll}}^2 = 0$ precisely when the chiral Jacobi identity holds.*

Proof. A two-step merger has two forms. Two disjoint pairs of blocks can be merged in either order. The coefficient maps commute, while the determinant orientations differ by a sign, so the terms cancel. Three blocks can be merged through three intermediate partitions. The resulting coefficient is the signed cyclic sum of iterated chiral brackets, which is the Jacobiator. These exhaust the rank-two intervals of the partition lattice. ◻

11.2 Low arity

For $I = \{1, 2\}$, the complex is

$$j_* j^! (\mathfrak{g} \boxtimes \mathfrak{g}) \xrightarrow{\mu} \Delta_! \mathfrak{g}.$$

For $I = \{1, 2, 3\}$, the length-two part is the sum of the three pairwise collisions; the final differential is the Jacobi sum on the total diagonal. Thus the first nontrivial square of the collision differential is exactly the chiral Jacobi identity.

11.3 Chevalley chains

The family $C_I^{\mathrm{coll}}(\mathfrak{g})$ has a factorisation coproduct obtained by restricting a partition to disjoint subsets. Symmetrisation over labels gives the chiral Chevalley chain coalgebra

$$\mathrm{CE}_*^{\mathrm{ch}}(\mathfrak{g}) = \mathrm{Sym}^{\star_{\mathrm{ch}}}(\mathfrak{g}[1])$$

after the standard completion.

Proposition 11.3.1 (Chevalley identification)**.** *The partition collision complex is the finite-set model of chiral Chevalley chains. Its primitives are* $\mathfrak{g}[1]$*, and its differential is the coderivation induced by the chiral bracket.*

Proof. A partition specifies a monomial in the cofree commutative coalgebra, one factor for each block. Merging two blocks is the coderivation determined by the binary bracket. The orientation shift is the standard Chevalley suspension. □

11.4 Fulton–MacPherson compactification

Let $\mathrm{FM}_I^{\mathbb{R}}(X^{\mathrm{an}})$ be the real oriented blowup compactification of configurations of I-labelled distinct points on the underlying real surface. A boundary stratum is indexed by a rooted forest of nested collision clusters. Each internal vertex carries a screen: the relative configuration of its incoming clusters in the tangent line, modulo translation and positive dilation.

The depth filtration of the boundary gives a spectral sequence. Its first page is

$$E^1_{p,q} = \bigoplus_{\substack{T \\ |E(T)|=p}} H_q(\mathrm{Screen}(T)) \otimes \mathrm{or}(T).$$

The d_1 differential contracts one forest edge.

Theorem 11.4.1 (Incidence–screen decomposition)**.** *The associated graded of the boundary-depth filtration of* $C_*(\mathrm{FM}_I^{\mathbb{R}}(X^{\mathrm{an}}))$ *is the forest incidence complex with coefficients in the screen-chain complexes. The constant-screen row is the partition collision complex. Higher rows retain configuration-space cohomology and period data.*

Proof. A collar of a codimension-p stratum is the product of the stratum with $[0,\varepsilon)^p$. The relative chain complex of consecutive filtration steps is the chain complex of the stratum tensored with the determinant of its p normal directions. Summing over strata gives the displayed associated graded. The connecting differential is the signed incidence map. □

11.5 The three-point screen

For the complex-algebraic screen on the affine complex line, quotient by complex translations and dilations; set $z_1 = 0$ and $z_2 = 1$. The remaining coordinate

$$\lambda = \frac{z_3 - z_1}{z_2 - z_1}$$

lies in $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its logarithmic one-forms

$$\omega_0 = d\log\lambda, \qquad \omega_1 = d\log(1-\lambda)$$

span H^1. The boundary points record the three binary collision channels.

For four points, the Arnold forms

$$\omega_{ij} = d\log(z_i - z_j)$$

satisfy

$$\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

This relation is the differential-form shadow of the codimension-two boundary identity and the flatness equation for the KZ connection.

11.6 Finite-window comparison

Let $P_{\leq N}$ truncate principal parts at pole order N, and let $C_{\leq H}$ truncate forest depth at H. Both yield finite filtrations.

Comparison Theorem 11.6.1 (Finite collision window)**.** *Assume that the screen coefficient system is locally constant on each stratum and that its cohomology is finite in the chosen window. Then the geometric Fulton–MacPherson complex and the algebraic forest complex have isomorphic spectral sequences from the first page onward. Their total complexes are quasi-isomorphic in that window.*

Proof. The first pages identify stratum by stratum by Theorem 11.4.1. The face restrictions agree by construction. Finite boundedness gives convergence, and the comparison theorem for spectral sequences gives the quasi-isomorphism. □

Chapter 12

The formal disc, vertex operations, and classical symbols

12.1 Translation-equivariant fields

Let V be a super vector space with translation operator T. A field is a formal distribution

$$Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1}$$

with lower truncation on every vector. Translation covariance is

$$[T, Y(a,z)] = \partial_z Y(a,z), \qquad Y(Ta,z) = \partial_z Y(a,z).$$

Locality asserts that for every a, b there is N with

$$(z-w)^N [Y(a,z), Y(b,w)] = 0.$$

The singular part is the formal-disc expression of the diagonal chiral operation.

12.2 Lie conformal form

The λ-bracket is

$$[a_\lambda b] = \sum_{n\geq 0} \frac{\lambda^n}{n!} a_{(n)} b.$$

It satisfies sesquilinearity,

$$[Ta_\lambda b] = -\lambda [a_\lambda b], \qquad [a_\lambda Tb] = (T+\lambda)[a_\lambda b],$$

skew-symmetry,

$$[a_\lambda b] = -(-1)^{|a||b|} [b_{-\lambda-T} a],$$

and Jacobi,

$$[a_\lambda [b_\mu c]] - (-1)^{|a||b|} [b_\mu [a_\lambda c]] = [[a_\lambda b]_{\lambda+\mu} c].$$

These equations are the polynomial coordinate form of the chiral Lie axioms.

Theorem 12.2.1 (Formal-disc equivalence)**.** *Translation-equivariant chiral Lie algebras on the formal disc are equivalent to Lie conformal algebras. After choosing a formal coordinate, unital chiral algebras with vacuum and translation operator are equivalent to vertex algebras in the sense of the Beilinson–Drinfeld reconstruction theorem [BD04, FBZ04, Kac98].*

Proof. Residue against $e^{\lambda(z-w)}$ converts a diagonal distribution into its λ-polynomial. The D-module relations become sesquilinearity. Permutation of the two formal points becomes skew-symmetry. The three partial diagonals in the formal three-disc become the displayed Jacobi identity. The unital factorisation product on the complement of the diagonal supplies the normally ordered product and vacuum; the reconstruction theorem for local fields gives the vertex operation. □

12.3 Borcherds identity

The coefficient form of the vertex Jacobi identity is

$$\begin{aligned}\sum_{i\geq 0}\binom{m}{i}(a_{(n+i)}b)_{(m+k-i)}c &= \sum_{i\geq 0}(-1)^i\binom{n}{i}a_{(m+n-i)}b_{(k+i)}c\\ &\quad - (-1)^{|a||b|+n}\sum_{i\geq 0}(-1)^i\binom{n}{i}b_{(n+k-i)}a_{(m+i)}c.\end{aligned}$$

It simultaneously contains the commutator formula, iterate formula, and associativity of normally ordered products.

Theorem 12.3.1 (Operadic Maurer–Cartan form of the vertex identity)**.** *Let $\mathcal{P}^{\mathrm{ch}}(V)$ be the translation-equivariant chiral operad of V, and let*

$$W^{\mathrm{ch}}(V) = \bigoplus_{n\geq -1}\left(\mathcal{P}^{\mathrm{ch}}(V)(n+1)\right)^{\mathfrak{S}_{n+1}}$$

be its associated graded Lie superalgebra. An odd element $X \in W_1^{\mathrm{ch}}(V)$ defines a vertex algebra structure on V exactly when

$$[X, X] = 0.$$

The resulting cochain differential is ad X *[BDSHK19].*

Proof. The operadic pre-Lie insertion sums the three compositions of the binary chiral operation along the partial diagonals of three formal points. Its graded commutator is the Lie bracket on $W^{\mathrm{ch}}(V)$. The equation $[X, X]/2 = 0$ is therefore the chiral Jacobi identity; expansion in formal modes is the Borcherds identity. The general operadic construction then makes ad X square to zero. □

12.4 The symbol operad

Apply the pole filtration of Chapter 10. The leading normal symbols of the chiral operations form the classical operad of translation-equivariant differential operations. The associated graded of a good filtered vertex algebra is a Poisson vertex algebra with commutative product and λ-bracket satisfying the Leibniz rule.

Theorem 12.4.1 (Chiral–classical operad map)**.** *The standard graph filtration on the chiral operad defines a Rees operad, and its associated graded is canonically isomorphic to the classical operad of Poisson vertex operations [BDSHK20]. Its special fibre is the symbol map of Theorem 10.6.1. Its generic fibre is the original chiral operation.*

Proof. The graph filtration is compatible with operadic substitution, so the Rees construction is operadic. The explicit residue/Fourier transform of [BDSHK20] identifies its associated graded with the classical operad. Specialization at one recovers the filtered chiral operation. ◻

12.5 Deformation theory

Let V be a Poisson vertex algebra. A chiral quantization is a formal Maurer–Cartan element

$$X_\hbar = X_0 + \hbar X_1 + \hbar^2 X_2 + \cdots$$

in the filtered chiral deformation complex, with X_0 the classical operation. At order r, the obstruction is the cohomology class of

$$\frac{1}{2} \sum_{i+j=r} [X_i, X_j]$$

in chiral Poisson cohomology. This is the order-by-order theory developed by Butson and Nair [BN26]. For affine symplectic varieties their result identifies the controlling groups with de Rham cohomology.

Chapter 13

Chiral Koszul duality and the two bars

13.1 The chiral Chevalley functor

The chiral convolution product makes the completed Ran category pro-nilpotent by Theorem 2.3.1. For a chiral Lie algebra $\mathfrak{g}$, its Chevalley coalgebra is

$$\mathrm{CE}^{\mathrm{ch}}_{*}(\mathfrak{g}) = \left(\mathrm{Sym}^{\star_{\mathrm{ch}}}(\mathfrak{g}[1]), d_{\mathfrak{g}} + d_{[-,-]}\right).$$

The factorisation condition says that this commutative chiral coalgebra has invertible coproduct on disjoint supports.

Theorem 13.1.1 (Chiral Koszul duality)**.** *On the nilcomplete chiral Lie side and the conilcomplete factorisation coalgebra side, chiral Chevalley chains and primitives form inverse equivalences. In the cardinality-complete Ran category this recovers the Francis–Gaitsgory equivalence [FG12, Heu24].*

Proof. The Lie operad and the shifted commutative cooperad are Koszul dual. The cardinality filtration makes reduced chiral convolution pro-nilpotent. Bar–cobar duality therefore converges on nilcomplete and conilcomplete objects. The unit identifies a chiral Lie algebra with the primitives of its Chevalley coalgebra; the counit reconstructs a conilcomplete factorisation coalgebra from its primitives. □

13.2 Two monoidal products, two dualities

The ordered associative bar uses the pointwise tensor:

$$A \longmapsto \overline{B}^{\mathrm{ord}}_{X}(A) = \Bbbk \otimes^{\mathbb{L}}_{A} \Bbbk.$$

The chiral Chevalley construction uses convolution:

$$\mathfrak{g} \longmapsto \mathrm{CE}^{\mathrm{ch}}_{*}(\mathfrak{g}).$$

The first resolves transverse fusion. The second assembles disjoint chiral supports. A comparison requires a functor preserving the relevant monoidal operations or a universal enveloping construction relating the source algebras.

13.3 Universal enveloping comparison

Let $\mathfrak{g}$ be a Lie algebra object in $\mathrm{FactD}(X)$ for the pointwise tensor. Its internal enveloping algebra $U(\mathfrak{g})$ is an ordered chiral algebra. PBW filters it by word length.

Theorem 13.3.1 (Bar–Chevalley comparison)**.** *Assume that $\mathfrak{g}$ is K-flat, positively graded, complete, and that every total weight has finitely many decompositions. Assume also that internal PBW holds for $U(\mathfrak{g})$. There is a natural quasi-isomorphism of conilpotent coalgebras in* $\mathrm{FactD}(X)$

$$\overline{B}_X^{\mathrm{ord}}\big(U(\mathfrak{g})\big) \xrightarrow{\sim} \mathrm{CE}_*(\mathfrak{g}).$$

It is compatible with the factorisation coefficient and with the PBW filtration.

Proof. Filter $U(\mathfrak{g})$ by PBW degree and both complexes by total word length. The associated graded enveloping algebra is $\mathrm{Sym}(\mathfrak{g})$. The bar of a symmetric algebra is the exterior, equivalently suspended symmetric, coalgebra on $\mathfrak{g}$, which is the associated graded of Chevalley chains. The comparison is an isomorphism on the first page. Finite decomposition gives a bounded exhaustive spectral sequence in each weight, and completeness then gives the quasi-isomorphism. The construction is built from internal tensor products and the bracket of $\mathfrak{g}$, so it remains in $\mathrm{FactD}(X)$. □

For a chiral Lie algebra $\mathfrak{g}_{\mathrm{ch}}$ satisfying the chiral PBW hypotheses, the same filtered argument in the cardinality-complete Ran category gives

$$\overline{B}^{\mathrm{ch}}(U^{\mathrm{ch}}\mathfrak{g}_{\mathrm{ch}}) \simeq \mathrm{CE}_*^{\mathrm{ch}}(\mathfrak{g}_{\mathrm{ch}}).$$

This is the comparison between associative bar language and chiral Chevalley language; the two monoidal products remain those specified at the start of the chapter.

13.4 Euler character

Let $\mathfrak{g} = \bigoplus_{\alpha\in\Gamma_+} \mathfrak{g}_\alpha$ be a graded Lie superalgebra with finite-dimensional pieces. The Chevalley chain object is $\mathrm{Sym}(\mathfrak{g}[1])$. Its graded Euler supercharacter is

$$\chi_\Gamma\, \mathrm{CE}_*(\mathfrak{g}) = \prod_{\alpha\in\Gamma_+} (1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_\alpha}.$$

By Theorem 13.3.1, this is also the Euler character of $\overline{B}^{\mathrm{ord}}(U\mathfrak{g})$. The formula will become the denominator identity in Chapter 29.

13.5 Commutator and classical shadows

An ordered algebra has the internal commutator

$$[a,b]_\perp = ab - (-1)^{|a||b|}ba.$$

A chiral algebra has the diagonal bracket encoded by its singular modes. A good filtration may compare them: if the ordered commutator lowers filtration and the leading chiral pole has the same symbol, the Rees special fibre has a single Poisson bracket controlling both. A specified filtered comparison map constitutes the mathematical datum of this identification.

Comparison Theorem 13.5.1 (Filtered bracket comparison)**.** *Let A carry a complete filtration preserved by the ordered multiplication and the chiral operation. Suppose the two commutators have equal leading symbol under a fixed map of Rees deformation complexes. Then the associated graded factorisation algebra carries one Poisson vertex bracket, and both quantum brackets are filtered quantizations of it.*

Proof. The common leading symbol satisfies Jacobi because each quantum bracket does, and it is a biderivation because the commutators lower filtration. Equality of the symbols identifies the two induced operations. □

Part IV

Exchange, quantum symmetry, and reduction

Chapter 14

From order to exchange

14.1 A second transverse direction

An oriented line has contractible ordered chambers. A plane has exchange paths. If two line defects occupy points of $\mathbb{R}^2$, moving one around the other gives a braid. Thus a second locally constant direction supplies the geometric passage

$$E_1 \text{ fusion} \longrightarrow E_2 \text{ exchange}.$$

Dunn additivity identifies locally constant factorisation algebras on $\mathbb{R}^2$ with E_2 algebras [Dun88, Lur17]. For a holomorphic–topological theory on $X \times \mathbb{R}^2$, the resulting E_2 structure varies factorisably over X.

Theorem 14.1.1 (Exchange-data theorem)**.** *Let $\mathcal{F}$ be a factorisation algebra on $X \times \mathbb{R}^2$ that is holomorphic on X and locally constant on $\mathbb{R}^2$. Restrict $\mathcal{F}$ to $X \times \ell$, where $\ell \subset \mathbb{R}^2$ is an oriented affine line, and denote the resulting ordered chiral algebra by A. Then:*

1. *the orientation of ℓ gives the transverse E_1 multiplication of A;*
2. *parallel transport in the fundamental groupoid of $\mathrm{Conf}_2(\mathbb{R}^2)$ gives exchange isomorphisms on the corresponding line or module category;*
3. *for three lines, the two composites*

$$\beta_{12}\beta_{23}\beta_{12} \quad \text{and} \quad \beta_{23}\beta_{12}\beta_{23}$$

coincide because they represent the same braid in $B_3 = \pi_1(\mathrm{Conf}_3(\mathbb{R}^2)/\Sigma_3)$.

Consequently the exchange operators satisfy the Yang–Baxter equation.

Proof. Restriction to the oriented line retains the factorisation maps for ordered disjoint intervals and therefore gives the E_1 product. Local constancy on $\mathbb{R}^2$ makes the coefficient of a configuration into a local system, so a path in $\mathrm{Conf}_2(\mathbb{R}^2)$ gives parallel transport between the two fusion orders. For three labelled discs, the two standard paths that exchange adjacent pairs represent the braid words $\sigma_1\sigma_2\sigma_1$ and $\sigma_2\sigma_1\sigma_2$. These words are equal in B_3. The fundamental-groupoid functor assigned by the local system therefore sends them to equal natural transformations. This equality is the categorical Yang–Baxter equation; evaluation on three modules gives its operator form. □

14.2 Algebra, module category, and centre

Three structures arise from this geometry.

First, the restriction to a line is an E_1 algebra. Second, a coproduct and exchange operator may make its module category braided. Third, the derived centre $Z^{\mathrm{der}}(A)$ is intrinsically E_2 by Theorem 9.1.1. A concrete field theory may identify these structures through comparison maps; the three types remain distinct until those maps are constructed.

Proposition 14.2.1 (Braided modules from quasi-triangular data)**.** *Let H be a complete bialgebra object with coproduct Δ and invertible $R \in H\widehat{\otimes}H$ satisfying*

$$R\Delta(h) = \Delta^{\mathrm{op}}(h)R,$$

$$(\Delta \otimes \mathrm{id})R = R_{13}R_{23}, \qquad (\mathrm{id} \otimes \Delta)R = R_{13}R_{12}.$$

Then the complete category of H-modules on which these series converge is braided monoidal, with

$$\beta_{M,N} = \sigma \circ R_{M,N}.$$

Proof. The coproduct gives the tensor-product action. The intertwining identity makes β a morphism of H-modules. The two coproduct identities are the two hexagon identities. Applying them to three modules gives the Yang–Baxter equation. □

14.3 Labelled and unlabelled configurations

The ordered theory naturally lives on labelled configurations. Forgetting labels gives a covering

$$\mathrm{Conf}_n(X) \longrightarrow \mathrm{Conf}_n(X)/\Sigma_n.$$

The fundamental groups are the pure braid group $P_n(X)$ and the full braid group $B_n(X)$, joined by

$$1 \longrightarrow P_n(X) \longrightarrow B_n(X) \longrightarrow \Sigma_n \longrightarrow 1.$$

A local system on labelled configurations descends precisely when its pure braid representation extends, with the required equivariance, to the full braid group. Symmetric descent asks in addition that the braid action factor through Σ_n.

Definition 14.3.1 (Pure-braid descent space)**.** Let M_n be a complex and let

$$\rho : BP_n(X) \longrightarrow B\,\mathrm{Aut}(M_n)$$

classify its labelled monodromy. The space of full-braid extensions of ρ is

$$\mathrm{Desc}_{B/P}(M_n, \rho) = \mathrm{hofib}_\rho\big(\mathrm{Map}\big(BB_n(X), B\,\mathrm{Aut}(M_n)\big) \longrightarrow \mathrm{Map}\big(BP_n(X), B\,\mathrm{Aut}(M_n)\big)\big)\,.$$

A point is a $B_n(X)$-action whose restriction is equivalent to ρ; paths and higher simplices are the coherent choices between extensions. The tangent obstruction theory is computed by the Lyndon–Hochschild–Serre spectral sequence for $1 \to P_n(X) \to B_n(X) \to \Sigma_n \to 1$ with coefficients in $\mathrm{End}(M_n)$ twisted by ρ.

This descent problem is independent of transverse commutativisation. An ordered multiplication can remain noncommutative while its spectral local system descends from labelled to unlabelled points.

14.4 Meromorphic exchange

Let X carry a local meromorphic spectral coordinate. A meromorphic line category consists of categories $\mathcal{L}_x$, fusion functors for pairwise distinct points, and meromorphic exchange isomorphisms

$$R_{M,N}(x-y) : M_x \otimes N_y \xrightarrow{\sim} N_y \otimes M_x,$$

together with unit, associativity, and factorisation compatibilities. The three-point compatibility is the spectral Yang–Baxter equation.

Recognition Theorem 14.4.1 (Configuration-space recognition of a line sector)**.** *Fix a collection of categories on the ordered configuration spaces of $\mathbb{R}^2$, locally constant in the transverse variables and meromorphic in the spectral variables on X. Suppose it is equipped with:*

1. *fusion maps for embeddings of disjoint discs;*
2. *parallel transport for the configuration-space fundamental groupoids;*
3. *coherent compatibility of fusion with restriction, composition, and transport.*

Then these data define the line sector of a holomorphic–topological factorisation category on product discs in $X \times \mathbb{R}^2$. Conversely, the line sector of such a factorisation category supplies exactly these data.

Proof. A product-disc factorisation category restricts to a locally constant coefficient on each transverse configuration space. Its factorisation maps give fusion, and local constancy gives parallel transport. Functoriality of factorisation supplies every compatibility. Conversely, assign to each product-disc configuration the given coefficient category and use fusion for the structure maps. The stated coherences are precisely the prefactorisation and descent diagrams on the product-disc basis. Weiss descent extends the construction to the generated open sets. □

14.5 Exchange and the centre

The braided line category acts on the ordered boundary category. Moving a bulk or exchange observable around a boundary insertion produces a coherent central action and hence a morphism

$$\mathrm{Obs}_{E_2} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ord,ch}}(A).$$

When the boundary satisfies the generation and detection hypotheses of Theorem 9.4.1, this morphism identifies the geometric exchange algebra with the universal derived centre.

Chapter 15

KZ transport and Tannaka reconstruction

15.1 The logarithmic connection

Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra with invariant nondegenerate form and split Casimir $\Omega \in \mathfrak{g}\otimes\mathfrak{g}$. On $V_1 \otimes \cdots \otimes V_n$ define

$$\nabla_{\mathrm{KZ}} = d - \hbar \sum_{1\le i<j\le n} \Omega_{ij}\, d\log(z_i - z_j).$$

The infinitesimal braid relations are

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{for disjoint pairs,}$$

$$[\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0 \quad \text{for distinct } i, j, k.$$

They follow from invariance of the form and the Jacobi identity.

Theorem 15.1.1 (Flatness of the KZ connection)**.** *The KZ connection is flat.*

Proof. Write $A = \hbar \sum_{i<j} \Omega_{ij}\omega_{ij}$ with $\omega_{ij} = d\log(z_i - z_j)$. Since $d\omega_{ij} = 0$, the curvature is $-A \wedge A$. Terms with disjoint index pairs vanish by the first infinitesimal braid relation. For each triple i, j, k, the remaining terms are the second braid relation multiplied by the Arnold identity

$$\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

Hence the curvature vanishes. □

A positively oriented small loop of $z_i - z_j$ around the origin has monodromy $\exp(2\pi i\hbar\Omega_{ij})$; a chosen half-twist gives the corresponding braid operator, up to the flip and associator convention. The rational operator $1 + \hbar\Omega/z + O(\hbar^2)$ belongs instead to the difference-type exchange geometry of four-dimensional Chern–Simons theory developed in Chapter 27. The KZ residues define the Gaudin Hamiltonians

$$H_i = \sum_{j\ne i} \frac{\Omega_{ij}}{z_i - z_j}.$$

Flatness implies $[H_i, H_j] = 0$.

15.2 Drinfeld–Kohno

The KZ associator is the regularized parallel transport of the three-point connection between the two asymptotic parenthesization chambers. It solves the pentagon equation. The monodromy around two points supplies the braid.

Theorem 15.2.1 (Drinfeld–Kohno)**.** *Let $\mathfrak{g}$ be complex semisimple and put $q = e^{\pi i \hbar}$. For generic $\hbar$, the braided tensor category obtained from KZ monodromy on finite-dimensional $\mathfrak{g}$-modules is braided equivalent to the corresponding category of type-one finite-dimensional $U_q(\mathfrak{g})$-modules. The tensor equivalence is expressed through a Drinfeld associator [Koh87, Dri90].*

The theorem compares braided tensor categories. A Hopf algebra presentation emerges after a fibre functor is chosen.

15.3 The coend

Let $\mathcal{L}$ be an essentially small $\Bbbk$-linear abelian rigid monoidal category with finite-dimensional Hom spaces, and let $F : \mathcal{L} \to \mathrm{Vect}_{\Bbbk}^{\mathrm{fd}}$ be a faithful exact strong monoidal functor. Define the coend

$$H = \int^{M \in \mathcal{L}} F(M)^* \otimes F(M).$$

Concretely, H is the quotient of the direct sum of matrix-coefficient spaces by the dinaturality relations

$$F(f)^* \phi \otimes v \sim \phi \otimes F(f) v.$$

The coalgebra maps are

$$\Delta[\phi \otimes v] = \sum_a [\phi \otimes e_a] \otimes [e^a \otimes v], \qquad \varepsilon[\phi \otimes v] = \phi(v).$$

Strong monoidality defines the product by sending the coefficients of M and N to the coefficient of $M \otimes N$. Duality defines the antipode. Every $F(M)$ has the canonical right coaction

$$v \longmapsto \sum_a e_a \otimes [e^a \otimes v],$$

and hence a strong monoidal comparison functor

$$K_F : \mathcal{L} \longrightarrow \mathrm{Comod}_{\mathrm{fd}}(H).$$

Theorem 15.3.1 (Tannaka reconstruction)**.** *The coend H is a Hopf algebra and K_F is faithful and strong monoidal, with $F = U \circ K_F$ for the forgetful functor U.*

Assume in addition that F admits a right adjoint R, that the Barr–Beck comparison for $F \dashv R$ is effective, and that the canonical matrix-coefficient map

$$V \otimes H \longrightarrow FR(V)$$

is an isomorphism for every finite-dimensional vector space V. Then K_F is an equivalence

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

If $\mathcal{L}$ is braided, its braiding defines a coquasitriangular form $r : H \otimes H \to \Bbbk$. A quasi-monoidal fibre functor reconstructs the corresponding coquasi-Hopf algebra, with coassociator equal to the transported associator.

Proof. Dinaturality makes Δ and ε basis independent. The monoidal constraints of F give an associative unital product and make Δ an algebra map. Evaluation and coevaluation for dual objects give the convolution inverse required for the antipode. The displayed coactions are dinatural, compatible with tensor products, and recover F after forgetting the coaction.

Under the additional hypotheses, Barr–Beck identifies $\mathcal{L}$ with the coalgebras for the comonad FR on finite-dimensional vector spaces. The matrix-coefficient isomorphism identifies this comonad with $V \mapsto V \otimes H$; its coalgebras are precisely finite-dimensional right H-comodules. For a braiding $\beta_{M,N}$, evaluation of $F(\beta_{M,N})$ on matrix coefficients defines r. The two hexagon identities become the two coquasitriangular identities. The same calculation with a quasi-monoidal associator gives the coquasi-Hopf axioms. □

15.4 Meromorphic Tannaka

When $F(M_x)$ varies meromorphically with $x \in X$, the same coend is formed over the field of meromorphic functions on configuration space. The coquasitriangular form becomes $r(x - y)$, and its dual operator on modules is the spectral R-matrix.

Recognition Theorem 15.4.1 (Quantum-group presentation)**.** *A named quantum group is recognized from a reconstructed Hopf algebra by the following data:*

1. *generators arising from matrix coefficients of a generating family of representations;*
2. *relations obtained from exchange and tensor decomposition;*
3. *a PBW theorem identifying the associated graded;*
4. *compatibility of the reconstructed coproduct and antipode with the presentation.*

Let $\psi : H_1 o H_2$ be a continuous filtered Hopf morphism between complete separated Hopf algebras with exhaustive bounded-below filtrations. If ψ is surjective on the chosen topological generators and $\mathrm{gr}\,\psi$ is an isomorphism, then ψ is an isomorphism: finite filtration quotients are isomorphic by induction, and completeness passes the inverse-limit isomorphism to H_1 and H_2.

15.5 Rational, trigonometric, and elliptic geometries

A meromorphic one-form on the spectral curve selects the local form of the exchange kernel. A rational curve with one double pole yields rational difference kernels and Yangians. A cylinder yields trigonometric kernels and quantum loop algebras. An elliptic curve yields elliptic dynamical or elliptic quantum groups. The line category and its fibre functor determine the exact algebra in each case.

Chapter 16

Zamolodchikov–Faddeev algebras and the RTT Yangian

16.1 The exchange algebra

Let V be a finite-dimensional vector space and let

$$R(u) \in \mathrm{End}(V \otimes V) \otimes \mathbb{k}(u)$$

be an invertible meromorphic operator. Work over the field obtained by localising the spectral variables away from all collision divisors. The Zamolodchikov–Faddeev algebra is generated by symbols $Z_i(u)$ with exchange relations

$$Z_i(u)Z_j(v) = \sum_{k,l} R_{ij}^{kl}(u-v)Z_l(v)Z_k(u).$$

The relation reverses the transverse order and transports the coefficient through the exchange operator.

Theorem 16.1.1 (Confluence equals Yang–Baxter)**.** *Fix a total order of the spectral labels and orient every exchange relation toward that order. Assume*

$$R(u)R^{21}(-u) = 1.$$

The oriented relations are confluent precisely when

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

When this equation holds, the ordered words in the $Z_i(u)$ form a basis over the localised spectral field.

Proof. Termination follows from the inversion number of a spectral word. Every critical ambiguity is obtained from three adjacent spectral labels. Reducing a three-letter word through the two reduced expressions of the longest element of Σ_3 gives the two displayed products of R-matrices. Their equality is the Yang–Baxter equation. Unitarity resolves the length-two ambiguity created by an exchange followed by its reverse. The diamond lemma now gives a unique ordered normal form for every word and hence the asserted basis. □

This is a concrete doubly noncommutative model. The spectral variable belongs to the curve, the word belongs to the ordered transverse direction, and the three-body consistency is the braid relation.

16.2 The rational matrix

Let $P(x \otimes y) = y \otimes x$ and
$$R(u) = 1 - \frac{\hbar P}{u}.$$
The permutation identities
$$P_{12}P_{13} = P_{23}P_{12}, \qquad P_{13}P_{23} = P_{12}P_{13}, \qquad P_{12}P_{23} = P_{13}P_{12}$$
give the Yang–Baxter equation after clearing the three spectral denominators. A scalar factor $f(u)$ with
$$f(u)f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1}$$
produces the unitary normalization $\widetilde{R}(u) = f(u)R(u)$.

16.3 RTT presentation

Let
$$T(u) = 1 + \sum_{r\geq 1} T^{(r)}u^{-r}, \qquad T^{(r)} = (t_{ij}^{(r)})_{1\leq i,j\leq n}.$$
The polynomial Rees RTT Yangian $Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)$ is the $\Bbbk[\hbar]$-algebra generated by the coefficients $t_{ij}^{(r)}$ subject to
$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$
Its $\hbar$-adic completion is denoted
$$\widehat{Y}_\hbar(\mathfrak{gl}_n) = \varprojlim_N Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)/(\hbar^N).$$
Equivalently, with $t_{ij}^{(0)} = \delta_{ij}$,
$$\begin{aligned}[] [t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] \\ = \hbar\left(t_{kj}^{(r)}t_{il}^{(s)} - t_{kj}^{(s)}t_{il}^{(r)}\right), \qquad r, s \geq 0.\end{aligned}$$
In particular,
$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}).$$
Thus the degree-one coefficient generates the Rees enveloping algebra of $\mathfrak{gl}_n$.

16.4 Hopf structure

Define
$$\Delta T(u) = T_{13}(u)T_{23}(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}.$$
In components,
$$\Delta t_{ij}(u) = \sum_a t_{ia}(u) \otimes t_{aj}(u).$$
Let
$$U_\hbar(\mathfrak{gl}_n) = T(\mathfrak{gl}_n)[\hbar] \Big/ \left(xy - yx - \hbar[x,y]_{\mathfrak{gl}_n}\right)$$
be the polynomial Rees enveloping algebra.

Theorem 16.4.1 (RTT Hopf algebra)**.** *The three displayed maps preserve the RTT ideal and make $Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)$ a Hopf algebra over $\Bbbk[\hbar]$; they extend continuously to the $\hbar$-adic completion. For every $a \in \Bbbk$, the formula*

$$\mathrm{ev}_a : T(u) \longmapsto 1 + \frac{E}{u-a}, \qquad E = (e_{ij})_{i,j=1}^n,$$

defines a homomorphism

$$Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n) \longrightarrow U_\hbar(\mathfrak{gl}_n), \qquad t_{ij}^{(r)} \longmapsto a^{r-1} e_{ij}.$$

Proof. Write the RTT equation in three auxiliary tensor factors. Applying it in each algebra tensor factor and using the Yang–Baxter equation moves the auxiliary R-matrices through $T_{13}(u)T_{23}(u)$, which proves that Δ preserves RTT. The counit is immediate. Matrix inversion reverses products and gives the antipode identities. Substitution of $1 + E/(u-a)$ reduces the coefficient relations to

$$[e_{ij}, e_{kl}] = \hbar(\delta_{kj} e_{il} - \delta_{il} e_{kj}),$$

which are precisely the defining relations of $U_\hbar(\mathfrak{gl}_n)$. □

16.5 Quantum determinant for $\mathfrak{gl}_2$

For $n = 2$ set

$$\mathrm{qdet}\, T(u) = t_{11}(u) t_{22}(u-\hbar) - t_{12}(u) t_{21}(u-\hbar).$$

Theorem 16.5.1 (Central group-like series)**.** *The coefficients of* $\mathrm{qdet}\, T(u)$ *are central and*

$$\Delta\, \mathrm{qdet}\, T(u) = \mathrm{qdet}\, T(u) \otimes \mathrm{qdet}\, T(u).$$

The quotient by $\mathrm{qdet}\, T(u) = 1$ *is the RTT Rees form of the Yangian of* $\mathfrak{sl}_2$.

Proof. At spectral difference $\hbar$, the matrix $R(\hbar) = 1 - P$ is twice the antisymmetrizer on $\mathbb{C}^2 \otimes \mathbb{C}^2$. Fuse two RTT equations and restrict to the one-dimensional antisymmetric representation. The fused monodromy is the displayed scalar series. A third RTT equation moves any $T(v)$ through this scalar, proving centrality. Applying the coproduct before the same fusion gives the group-like identity. □

16.6 PBW theorem

Give $t_{ij}^{(r)}$ loop degree $r-1$. Give $e_{ij} t^{r-1}$ the same degree in the current Rees algebra

$$U_\hbar(\mathfrak{gl}_n[t]) = T(\mathfrak{gl}_n[t])[\hbar] \Big/ \big(xy - yx - \hbar[x,y]\big).$$

Theorem 16.6.1 (Yangian PBW)**.** *For every total order on the generators $t_{ij}^{(r)}$, their ordered monomials form a $\Bbbk[\hbar]$-basis of $Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)$. The loop filtration has associated graded algebra*

$$\mathrm{gr}_{\mathrm{loop}}\, Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n) \cong U_\hbar(\mathfrak{gl}_n[t]), \qquad \mathrm{gr}(t_{ij}^{(r)}) \longmapsto e_{ij} t^{r-1}.$$

Consequently

$$Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)/(\hbar - 1) \cong Y(\mathfrak{gl}_n), \qquad Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n)/\hbar \cong \mathrm{Sym}(\mathfrak{gl}_n[t]).$$

The $\hbar$-adic completion is topologically free over $\Bbbk[[\hbar]]$ with the same ordered monomial basis.

Proof. The coefficient relations move an inverted pair into the chosen order and replace it by ordered terms of no larger loop degree. Their top loop-degree part is the Rees current relation, which gives a filtered surjection

$$U_\hbar(\mathfrak{gl}_n[t]) \twoheadrightarrow \mathrm{gr}_{\mathrm{loop}} Y_\hbar^{\mathrm{Rees}}(\mathfrak{gl}_n).$$

The RTT PBW theorem proves linear independence of the ordered monomials and identifies this surjection with an isomorphism [Mol07, Theorems 1.22 and 1.26]. Specialising the polynomial Rees parameter at $\hbar = 1$ gives the ordinary Yangian. Specialising at $\hbar = 0$ gives the symmetric current algebra. Completion preserves the basis because the polynomial Rees algebra is free over $\Bbbk[\hbar]$. □

16.7 Ordered chiral interpretation

The matrix coefficients $T_{ij}(u)$ form a meromorphic coefficient on the spectral curve. RTT is the transverse E_1 multiplication. The coproduct acts on tensor products of line representations. The rational R-matrix gives their exchange. The Yangian is therefore an ordered meromorphic algebra, while its module category and derived centre carry the exchange and bulk structures.

Chapter 17

BRST descent and Drinfeld–Sokolov reduction

17.1 A compatible differential

Let A be an ordered chiral algebra and let $Q : A \to A[1]$ be a square-zero derivation of the internal multiplication and of every factorisation map. It acts on a bar word by

$$Q_B[a_1|\cdots|a_r] = \sum_{i=1}^{r}(-1)^{|a_1|+\cdots+|a_{i-1}|+i-1}[a_1|\cdots|Qa_i|\cdots|a_r].$$

Proposition 17.1.1 (BRST action on the bar)**.** *The operator Q_B satisfies*

$$Q_B^2 = 0, \qquad Q_B d_B + d_B Q_B = 0.$$

Hence $(\overline{B}_X^{\mathrm{ord}}(A), d_B, Q_B)$ is a bicomplex in factorisation coefficients.

Proof. The first identity is the standard tensor-product differential calculation. For the second, each interior bar face multiplies adjacent letters. The two ways of applying Q before or after that face agree by the derivation rule, with the totalization signs giving the anticommutator. Outer faces agree because the augmentation is Q-closed. □

17.2 Comparison spectral sequence

Filter the total complex by bar length. The first differential is Q_B and the next is the bar differential of the induced product on BRST cohomology.

Theorem 17.2.1 (Bar–BRST comparison)**.** *Assume that tensor products preserve Q-quasi-isomorphisms and that the bar filtration is complete and bounded below in every total weight. Let*

$$\mathrm{Tot}_Q\, \overline{B}_X^{\mathrm{ord}}(A) = \big(\overline{B}_X^{\mathrm{ord}}(A), d_B + Q_B\big).$$

There is a convergent spectral sequence whose first page, equipped with its first induced differential, is

$$E_1 \cong \overline{B}_X^{\mathrm{ord}}\big(H_Q(A)\big) \Longrightarrow H^*\big(\mathrm{Tot}_Q\, \overline{B}_X^{\mathrm{ord}}(A)\big).$$

If $H_Q(A)$ is concentrated in cohomological degree zero, the canonical comparison is a quasi-isomorphism

$$\overline{B}_X^{\mathrm{ord}}\big(H_Q^0(A)\big) \xrightarrow{\simeq} \mathrm{Tot}_Q\, \overline{B}_X^{\mathrm{ord}}(A).$$

Proof. Filter the total complex by BRST cohomological degree and then by bar length. The associated vertical complex is the tensor product of copies of (A, Q). The Künneth hypothesis identifies its cohomology with tensor powers of $H_Q(A)$, and the induced horizontal differential is the bar differential of the cohomology algebra. Completeness and boundedness give strong convergence. When $H_Q(A)$ lies in degree zero, only one vertical row survives. The spectral sequence collapses and the edge morphism is the displayed quasi-isomorphism. □

17.3 Hamiltonian reduction

Let $V_k(\mathfrak{g})$ be an affine vertex algebra, let $\mathfrak{n} \subset \mathfrak{g}$ be the nilpotent subalgebra selected by a good grading, and introduce the charged and neutral ghost systems required by the moment-map constraints. The BRST current is the sum of:

1. the current-valued moment map paired with the ghost;
2. the cubic ghost term encoding the Lie bracket of $\mathfrak{n}$;
3. the character term specifying the nilpotent element;
4. the neutral-fermion correction for half-integer grading.

Its residue is a square-zero derivation. The quantum Drinfeld–Sokolov algebra is

$$\mathcal{W}_k(\mathfrak{g}, f) = H^0_Q\big(V_k(\mathfrak{g}) \otimes \text{Ghosts}\big).$$

Theorem 17.3.1 (Quantum Drinfeld–Sokolov concentration)**.** *For principal f and generic level, the BRST cohomology is concentrated in degree zero and equals the principal $\mathcal{W}$-algebra. The bar–BRST comparison of Theorem 17.2.1 therefore identifies the ordered bar of the principal $\mathcal{W}$-algebra with the BRST cohomology of the affine-ghost bar.*

The concentration theorem is the representation-theoretic input of quantum Hamiltonian reduction; see [FF92, Ara04] and the modern treatments of principal $\mathcal{W}$-algebras.

17.4 The $\mathfrak{sl}_2$ calculation

Let e, h, f be affine $\mathfrak{sl}_2$ currents at level k, with

$$h(z)h(w) \sim \frac{2k}{(z-w)^2}, \quad h(z)e(w) \sim \frac{2e(w)}{z-w}, \quad e(z)f(w) \sim \frac{k}{(z-w)^2} + \frac{h(w)}{z-w}.$$

Introduce fermionic ghosts b, c with $b(z)c(w) \sim (z-w)^{-1}$. Since the positive nilpotent is one-dimensional, the BRST current is

$$j_{\mathrm{BRST}}(z) = (e(z) - 1)c(z), \qquad Q = \mathrm{Res}_{z=0}\, j_{\mathrm{BRST}}(z).$$

The e–e and c–c OPEs are regular, so $Q^2 = 0$.

A Q-closed stress tensor is obtained by improving the Sugawara field and adding the ghost stress tensor. Its cohomology class generates the reduced vertex algebra. Wick contraction gives

$$c_{\mathrm{DS}}(k) = 13 - 6(k+2) - \frac{6}{k+2} = 1 - \frac{6(k+1)^2}{k+2}.$$

Theorem 17.4.1 (Principal $\mathfrak{sl}_2$ reduction)**.** *For $k \neq -2$, the BRST cohomology of the principal $\mathfrak{sl}_2$ complex is the Virasoro vertex algebra of central charge $c_{\mathrm{DS}}(k)$, concentrated in degree zero at generic level.*

Proof. Filter by the number of f and b modes. The leading differential pairs $e - 1$ with b and pairs the non-Cartan current modes into contractible quartets. The remaining complex is the Heisenberg algebra generated by the Cartan current with the improved background charge. Its Feigin–Fuchs stress tensor has central charge $1 - 6(k+1)^2/(k+2)$. The Miura map identifies its invariant subalgebra with Virasoro, and the filtration spectral sequence is concentrated in degree zero. □

17.5 Structures transported by reduction

Because Q is a derivation of the factorisation maps, BRST cohomology remains a factorisation coefficient. Compatible coproducts, exchange operators, traces, and module actions descend whenever their defining maps commute with Q. The bar bicomplex makes each descent a chain-level statement and records the higher homotopies when strict commutation is replaced by a specified null-homotopy.

Part V

Two sewings and quantum anomalies

Chapter 18

Cyclic traces and ribbon sewing

18.1 Closing the interval

Let A be a proper ordered chiral algebra and write $A^\vee = \mathrm{RHom}_{\Bbbk}(A, \Bbbk)$ for its coefficientwise linear dual. A degree-d proper Calabi–Yau structure is a negative-cyclic class whose induced bimodule morphism

$$A \longrightarrow A^\vee[d]$$

is an equivalence. On a strict finite model it is represented by a cyclic trace

$$\mathrm{Tr}_A : CC_d(A) \longrightarrow \Bbbk$$

with perfect pairing

$$\langle a, b\rangle = \mathrm{Tr}_A(ab).$$

Cyclicity identifies cyclic rotations of a word. Perfection gives the coevaluation tensor that contracts the two flags of an internal ribbon edge.

The circle value of the ordered theory is

$$\int_{S^1} A \simeq \mathrm{HH}_*(A).$$

The annulus carries the chain–cochain pairing. Surface operations require the cyclic A_∞ structure and its nondegenerate pairing, rather than the associative multiplication alone.

Definition 18.1.1 (Cyclic ordered chiral algebra)**.** A cyclic ordered chiral algebra is an ordered chiral algebra A, presented by a finite perfect cyclic A_∞ object in $\mathrm{FactD}(X)$, together with a fixed-degree nondegenerate cyclic pairing. Every higher multiplication and the pairing is a morphism of factorisation coefficients.

18.2 Ribbon graphs

A ribbon graph is a graph equipped with a cyclic order of half-edges at every vertex. A valence-r vertex is evaluated by the corresponding cyclic A_∞ operation. An internal edge is evaluated by the coevaluation of the perfect pairing. Boundary cycles of the thickened graph carry Hochschild inputs and outputs.

Let $C_*^{\mathrm{rib}}(h; b_-, b_+)$ denote the stable ribbon-graph chain complex for oriented bordered surfaces of genus h with incoming and outgoing boundary components, in the positive-boundary convention. Its differential contracts one admissible internal edge.

Theorem 18.2.1 (Ribbon-graph action)**.** *Let A be a finite perfect cyclic A_∞ algebra object in* $\mathrm{FactD}(X)$*. There is a dg PROP morphism*

$$C_*^{\mathrm{rib}} \longrightarrow \mathrm{End}\big(CC_*(A)\big).$$

For every stable ribbon graph G, the image is obtained by placing the cyclic higher multiplication at each vertex and the inverse pairing at each internal edge. Composition in the PROP agrees with gluing boundary circles. After the standard orbi-cellular identification of the ribbon-graph complex, this gives the positive-boundary open topological conformal field theory of A.

Proof. Assign to a graph G the tensor contraction obtained from its vertex operations and edge coevaluations. The cyclic order at a vertex fixes the order of the inputs to the cyclic operation. Contracting an internal edge composes the two adjacent cyclic operations; summing all such contractions with determinant-line signs is precisely the cyclic A_∞ identity. Therefore the graph assignment intertwines the ribbon differential with the Hochschild differential. Disjoint union is tensor product, and gluing an outgoing boundary to an incoming boundary inserts the coevaluation and hence agrees with PROP composition. Every operation is internal to $\mathrm{FactD}(X)$, so the construction varies factorisably over X. The identification with the positive-boundary open TCFT is the ribbon-graph model of Costello [Cos07]. □

The ribbon genus counts handles formed by the transverse cyclic theory. It is independent of the genus of the holomorphic curve considered in Chapter 19.

18.3 The annulus

The normalized cyclic bar complex is

$$CC_*(A) = \bigoplus_{r\geq 0} A \otimes^! (s\bar{A})^{\otimes^! r}$$

with Hochschild differential b and Connes operator B. The mixed-complex relations are

$$b^2 = B^2 = bB + Bb = 0.$$

The geometric rotation of the annulus represents B. Homotopy fixed points and the Tate construction give negative and periodic cyclic homology.

Proposition 18.3.1 (Annulus pairing)**.** *For a smooth proper degree-d Calabi–Yau ordered chiral algebra represented by a finite perfect cyclic model, the Calabi–Yau equivalence of diagonal bimodules induces*

$$\mathrm{HH}^*(A, A) \simeq \mathrm{HH}_{d-*}(A)^*.$$

Under the two-sided Koszul–Morita equivalence this pairing becomes the coHochschild pairing of $C = \overline{B}^{\mathrm{ord}}(A)$.

Proof. The Calabi–Yau class identifies the diagonal bimodule A with its bimodule dual shifted by d. Applying derived A^e-Hom gives Hochschild duality. The identification with coHochschild theory is Theorem 8.5.1. □

18.4 Connected amplitudes

Let Z be the formal sum of all ribbon-graph amplitudes and write $F = \log Z$. The exponential formula expresses Z as the symmetric algebra on connected graphs and F as the connected cumulant. For a connected ribbon graph,

$$\chi = V - E + F = 2 - 2h - b,$$

where h is the ribbon genus and b the number of boundary components.

For a matrix extension $A \otimes \mathrm{Mat}_N$, index contraction gives one factor of N for every closed face. With one factor of N at an interaction vertex and N^{-1} at a propagator, the graph weight is

$$N^{V-E+F} = N^{2-2h-b}.$$

This is the topological large-N expansion derived directly from the ribbon surface.

18.5 Koszul transport of a cyclic theory

A two-sided Koszul equivalence transports bimodules and their compositions. A cyclic enhancement additionally identifies the Calabi–Yau class of the diagonal bimodule with a cotrace on the diagonal bicomodule.

Theorem 18.5.1 (Cyclic Koszul transport)**.** *Let A be nilcomplete and let $C = \overline{B}^{\mathrm{ord}}(A)$ be conilcomplete. Suppose the two-sided Koszul equivalence of Theorem 8.3.1 is equipped with an identification of the degree-d Calabi–Yau class of A with a perfect cyclic cotrace on the diagonal C-bicomodule. Then the ribbon-graph action of A and the ribbon-cograph action of C are intertwined by the Hochschild–coHochschild equivalence. Corresponding surface amplitudes are quasi-isomorphic.*

Proof. Every ribbon amplitude is an iterated composite of three morphisms: higher multiplication, diagonal coevaluation, and trace. The two-sided Koszul functor carries the first to the corresponding bicomodule operation and the diagonal bimodule to the diagonal bicomodule. The cyclic enhancement carries the remaining coevaluation and trace to the specified cotrace data. Hence it identifies the evaluation of every ribbon graph. Compatibility with the graph differential follows from functoriality, and completion follows by passing through finite weight windows. □

Chapter 19

Conformal blocks and chiral sewing

19.1 Coinvariants and blocks

Let V be a conformal vertex algebra and let $(C; p_1, \ldots, p_n; z_1, \ldots, z_n)$ be a pointed smooth curve with formal coordinates. For V-modules $M_1, \ldots, M_n$, the global current algebra on the punctured curve acts on $\bigotimes_i M_i$. The coinvariants are the quotient by this action; conformal blocks are their dual.

The stress tensor differentiates the family with respect to deformations of the pointed curve. Its central term makes the resulting connection projectively flat. The projective curvature is the curvature of a power of the Hodge determinant line.

19.2 Sewing a node

Let C_1 and C_2 have marked points with local coordinates x and y. Plumbing with parameter q identifies

$$xy = q.$$

Insert a module M at one branch and its contragredient M' at the other. The sewing contraction is the formal sum over a homogeneous dual basis, weighted by $q^{L_0 - c/24}$.

For rational theories this sum converges in a punctured disc and extends the conformal block to the plumbed family. Algebraically it is a coend over the module category.

Theorem 19.2.1 (Sewing and factorisation of conformal blocks)**.** *Let V be strongly rational. The vector bundles of conformal blocks on the moduli of pointed stable curves satisfy factorisation at every separating and nonseparating node. They form a modular functor, and their genus-zero module category is a modular fusion category.*

This theorem combines the algebraic factorisation results for conformal blocks with the modular-functor construction of Damiolini–Woike [DGT21, DGT24, DW25]. Analytic sewing for broader C_2-cofinite classes is developed through convergent coend formulas such as [GZ25].

19.3 The anomaly line

Let $\lambda = \det \pi_* \omega_\pi$ be the Hodge line on a family of curves. For a conformal theory of central charge c, the projective connection lifts to a flat connection after tensoring by the appropriate fractional power of λ. In differential form,

$$F_{\nabla^{\text{blocks}}} = \frac{c}{2} F_{\nabla^\lambda} + \sum_i h_i \, F_i,$$

with the puncture contributions determined by the conformal weights.

Proposition 19.3.1 (Cancellation by an anomaly line)**.** *Let E be a projectively flat bundle with scalar curvature $F_E = \omega \mathrm{id}_E$. If a line bundle L has curvature $F_L = -\omega$, then $E \otimes L$ is flat.*

Proof. The curvature of the tensor product is $F_{E\otimes L} = F_E \otimes 1 + 1 \otimes F_L = 0$. □

This simple identity is the geometric form of anomaly cancellation. The line, its connection, and its curvature are the primary data; the central charge is their scalar coefficient in a chosen normalization.

19.4 Genus one

On an elliptic curve of modulus τ, a module character is

$$\chi_M(\tau) = \mathrm{Tr}_M\, q^{L_0 - c/24}, \qquad q = e^{2\pi i \tau}.$$

For a rational C_2-cofinite vertex algebra, the characters span a finite-dimensional representation of $\mathrm{SL}_2(\mathbb{Z})$ by Zhu's theorem [Zhu96]. The S-matrix diagonalizes fusion, and the Verlinde formula expresses the fusion coefficients through its matrix entries.

19.5 Free boson determinant

For one chiral boson, oscillator quantization gives

$$\mathrm{Tr}\, q^{L_0 - 1/24} = \eta(\tau)^{-1}, \qquad \eta(\tau) = q^{1/24} \prod_{n\geq 1} (1 - q^n).$$

The holomorphic oscillator determinant section is $\eta(\tau)$; its Quillen norm includes the standard nonholomorphic zero-mode factor. The modular transformation of η is the transition function of the holomorphic determinant line. Thus the oscillator trace and the determinant-line section are two realizations of one anomaly object.

19.6 Lattice theories

For an even positive-definite lattice L, the lattice vertex algebra has character

$$\chi_{V_L}(\tau, z) = \frac{\Theta_L(\tau, z)}{\eta(\tau)^{\mathrm{rank}\, L}}, \qquad \Theta_L(\tau, z) = \sum_{\ell \in L} q^{(\ell,\ell)/2} e^{2\pi i (\ell, z)}.$$

At higher genus, the numerator becomes the Siegel theta series of L and the denominator is the determinant of the chiral boson. The stable-curve sewing contracts lattice momenta and oscillator modes simultaneously.

19.7 Chiral genus

The genus produced here is the genus of the complex curve. It is independent of the ribbon genus produced by the transverse trace. A theory carrying both structures has a pair of genus variables, developed in Chapter 20.

Chapter 20

The two-edge master equation

20.1 Bicoloured stable graphs

A bicoloured stable graph has two types of internal edge:

- a transverse ribbon edge, created by the Calabi–Yau pairing of the ordered algebra;
- a chiral nodal edge, created by the sewing pairing of conformal blocks.

At every vertex, the incident half-edges of transverse type carry a cyclic order; the chiral flags carry the markings of a stable curve component. The orientation line is

$$\mathrm{or}(\Gamma) = \det(E_\perp(\Gamma)) \otimes \det(E_X(\Gamma)).$$

Let $\mathfrak{F}^{(2)}$ be the completed free graph complex generated by bicoloured stable graphs. Its differential is

$$d_{\mathfrak{F}} = d_0 + \partial_\perp + \partial_X,$$

where the last two terms contract one edge of the indicated colour.

Proposition 20.1.1 (Independent graph boundaries)**.** *The two edge-contraction operators satisfy*

$$\partial_\perp^2 = 0, \qquad \partial_X^2 = 0, \qquad \partial_\perp \partial_X + \partial_X \partial_\perp = 0.$$

Proof. Contracting two edges of one colour in opposite orders gives opposite orientations. Contracting one edge of each colour gives the same final graph, and swapping their determinant factors changes the sign. This is the graph version of Theorem 5.2.2. □

20.2 Brackets and loop operators

Gluing two distinct vertices along an edge defines odd brackets $[-,-]_\perp$ and $[-,-]_X$. Gluing two flags of the same vertex defines loop operators $\Delta_\perp$ and Δ_X. The product orientation gives

$$\Delta_\perp \Delta_X + \Delta_X \Delta_\perp = 0.$$

Each operator is second order with respect to disjoint union of graphs, and its derived bracket is the corresponding gluing bracket.

20.3 The master element

Fix a dg collection V indexed by the two genera, the two sets of flags, and the relevant symmetric and cyclic actions. Let $\mathbb{F}^{(2)}(V)$ be its completed free bicoloured modular envelope. An endomorphism-valued action is a continuous dg morphism from $\mathbb{F}^{(2)}(V)$ to the endomorphism graph complex of a coefficient object. Let Θ be the image of the connected corollas and introduce independent loop parameters $\hbar_\perp$ and $\hbar_X$.

Theorem 20.3.1 (Two-edge master equation)**.** *Continuous dg actions of $\mathbb{F}^{(2)}(V)$ are in bijection with continuous corolla maps whose formal sum Θ satisfies*

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_\perp + \frac{1}{2}[\Theta, \Theta]_X + \hbar_\perp \Delta_\perp \Theta + \hbar_X \Delta_X \Theta = 0.$$

The operator defined by the left-hand side is square zero.

Proof. The completed free envelope is generated by corollas and by the two edge contractions. A continuous morphism is therefore determined by its corolla map. Compatibility with the differential on a one-vertex graph gives the internal differential term. Contracting an edge between two vertices gives the corresponding bracket, and contracting a loop at one vertex gives the corresponding Δ-operator. The factor $1/2$ identifies the two orientations of a separating edge. Conversely, the displayed equation makes the corolla map commute with every generating differential and hence extends it to a dg morphism on the free envelope. The relations in Theorem 20.1.1 show that the two graph BV operators and their derived brackets form a pair of supercommuting BV structures, so the total operator squares to zero. □

20.4 Two genus gradings

For a connected bicoloured graph, let h be the ribbon genus of the transverse thickening and let g be the arithmetic genus of the chiral stable curve. The amplitude decomposes as

$$\Theta = \sum_{g,h \geq 0} \hbar_X^g \hbar_\perp^h \Theta_{g,h}.$$

The two-edge equation recursively determines the boundary of $\Theta_{g,h}$ from lower pairs in the partial order generated by edge contraction. This is the universal double expansion of a theory carrying open-string colour loops and worldsheet handles.

20.5 Existence from a bicoloured sewing datum

Recognition Theorem 20.5.1 (Geometric two-edge realization)**.** *Let V be a perfect dg collection equipped with:*

1. *cyclic transverse gluing maps and their perfect pairing;*
2. *chiral clutching maps and their perfect sewing pairing;*
3. *common corolla amplitudes carrying both families of flags;*
4. *the one-colour associativity relations;*
5. *the mixed interchange square equating the two orders of one transverse and one chiral gluing, with the determinant-line sign.*

These data define a unique continuous dg action of $\mathbb{F}^{(2)}(V)$. Its master element satisfies Theorem 20.3.1.

Proof. The common corollas define the images of the generators. The two gluing families evaluate every edge of the corresponding colour. One-colour associativity evaluates the codimension-two faces with equal colours, and the mixed interchange square evaluates the faces with distinct colours. Thus every relation in the bicoloured graph differential is respected. Freeness gives existence and uniqueness of the action, and Theorem 20.3.1 gives its Maurer–Cartan equation. □

20.6 Graphwise duality

If every coefficient space is perfect, linear duality reverses every edge contraction. The transpose of a two-edge master element therefore satisfies the dual master equation, with the determinant orientation shifted by the duality degree. Under Koszul duality, transverse tensor contractions become cotensor contractions, while the chiral sewing coefficient is transported by Verdier or contragredient duality. This is the exact graphwise meaning of duality for the two sewing geometries.

Chapter 21

Perturbative BV realization

21.1 Local cyclic L_∞ data

A classical perturbative field theory on a manifold M is described by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and a local action

$$S(\phi) = \sum_{n\geq 1} \frac{1}{(n+1)!}\langle \phi, \ell_n(\phi^{\otimes n})\rangle.$$

The classical master equation is

$$QS + \frac{1}{2}\{S,S\} = 0.$$

Classical observables on an open set U are Chevalley cochains

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE}}(\mathcal{L}_c(U)).$$

Disjoint supports give the prefactorisation product.

A gauge fixing produces a heat kernel and a propagator. A renormalisation scheme gives effective interactions $I[L]$. At scale L, the quantum master equation is

$$(Q + \hbar\Delta_L)e^{I[L]/\hbar} = 0,$$

and the quantum observable differential is

$$Q_L = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

A factorising renormalisation scheme further requires the effective weights to respect restriction to disjoint opens and to factor on every boundary stratum of the compactified configuration spaces. This is the locality input that turns the scale-dependent BV complex into a factorisation algebra [CG17, CG21].

21.2 Holomorphic–topological theories

Let $M = X^{\mathrm{an}} \times \mathbb{R}$. Suppose the elliptic complex is holomorphic in X and de Rham in $\mathbb{R}$, and suppose translation along $\mathbb{R}$ is homotopically trivialized by topological descent. Then the quantum observables are holomorphic in the curve direction and locally constant in the interval direction.

Definition 21.2.1 (Factorising BV quantization)**.** A factorising BV quantization on $X^{\mathrm{an}} \times \mathbb{R}$ consists of:

1. effective interactions satisfying homotopy RG flow and QME;
2. compatible extension of every graph weight to the chosen compactified configuration spaces;
3. the contraction identity on each collision face;
4. Weiss descent and disjoint-union multiplicativity for the resulting observable complexes.

Theorem 21.2.2 (BV realization of ordered chirality)**.** *Let a factorising perturbative BV quantization on $X^{\mathrm{an}} \times \mathbb{R}$ satisfy:*

1. *holomorphic algebraisation of its observable coefficient on X as a factorisation right D-module;*
2. *local constancy along $\mathbb{R}$;*
3. *a boundary or defect condition whose observable assignment is closed under the factorisation maps.*

Then its boundary observables form an ordered chiral algebra

$$\mathrm{Obs}^{\mathrm{q}}_{\partial} \in \mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X), \otimes^{!}\big).$$

The E_1 product is ordered interval fusion. The chiral coefficient is the algebraised renormalised distribution on the holomorphic collision diagonals.

Proof. The factorising quantization supplies a factorisation algebra of quantum observables. Holomorphic algebraisation identifies its curve coefficient with an object of $\mathrm{FactD}(X)$. Local constancy on the ordered configuration spaces of intervals gives the Segal E_1 operations. The collision-face contraction identity gives compatibility of these operations with the factorisation coefficient, and QME gives the square-zero quantum differential. The boundary condition restricts this rectangular structure to the boundary observable object. The rectangular recognition theorem Theorem 4.2.1 now gives the stated E_1 algebra in $\mathrm{FactD}(X)$. □

21.3 Compactified graph configurations

For a graph Γ with vertex set I, the unrenormalised weight is a distribution on $\mathrm{Conf}_I(M)$. Resolve simultaneous collisions by a Fulton–MacPherson or equivalent wonderful compactification. A factorising renormalisation extends the weight to its boundary and satisfies, on the face where a connected subgraph γ collapses,

$$W_\Gamma|_{F_\gamma} = W_\gamma \circ W_{\Gamma/\gamma},$$

where W_γ includes its local counterterm.

With the determinant-line orientation of the normal directions, Stokes' theorem gives

$$dW_\Gamma = \sum_{\gamma \subset \Gamma} \pm W_\gamma \circ W_{\Gamma/\gamma}.$$

The left-hand side contains the de Rham and BV differentials; the boundary sum contains the interaction bracket, BV loop contraction, and counterterm faces. Thus the master equation is the chain-map condition for the graph weight assignment. This proves the mixed-forest morphism of Theorem 5.4.1 in every factorising renormalisation scheme.

21.4 Anomaly classes

At order $\hbar^r$, the defect of QME is a closed local functional

$$\mathfrak{A}_r \in H^1_{\mathrm{loc}}(\mathcal{L}).$$

A local counterterm changes it by an exact class. The obstruction to extending the quantization is therefore $[\mathfrak{A}_r]$.

Theorem 21.4.1 (Locality of the BV anomaly)**.** *The obstruction to extending a factorising quantization from order $\hbar^{r-1}$ to order $\hbar^r$ is a class in the local deformation cohomology of the bulk theory. Restriction to a boundary or defect maps this class to the obstruction for the induced ordered chiral quantization.*

Proof. The lower-order master equation and the Jacobi identity make the order-r defect closed for the linearized QME differential. Renormalisation locality expresses the defect by a local functional. Changing the effective interaction by an order-r counterterm changes the defect by a coboundary. Boundary restriction intertwines the local brackets, differentials, and factorisation maps, and hence defines the asserted map of obstruction complexes. □

21.5 Invertible anomaly theories

A determinant line with connection, a projective central extension, and an invertible field theory become equivalent descriptions after the appropriate transgression maps are supplied. Tensoring by an inverse invertible theory cancels the class. Comparisons among BRST, Quillen, framing, and BV anomalies are therefore morphisms between their deformation complexes, followed by the relevant transgressions.

21.6 Bulk action

A bulk observable can be moved to the boundary through a half-disc. The resulting operations commute coherently with ordered boundary insertions and give the open–closed morphism

$$\mathrm{Obs}^{\mathrm{q}}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ord,ch}}(\mathrm{Obs}^{\mathrm{q}}_{\partial}).$$

Under the generator, bulk-detection, and central-representability hypotheses of Theorem 9.4.1, this morphism is an equivalence. The physical bulk is then identified with the universal derived centre by a constructed open–closed comparison.

Part VI

Exact noncommutative models

Chapter 22

Free, square-zero, quiver, and matrix models

22.1 The free ordered chiral algebra

Let $M \in \mathrm{FactD}(X)$. The free augmented ordered chiral algebra is

$$T_X(M) = \Bbbk \oplus M \oplus M^{\otimes^! 2} \oplus M^{\otimes^! 3} \oplus \cdots$$

with concatenation and augmentation onto the empty word.

Theorem 22.1.1 (Bar of the free algebra)**.** *Assume that tensor product is exact on the powers of M. Then*

$$\overline{B}_X^{\mathrm{ord}}(T_X(M)) \simeq \Bbbk \oplus sM$$

as conilpotent coalgebras, with primitive reduced part sM.

Proof. Every nonempty tensor word has a unique last letter. Hence there is an exact sequence of left $T_X(M)$-modules

$$T_X(M) \otimes^! M \xrightarrow{\text{mult}} T_X(M) \longrightarrow \Bbbk.$$

The first map identifies $T_X(M) \otimes^! M$ with the positive-word summand. This is a two-term free resolution of the augmentation. Tensoring over $T_X(M)$ with $\Bbbk$ gives the complex $M \to \Bbbk$ with zero reduced differential and suspension sM. Equivalently, the normalized bar has an extra degeneracy that contracts every bar word with an internal cut and leaves the one-letter primitive. □

If M is dualisable, the augmentation-boundary algebra is the square-zero extension

$$T_X(M)^! \simeq \Bbbk \oplus M^*[-1].$$

Free noncommutative composition and first-order infinitesimal deformation are Koszul dual.

22.2 Square-zero algebra

Let

$$A = \Bbbk \oplus M, \qquad M \otimes^! M \xrightarrow{0} M.$$

The multiplication component of the reduced bar differential vanishes. Thus

$$\overline{B}_X^{\mathrm{ord}}(A) = T^c(sM)$$

with deconcatenation, and its continuous dual is

$$A^! \simeq \widehat{T}_X(M^*[-1]).$$

This pair is the universal local model of an infinitesimal boundary coupling and its algebra of iterated defect insertions.

22.3 Internal quiver path algebras

Let $Q = (Q_0, Q_1)$ be a finite quiver. Put

$$S = \bigoplus_{i \in Q_0} \Bbbk e_i, \qquad e_i e_j = \delta_{ij} e_i.$$

Assign $M_a \in \mathrm{FactD}(X)$ to each arrow $a : i \to j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i.$$

The internal path algebra is

$$T_S(M) = S \oplus M \oplus (M \otimes^!{}_S M) \oplus \cdots .$$

A path contributes exactly when the target idempotent of one arrow equals the source idempotent of the next.

Let $R \hookrightarrow M \otimes^!{}_S M$ be a factorisation subobject of quadratic relations. The quotient

$$A_Q = T_S(M)/(R)$$

is a concrete ordered chiral algebra. Its weight-two bar differential is path concatenation, and its degree-two Koszul cogenerators are $R[2]$.

Example 22.3.1 (A_2 path algebra)**.** Take one arrow $a : 1 \to 2$ with coefficient $M_a = \Bbbk$. The path algebra has basis e_1, e_2, a and products

$$e_2 a = a = a e_1, \qquad e_1 a = a e_2 = 0.$$

Its simple left modules S_1, S_2 have a single nontrivial extension

$$0 \longrightarrow S_2 \longrightarrow M \longrightarrow S_1 \longrightarrow 0.$$

The derived endomorphism algebra of $S_1 \oplus S_2$ is the opposite Ext quiver, with the extension class in degree one and square zero. This is the quiver form of the free/square-zero Koszul pair.

22.4 A quadratic quiver calculation

Let $a : 1 \to 2$ and $b : 2 \to 1$, and consider the radical-square-zero algebra

$$A_{\mathrm{cyc}} = \Bbbk Q/(ab, ba).$$

It has basis e_1, e_2, a, b. Write $S = \Bbbk e_1 \oplus \Bbbk e_2$ and let M be the S-bimodule spanned by a, b. Every composable quadratic word is a relation, so

$$R = M \otimes_S M.$$

For the quadratic dual

$$A^!_{\mathrm{cyc}} = T_S(M^*[-1])/(R^\perp)$$

one has $R^{\perp} = 0$. Hence $A^!_{\mathrm{cyc}}$ is the full path algebra of the opposite two-cycle, with no quadratic relations.

The algebra A_{cyc} is Koszul. If $P_i = Ae_i$ and S_i is its simple quotient, then the minimal resolution of S_1 is

$$\cdots \xrightarrow{\cdot b} P_2 \xrightarrow{\cdot a} P_1 \xrightarrow{\cdot b} P_2 \xrightarrow{\cdot a} P_1 \longrightarrow S_1 \longrightarrow 0,$$

where the displayed maps alternate after the augmentation; the resolution of S_2 is obtained by interchanging a and b. Exactness follows from

$$\operatorname{rad} P_1 = \Bbbk a \cong S_2, \qquad \operatorname{rad} P_2 = \Bbbk b \cong S_1, \qquad \operatorname{rad}^2 A_{\mathrm{cyc}} = 0.$$

Thus

$$\operatorname{Ext}^*_{A_{\mathrm{cyc}}}(S_1 \oplus S_2, S_1 \oplus S_2) \cong A^!_{\mathrm{cyc}}$$

as the graded opposite-cycle path algebra. Tensoring the two arrow spaces with dualisable factorisation coefficients gives the same periodic Koszul resolution internally in $\mathrm{FactD}(X)$.

22.5 Matrix and Chan–Paton extension

For a finite-dimensional vector space W, put

$$A_W = A \otimes^! \operatorname{End}(W).$$

Multiplication is the product of A and matrix composition. The factorisation coefficient is unchanged up to the constant endomorphism system.

Proposition 22.5.1 (Morita matrix invariance)**.** *The bimodules $A \otimes W$ and $A \otimes W^*$ define a Morita equivalence between A and A_W. Hence*

$$Z^{\mathrm{der}}(A_W) \simeq Z^{\mathrm{der}}(A), \qquad \mathrm{HH}_*(A_W) \simeq \mathrm{HH}_*(A).$$

Proof. Matrix algebra is Morita equivalent to the ground field. Tensor the standard matrix Morita context with A inside $\mathrm{FactD}(X)$. Apply Theorem 9.5.1 and Morita invariance of Hochschild chains. □

For $W = \mathbb{C}^N$, ribbon contraction of matrix indices produces the large-N weights derived in Chapter 18. Matrix extension therefore changes the topological expansion while preserving the local Morita-invariant bulk.

22.6 Enveloping ordered chiral algebras

Let $\mathfrak{g}$ be a Lie object in $\mathrm{FactD}(X)$. Its enveloping algebra is the coequalizer

$$U_X(\mathfrak{g}) = T_X(\mathfrak{g}) / \big(xy - (-1)^{|x||y|}yx - [x, y]\big).$$

PBW gives

$$\operatorname{gr} U_X(\mathfrak{g}) \simeq \operatorname{Sym}_X(\mathfrak{g})$$

under the usual flatness hypotheses. Its bar is the Chevalley coalgebra by Theorem 13.3.1. This model is the universal bridge from a chiral Lie coefficient to a strictly noncommutative transverse algebra.

Chapter 23

The quantum plane, Weyl algebra, and curved Koszul duality

23.1 Quantum plane

Fix $q \in \Bbbk^\times$ and define

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy).$$

Orient the relation as $yx \to qxy$ and order words with every x before every y.

Theorem 23.1.1 (Diamond lemma for the quantum plane)**.** *The monomials $x^i y^j$, $i, j \geq 0$, form a basis of A_q. Its Hilbert series is*

$$H_{A_q}(t) = \frac{1}{(1-t)^2}.$$

Proof. Every inclusion overlap of the single quadratic relation is trivial. Each inversion yx is replaced by qxy, and the number of inversions decreases. The process terminates at $x^i y^j$. Any two reduction sequences exchange the same set of inversions and produce the same factor $q^{\#\mathrm{inv}}$, so the normal form is unique. Bergman's diamond lemma gives the basis [Ber78]. Counting normal monomials gives the Hilbert series. □

Let $\xi = x^*$ and $\eta = y^*$. The orthogonal complement of $y \otimes x - qx \otimes y$ is generated by

$$\xi^2, \qquad \eta^2, \qquad \xi\eta + q\eta\xi.$$

Hence

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Its Hilbert series is $1 + 2t + t^2$ and $H_{A_q}(t)H_{A_q^!}(-t) = 1$.

23.2 Explicit Koszul complex

The left Koszul resolution of the trivial module is

$$0 \longrightarrow A_q \otimes \Bbbk\xi\eta \xrightarrow{d_2} A_q \otimes (\Bbbk\xi \oplus \Bbbk\eta) \xrightarrow{d_1} A_q \longrightarrow \Bbbk \longrightarrow 0,$$

with

$$d_1(a \otimes \xi) = ax, \qquad d_1(a \otimes \eta) = ay,$$

$$d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Then

$$d_1 d_2(a \otimes \xi\eta) = a(yx - qxy) = 0.$$

Exactness follows from the PBW basis by separating the lowest y-degree in a cycle. Thus A_q is Koszul and its derived augmentation algebra is the quadratic dual above.

Tensoring x and y with any pair of factorisation coefficient objects gives a quantum-plane ordered chiral algebra. The relation is imposed inside the pointwise tensor category, so every chiral singular coefficient is retained.

23.3 The Weyl algebra

Let

$$A_\hbar = \Bbbk[[\hbar]]\langle x, y\rangle/(yx - xy - \hbar).$$

Filter by total word degree. The constant term lowers filtration by two.

Theorem 23.3.1 (PBW for the Weyl algebra)**.** *The monomials $x^i y^j$ form a $\Bbbk[[\hbar]]$-basis, and*

$$\operatorname{gr} A_\hbar \simeq \Bbbk[x, y][[\hbar]].$$

Proof. Orient $yx \longrightarrow xy + \hbar$. Every reduction decreases the inversion number; the constant term also decreases word length. The overlaps are disjoint applications of the same relation, and they commute. Unique normal forms follow. The leading relation is $yx - xy$, giving the commutative associated graded. □

The quadratic dual of the associated graded is $\Lambda(\xi, \eta)$. The constant term becomes curvature.

Theorem 23.3.2 (Curved dual of the Weyl algebra)**.** *The Koszul dual of $A_\hbar$ is the curved algebra*

$$\big(\Lambda(\xi, \eta)[[\hbar]], d = 0, h = \hbar\,\xi\eta\big).$$

The curved module equation $d_M^2 = h \cdot -$ is the dual form of the canonical commutation relation.

Proof. The relation has quadratic part $yx - xy$, linear part zero, and constant part $\hbar$. Apply Theorem 7.6.1. Evaluation of the constant functional on the dual relation produces $\hbar\xi\eta$. □

23.4 Clifford algebra

Let (V, q) be a finite-dimensional quadratic space. The Clifford algebra is

$$\mathrm{Cl}_\hbar(V, q) = T(V)[[\hbar]]/\big(vw + wv - 2\hbar q(v, w)\big).$$

Its associated graded is $\Lambda(V)$, and its curved Koszul dual is the symmetric algebra $\mathrm{Sym}(V^*[-1])$ with curvature given by the quadratic form $q^\vee$.

Proposition 23.4.1 (Clifford PBW and curved dual)**.** *For an ordered basis of V, monomials with strictly increasing indices form a basis of* $\mathrm{Cl}_\hbar(V, q)$*. The curved dual is*

$$\big(\mathrm{Sym}(V^*[-1])[[\hbar]], 0, \hbar q^\vee\big).$$

Proof. The standard Clifford rewriting exchanges inverted generators and replaces a repeated generator by its scalar square. The degree-three overlaps are the symmetry identities of q. PBW and the curved dual follow as in the Weyl case. □

23.5 Chiral Weyl and Clifford systems

Let M be a factorisation coefficient with a central pairing

$$\omega : M \otimes^! M \to \Bbbk[d].$$

The relations

$$xy - (-1)^{|x||y|} yx = \hbar\omega(x, y)$$

define a Weyl or Clifford ordered chiral algebra according to parity. The pairing may itself be a diagonal chiral distribution. The PBW theorem holds when the associated graded symmetric or exterior algebra is flat. Its curved bar dual records the central contraction as curvature. Free first-order systems and chiral differential operators arise from this construction.

Chapter 24

The A_2 Hall algebra in complete detail

24.1 The category

Let Q be the quiver $1 \to 2$ over $\mathbb{F}_q$. Its indecomposable representations are

$$S_1 = (\mathbb{F}_q \to 0), \qquad S_2 = (0 \to \mathbb{F}_q), \qquad M = (\mathbb{F}_q \xrightarrow{1} \mathbb{F}_q).$$

There is one nonsplit extension

$$0 \longrightarrow S_2 \longrightarrow M \longrightarrow S_1 \longrightarrow 0.$$

The Euler form on dimension vectors is

$$\langle (a,b),(c,d) \rangle = ac + bd - ad.$$

Let $[A] * [B]$ count subobjects isomorphic to B with quotient isomorphic to A:

$$[A] * [B] = \sum_C g_{A,B}^C [C].$$

Put $e_i = [S_i]$.

24.2 Binary products

The extension above gives

$$e_1 * e_2 = [S_1 \oplus S_2] + [M], \qquad e_2 * e_1 = [S_1 \oplus S_2].$$

A two-dimensional vector space has $q+1$ lines, hence

$$e_1 * e_1 = (q+1)[2S_1].$$

There are two isomorphism classes of representations of dimension vector $(2,1)$: the zero arrow

$$X = 2S_1 \oplus S_2$$

and a rank-one arrow

$$Y = S_1 \oplus M.$$

24.3 The three products

First,

$$e_1^2 e_2 = (q+1)([X]+[Y]).$$

Indeed, in both X and Y the subrepresentation S_2 is unique and the quotient is $2S_1$.

Second,

$$e_1 e_2 e_1 = (q+1)[X]+[Y].$$

For X, every line in the two-dimensional vertex-1 space is a submodule S_1, and the quotient is $S_1 \oplus S_2$; this gives $q+1$. For Y, the kernel of the rank-one arrow is the unique submodule S_1 whose quotient is M; this gives the coefficient one.

Third,

$$e_2 e_1^2 = (q+1)[X].$$

A subrepresentation $2S_1$ fills the vertex-1 space and has zero vertex-2 component. Stability forces the arrow to vanish; the contribution is therefore X.

Theorem 24.3.1 (Untwisted Hall–Serre relation)**.** *The Hall products satisfy*

$$e_1^2 e_2 - (q+1)e_1 e_2 e_1 + q e_2 e_1^2 = 0.$$

Proof. Substitute the three formulas. The coefficient of $[Y]$ is $(q+1)-(q+1)=0$. The coefficient of $[X]$ is

$$(q+1)-(q+1)^2+q(q+1)=0.$$

□

24.4 The companion dimension vector

There are two isomorphism classes of representations of dimension vector $(1,2)$: the zero arrow

$$X' = S_1 \oplus 2S_2$$

and the rank-one arrow

$$Y' = S_2 \oplus M.$$

The same flag count gives

$$e_2^2 e_1 = (q+1)[X'],$$

because a subrepresentation S_1 exists precisely for the zero arrow. Next,

$$e_2 e_1 e_2 = (q+1)[X']+[Y'].$$

For X', every line in the two-dimensional vertex-2 space gives a submodule S_2 with quotient $S_1 \oplus S_2$. For Y', the image line of the rank-one arrow is the unique such submodule. Finally,

$$e_1 e_2^2 = (q+1)([X']+[Y']),$$

because the full vertex-2 space is the unique submodule $2S_2$ in either representation, and its quotient is S_1.

Theorem 24.4.1 (Second untwisted Hall–Serre relation)**.** *The companion products satisfy*

$$q e_2^2 e_1 - (q+1)e_2 e_1 e_2 + e_1 e_2^2 = 0.$$

Proof. The coefficient of $[X']$ is

$$q(q+1) - (q+1)^2 + (q+1) = 0,$$

and the coefficient of $[Y']$ is

$$0 - (q+1) + (q+1) = 0.$$

□

24.5 Twisted Hall product

Let $v^2 = q$ and define

$$[A] \diamond [B] = v^{\langle \dim A, \dim B \rangle} [A] * [B].$$

Associativity follows from bilinearity of the Euler form.

The total twisting factors of the three triple products are

$$v^{-1}, \qquad 1, \qquad v,$$

respectively. Since $v^{-1}(q+1) = v + v^{-1}$, the untwisted relation becomes:

Theorem 24.5.1 (Quantum Serre relations)**.** *In the twisted Hall algebra,*

$$e_1 \diamond e_1 \diamond e_2 - (v + v^{-1}) e_1 \diamond e_2 \diamond e_1 + e_2 \diamond e_1 \diamond e_1 = 0,$$

and

$$e_2 \diamond e_2 \diamond e_1 - (v + v^{-1}) e_2 \diamond e_1 \diamond e_2 + e_1 \diamond e_2 \diamond e_2 = 0.$$

The subalgebra generated by e_1, e_2 is $U_v^+(\mathfrak{sl}_3)$.

Proof. For the first relation, the total twisting factors of $e_1^2 e_2, e_1 e_2 e_1, e_2 e_1^2$ are $v^{-1}, 1, v$. Multiplying Theorem 24.3.1 by v gives the first displayed identity. For the second relation, the twisting factors of $e_2^2 e_1, e_2 e_1 e_2, e_1 e_2^2$ are $v, 1, v^{-1}$. Substitution into Theorem 24.4.1 and division by v gives the second identity.

Let $Z = S_1 \oplus S_2$. Since

$$e_1 \diamond e_2 = v^{-1}([Z] + [M]), \qquad e_2 \diamond e_1 = [Z],$$

the positive-root vector is

$$e_{12} = e_1 \diamond e_2 - v^{-1} e_2 \diamond e_1 = v^{-1}[M].$$

Every representation of Q has a unique decomposition $S_2^a \oplus M^b \oplus S_1^c$. Expanding the divided-power monomial

$$e_2^{(a)} e_{12}^{(b)} e_1^{(c)}$$

in the isomorphism-class basis gives a nonzero coefficient on $[S_2^a \oplus M^b \oplus S_1^c]$ and terms with smaller rank of the arrow. Ordering strata by arrow rank makes this transition matrix triangular with invertible diagonal. These monomials form a basis and satisfy precisely the two quantum Serre relations, which identifies the generated algebra with $U_v^+(\mathfrak{sl}_3)$. □

24.6 Extension correspondence

The Hall product is the pull–push operation along the stack of short exact sequences

$$\mathcal{M}_A \times \mathcal{M}_B \xleftarrow{(q,s)} \mathcal{M}_{A\subset C} \xrightarrow{p} \mathcal{M}_C.$$

The calculation above is the finite-field shadow of the same correspondence in cohomology, Borel–Moore homology, or vanishing-cycle sheaves. Associativity is the equivalence between the two stacks of length-two filtrations. The quantum Serre relation is the first nontrivial cancellation among their strata.

24.7 Ordered chiral Hall coefficients

Let the representations vary in a factorisation family over X. Extension correspondences at a fixed chiral support define the transverse Hall product; disjoint supports define the factorisation coefficient. Whenever the pull–push maps preserve the factorisation system, the Hall algebra is an ordered chiral algebra. The A_2 calculation supplies a complete local model for the relation and its structure constants.

Chapter 25

The formal mixed holomorphic–topological trace algebra

25.1 Hamiltonians on the formal symplectic disc

Let

$$R = \mathbb{C}[[z_1, z_2]], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$

and let

$$\mathfrak{h} = R/\mathbb{C} \cdot 1$$

be the reduced Hamiltonian Lie algebra. Its shifted cotangent extension is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1].$$

The formal closed observables of Hamiltonian BF theory are the continuous Chevalley cochains $C^*_{\mathrm{CE,cont}}(\mathfrak{g})$.

This algebra is the local closed sector of the mixed holomorphic–topological model on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$. A Dirac brane supported on a topological line has two transverse matrix coordinates.

25.2 Derived commuting matrices

For N branes let $X, Y \in \mathfrak{gl}_N$ be even matrices and let $\psi \in \mathfrak{gl}_N[1]$ be odd. The Koszul differential

$$QX = QY = 0, \qquad Q\psi = [X, Y]$$

resolves the commuting equation. Including the conjugation ghost gives the function algebra of the derived quotient

$$[\mu^{-1}_{\mathrm{der}}(0)/\mathrm{GL}_N], \qquad \mu(X, Y) = [X, Y].$$

The gauge-invariant trace observables are

$$J_N(f) = \mathrm{Tr}\, f(X, Y).$$

25.3 Necklace Hamiltonians

Let $F = \mathbb{C}\langle x, y\rangle$ and $F_{\mathrm{cyc}} = F/[F, F]$. For a cyclic word f, define cyclic derivatives by cutting the word after every occurrence of the indicated letter:

$$\partial^{\mathrm{cyc}}_x [a_1 \cdots a_r] = \sum_{a_i = x} a_{i+1} \cdots a_r a_1 \cdots a_{i-1},$$

and similarly for y.

Define the Hamiltonian derivation

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f.$$

The cyclic Euler identity is

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0.$$

It implies

$$\delta_f([x, y]) = 0.$$

Thus necklace Hamiltonians act on the derived commuting equation.

The necklace bracket is

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f\, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}.$$

Theorem 25.3.1 (Necklace Lie algebra and trace action)**.** *The bracket* $\{-,-\}_{\mathrm{neck}}$ *makes* $F_{\mathrm{cyc}}/\mathbb{C}$ *a Lie algebra,*

$$[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}},$$

and the finite-rank trace map satisfies

$$\delta_f J_N(g) = J_N(\{f, g\}_{\mathrm{neck}}).$$

Proof. The Hamiltonian derivations preserve the noncommutative symplectic form $dx\, dy$ and hence their commutator is Hamiltonian. Evaluating the commutator on x and y gives the displayed necklace bracket and proves Jacobi modulo commutators. For the trace identity, cyclic differentiation gives

$$\begin{aligned}\delta_f \operatorname{Tr} g(X, Y) &= \operatorname{Tr}\left(\partial_x^{\mathrm{cyc}} g\, \delta_f X + \partial_y^{\mathrm{cyc}} g\, \delta_f Y\right)\\ &= \operatorname{Tr}\left(-\partial_x^{\mathrm{cyc}} g\, \partial_y^{\mathrm{cyc}} f + \partial_y^{\mathrm{cyc}} g\, \partial_x^{\mathrm{cyc}} f\right).\end{aligned}$$

Cyclicity of trace turns this into $J_N(\{f, g\}_{\mathrm{neck}})$. □

This theorem is a finite-N open–closed map at the level of Hamiltonian operations. It is an exact noncommutative calculation, independent of a large-N limit.

25.4 Stable multisymmetric algebra

On the semisimple commuting locus, simultaneously diagonalize

$$X = \operatorname{diag}(x_1, \ldots, x_N), \qquad Y = \operatorname{diag}(y_1, \ldots, y_N).$$

Define power sums

$$p_{ab} = \sum_{i=1}^{N} x_i^a y_i^b, \qquad a, b \geq 0, \quad a + b > 0.$$

The inverse limit of multisymmetric invariant rings over N is the polynomial algebra

$$\Lambda_2 = \mathbb{C}[p_{ab} : a + b > 0].$$

Equip the particle coordinates with $\{x_i, y_j\} = \delta_{ij}$.

Theorem 25.4.1 (Stable trace Poisson algebra)**.** *The stable power sums satisfy*

$$\boxed{\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1,b+d-1}.}$$

This is the commutative reduction of the necklace bracket.

Proof. Different particles Poisson commute. For one particle,

$$\{x^a y^b, x^c y^d\} = (ad - bc)x^{a+c-1}y^{b+d-1}.$$

Sum over particles. The trace of a cyclic monomial reduces to the corresponding power sum on the commuting locus, so the formula is also the commutative shadow of Theorem 25.3.1. □

25.5 Two branes and the A_1 cone

For $N = 2$, remove centre-of-mass coordinates and put

$$x = x_1 - x_2, \qquad y = y_1 - y_2, \qquad \{x, y\} = 2.$$

The Weyl-invariant relative coordinates are

$$A = \frac{x^2}{2}, \qquad B = \frac{xy}{2}, \qquad C = \frac{y^2}{2}.$$

They satisfy

$$B^2 = AC$$

and

$$\{A, B\} = 2A, \qquad \{B, C\} = 2C, \qquad \{A, C\} = 4B.$$

After a linear rescaling this is the Kirillov–Kostant bracket on the nilpotent cone of $\mathfrak{sl}_2$. Thus the first interacting finite-rank trace sector already contains a singular symplectic surface and a nonabelian Lie bracket.

25.6 Relation to the ordered bar

The boundary algebra is E_1 along the brane worldline. Its ordered bar resolves iterated trace insertions. The closed Hamiltonian CE algebra acts on that bar through the necklace derivations. The full mixed holomorphic–topological Deligne comparison asks for a quasi-isomorphism from the closed factorisation algebra to the derived centre of the boundary category; Theorem 25.3.1 constructs its formal trace-sector coordinate map.

Chapter 26

Exact chiral coefficient systems

26.1 Heisenberg current

Let a be an even field with

$$a(z)a(w) \sim \frac{k}{(z-w)^2}, \qquad [a_\lambda a] = k\lambda.$$

The Jacobi identity is immediate because the bracket is central. For $k \neq 0$, the Sugawara field

$$T = \frac{1}{2k} : aa :$$

satisfies the Virasoro OPE with central charge 1 and makes a primary of weight one. The coefficient system has one nonzero normal symbol: the second symbol given by k.

Tensoring the Heisenberg coefficient with a noncommutative algebra B gives an ordered chiral algebra $B \otimes V_{\mathrm{Heis}}$. The chiral current remains Heisenberg, while the transverse product is the multiplication of B. Matrix-valued free bosons and Chan–Paton currents are of this form.

26.2 Affine currents

Let $\mathfrak{g}$ have invariant form $(,-)$. The affine λ-bracket is

$$[J^a_\lambda J^b] = J^{[a,b]} + k(a,b)\lambda.$$

Proposition 26.2.1 (Affine Jacobi identity)**.** *The affine bracket satisfies the Lie conformal Jacobi identity precisely because* $[-,-]$ *satisfies Jacobi and* $(,-)$ *is invariant.*

Proof. The coefficient of $\lambda^0\mu^0$ is the Lie Jacobiator. The terms linear in λ and μ combine to $([a,b],c) - (a,[b,c])$, which vanishes by invariance. All higher terms vanish. □

At $k \neq -h^\vee$, the Sugawara field is

$$T_{\mathrm{Sug}} = \frac{1}{2(k+h^\vee)} \sum_a : J^a J_a :,$$

with central charge

$$c = \frac{k \dim \mathfrak{g}}{k+h^\vee}.$$

The KZ connection of Chapter 15 is the flat connection on affine conformal blocks induced by this stress tensor.

26.3 Virasoro

The Virasoro λ-bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

Substitution into the Lie conformal Jacobi identity gives equality of the coefficients of T, ∂T, and the central polynomial. The cubic central term is the fourth normal symbol of the diagonal OPE. The lower symbols form the projective-connection extension described in Chapter 10.

26.4 First-order systems

A bosonic $\beta\gamma$ pair has

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim -\frac{1}{z-w}.$$

Choose weights $(\lambda, 1-\lambda)$ and stress tensor

$$T_{\beta\gamma} = -\lambda : \beta\partial\gamma : -(1-\lambda) : \partial\beta\,\gamma : .$$

A Wick calculation gives

$$c_{\beta\gamma} = 2(6\lambda^2 - 6\lambda + 1).$$

For a fermionic bc pair with the same weights, the statistics reverse the double contraction and

$$c_{bc} = -2(6\lambda^2 - 6\lambda + 1) = 1 - 3(2\lambda - 1)^2.$$

These systems are chiral Weyl and Clifford algebras. Their singular first symbol is the perfect pairing; their ordered bar has the curved duality structure of Chapter 23 when a central deformation is included.

26.5 Lattice vertex algebras

Let L be an even lattice. The lattice vertex algebra is generated by the Heisenberg fields and exponentials e^α, $\alpha \in L$, with

$$Y(e^\alpha, z)Y(e^\beta, w) \sim (z-w)^{(\alpha,\beta)}\varepsilon(\alpha,\beta)Y(e^{\alpha+\beta}, w).$$

The cocycle ε satisfies the group-cohomology equation required by associativity. Analytic continuation gives the exchange phase $e^{\pi i(\alpha,\beta)}$. The genus-one character is the lattice theta series divided by $\eta^{\operatorname{rank} L}$.

26.6 Principal $\mathcal{W}$-algebras

Quantum Drinfeld–Sokolov reduction of $V_k(\mathfrak{g})$ produces the principal $\mathcal{W}$-algebra. At generic level its strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$. Their OPEs are reconstructed from the BRST complex and the requirement that the generators are primary. For $\mathfrak{sl}_2$ the result is Virasoro, as computed in Chapter 17. For $\mathfrak{sl}_3$ the generators are T and a weight-three field W. After the normalization of W is fixed, conformal covariance, the central charge, and the Jacobi identity determine the coefficients in the WW OPE.

26.7 Strictly ordered extensions

The coefficient systems above support several exact noncommutative constructions.

1. For every associative algebra B, the tensor product $B \otimes V$ is an ordered chiral algebra.
2. If a group G acts by automorphisms of both B and V and preserves the factorisation maps, the crossed product $(B \otimes V) \rtimes G$ is an ordered chiral algebra.
3. A collection of V-module coefficients on the arrows of a quiver gives the internal path algebra and its quotients of Chapter 22.
4. A meromorphic operator $R(u)$ gives a ZF or RTT extension precisely when it preserves the coefficient system, satisfies the Yang–Baxter relation, and obeys the locality compatibility with every chiral factorisation map.

In each construction the chiral OPE and the transverse multiplication remain separate structure maps. Their common constraint is the rectangular locality equation.

Chapter 27

Gauge-theoretic realizations

27.1 Free holomorphic–topological BF theory

Let V be a finite-dimensional complex vector space. On $X \times \mathbb{R}$ take fields

$$A \in \Omega^{0,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V[1], \qquad B \in \Omega^{1,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V^*[-1],$$

with action

$$S_{\mathrm{BF}} = \int_{X\times\mathbb{R}} \langle B, (\bar{\partial} + d_t)A\rangle.$$

The theory is Gaussian. Its observable differential is induced by $\bar{\partial} + d_t$. The de Rham homotopy on an interval proves local constancy in the t direction. The Dolbeault complex supplies the holomorphic coefficient on X.

Complementary linear boundary conditions select a polarization of $V \oplus V^*$. Boundary fields in dual polarizations have the propagator pairing, and their quantized boundary algebra is the chiral Weyl or Clifford algebra according to parity. The exact PBW and curved-dual calculations are those of Chapter 23.

27.2 Three-dimensional Chern–Simons and affine currents

Let $M = \Sigma \times \mathbb{R}_{\geq 0}$ and

$$S_{\mathrm{CS}}(A) = \frac{k}{4\pi}\int_M \mathrm{Tr}\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right).$$

The boundary variation is

$$\delta S_{\mathrm{CS}}|_{\partial M} = \frac{k}{4\pi}\int_\Sigma \mathrm{Tr}(A \wedge \delta A).$$

A chiral polarization fixes $A_{\bar{z}}$ and leaves A_z dynamical. The boundary symplectic form gives the current Poisson bracket

$$\{J^a(z), J^b(w)\} = f^{ab}{}_c J^c(w)\delta(z-w) + \frac{k}{2\pi}\kappa^{ab}\partial_w\delta(z-w).$$

Quantization gives the affine OPE of Chapter 26.

Wilson lines in the three-dimensional bulk end on affine primary fields. The flatness constraint gives the WZW conformal blocks, and the Ward identity produces the KZ connection. Thus the Chern–Simons/WZW correspondence realizes diagonal current algebra, braid transport, ribbon trace, and modular sewing in one gauge theory [Wit89].

27.3 Four-dimensional Chern–Simons

Let Σ be an oriented real surface, C a complex curve with meromorphic one-form ω, and $M = \Sigma \times C$. Four-dimensional Chern–Simons theory has action

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma\times C} \omega \wedge \operatorname{Tr}\left(A \wedge dA + \frac{2}{3}A^3\right),$$

where A has components along Σ and the antiholomorphic direction of C. The theory is topological on Σ and holomorphic on C.

A Wilson line along a curve in Σ at spectral point $z \in C$ is a line object. Fusion follows the order of parallel lines. Exchange in Σ gives the R-matrix. For $C = \mathbb{P}^1$ and $\omega = dz$, the tree-level exchange of two lines is

$$R(z_1 - z_2) = 1 + \hbar\frac{\Omega}{z_1 - z_2} + O(\hbar^2).$$

Proposition 27.3.1 (Classical Yang–Baxter from gauge invariance)**.** *The tree-level kernel*

$$r(z) = \frac{\Omega}{z}$$

satisfies the spectral classical Yang–Baxter equation.

Proof. The coefficient of the three-line gauge variation is

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})].$$

Invariance of the Casimir gives the infinitesimal braid relations, and the partial-fraction identity

$$\frac{1}{z_{12}z_{13}} - \frac{1}{z_{12}z_{23}} + \frac{1}{z_{13}z_{23}} = 0$$

annihilates the sum. □

Perturbative quantization fixes the higher terms by the quantum master equation and the framing correction. For gauge algebra $\mathfrak{gl}_N$ with the standard Wilson-line family, the completed rational line-operator algebra is the $\hbar$-adic Yangian; its RTT relations are the algebraic form of line exchange. For other simple gauge algebras, the same reconstruction includes the self-duality, framing, and quantum-junction data of the chosen representation family. Trigonometric and elliptic choices of (C, ω) give the corresponding quantum loop and elliptic exchange structures [CWY18a, CWY18b].

27.4 Boundary and centre

A boundary condition in a gauge theory gives a factorisation category of boundary lines and local boundary-changing operators. A factorisation-compatible compact generator b gives the ordered chiral endomorphism algebra A_b. Bulk local operators act centrally and define

$$\mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}(A_b).$$

When the generator, detection, and central representability hypotheses of Theorem 9.4.1 hold, this map is an equivalence of internal E_2 algebras.

27.5 Omega backgrounds

An Omega background couples a rotation vector field V_Ω to the cohomological supercharge:

$$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = \mathcal{L}_{V_\Omega}.$$

After localization, equivariant weights become deformation parameters in the operator algebra. In geometric representation theory they appear in Hall shuffle kernels, stable envelopes, and Yangian structure functions. The fixed-locus geometry supplies the exact parameters; the ordered chiral formalism supplies the common target category.

Part VII

Higher sources and arithmetic shadows

Chapter 28

Higher-dimensional sources and factorisation pushforward

28.1 Pushforward along an actual map

Let $p : Y \to X$ be a continuous map. For a prefactorisation algebra $\mathcal{F}$ on Y, define

$$(p_*\mathcal{F})(U) = \mathcal{F}(p^{-1}U).$$

Inverse image preserves inclusions and disjointness, so every prefactorisation operation pushes forward.

Theorem 28.1.1 (Factorisation pushforward)**.** *Let $p : Y \to X$ be continuous and let $\mathcal{F}$ be a factorisation algebra on Y valued in a presentable symmetric monoidal category. Then $p_*\mathcal{F}$ is a factorisation algebra on X.*

Suppose in addition that p is locally a product projection $U \times F \to U$, that F is compact, and that $\mathcal{F}$ is locally constant in the F direction. Then, on each such product chart,

$$(p_*\mathcal{F})(U) \simeq \int_F \mathcal{F}|_{U\times F}.$$

If X and Y are complex manifolds, p is a holomorphic submersion, and the fibrewise pushforward of the holomorphic observable complex is coherent and compatible with base change, then this analytic pushforward admits the corresponding holomorphic factorisation coefficient on X.

Proof. Let $\{U_i\}$ be a Weiss cover of $U \subset X$. Every finite subset of $p^{-1}U$ has finite image in U, hence lies in $p^{-1}U_i$ for some i. Thus $\{p^{-1}U_i\}$ is a Weiss cover of $p^{-1}U$. Descent for $\mathcal{F}$ therefore gives descent for $p_*\mathcal{F}$. The local product formula is the Fubini/excision theorem for factorisation homology. In the holomorphic case, the stated coherence and base-change hypotheses identify the analytic pushforward with the de Rham realization of a factorisation D-module on X. □

An ordered chiral algebra appears when the pushforward retains one oriented locally constant direction. The residual structure is determined by the map p, the fibre theory, and the chosen boundary or defect.

28.2 Categorical source data

A smooth proper d-Calabi–Yau category has two universal categorical outputs:

1. its Hochschild cochains carry the framed E_2 structure of the cyclic Deligne theorem [BR23];

2. its derived moduli stack of objects carries a $(2-d)$-shifted symplectic structure under the geometric finiteness hypotheses of shifted symplectic geometry [PTVV13].

These outputs describe the categorical geometry. A local field-theoretic source additionally contains a space Y, fields or moduli assigned to opens of Y, a local action or shifted Poisson structure, a quantization, and anomaly control. A boundary condition then produces the endomorphism object of Theorem 1.4.1.

Theorem 28.2.1 (Categorical boundary source)**.** *Let C be a stable category enriched in* $\mathrm{FactD}(X)$*, with factorising composition. Let b be a factorisation-compatible compact dualisable generator: its restrictions to disjoint opens factorise, and the resulting local Morita equivalences satisfy descent. Then*

$$A_b = \mathrm{RHom}_C(b, b)$$

is an ordered chiral algebra and $C \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$ as factorisation categories. A nondegenerate cyclic Calabi–Yau trace on the full subcategory generated by b induces the perfect cyclic structure used in Theorem 18.2.1.

Proof. Factorising composition gives the internal E_1 multiplication on A_b, and factorisation compatibility of b gives its coefficient maps. The local compact-generator Morita theorem identifies each local category with A_b-modules; the descent hypothesis glues these equivalences. The categorical trace restricts to a cyclic trace on endomorphisms of b. Dualisability identifies the categorical diagonal with a perfect A_b-bimodule, so nondegeneracy becomes perfection of the induced pairing. □

28.3 Shifted symplectic quantization

Let $\mathcal{M} \to X$ be a factorising derived moduli problem equipped with a P_n-structure, equivalently a bracket of cohomological degree $1-n$ obeying the shifted Poisson identities. An E_n deformation quantization is a filtered E_n algebra whose associated graded P_n algebra is the function algebra of $\mathcal{M}$. Such a quantization varies factorisably over X and therefore gives an E_n algebra in the coefficient category. Restriction to one oriented residual direction gives an E_1 algebra in factorisation coefficients. The construction is functorial for quantized morphisms; its existence and uniqueness are computed by the deformation complex of the specified moduli problem.

28.4 Hall multiplication

Let $\mathcal{M}$ be a moduli stack in a stable category and let $\mathcal{M}^{(2)}$ classify exact triangles

$$E_1 \longrightarrow E \longrightarrow E_2 \longrightarrow E_1[1].$$

The correspondence

$$\mathcal{M} \times \mathcal{M} \xleftarrow{(E_1,E_2)} \mathcal{M}^{(2)} \xrightarrow{E} \mathcal{M}$$

defines Hall multiplication whenever the pullback, coefficient twist, and pushforward are defined. In a critical theory, the coefficient is a vanishing-cycle complex in the cohomological Hall framework [KS11] and the orientation datum supplies its multiplicative square root.

Theorem 28.4.1 (Hall associativity from flags)**.** *Suppose the extension correspondence admits the required proper or localized pushforwards, all Beck–Chevalley maps for the flag squares are equivalences, and the coefficient system is multiplicative under extensions. Then the Hall product is associative. The associator is the canonical identification through the stack of length-two filtrations*

$$0 \subset E_1 \subset E_{12} \subset E.$$

Proof. The left parenthesization first chooses $E_1 \subset E_{12}$ and then $E_{12} \subset E$. The right parenthesization first chooses the quotient filtration and then pulls it back. Both constructions present the same derived flag stack. Beck–Chevalley identifies the two pull–push functors, and multiplicativity identifies their coefficient complexes. The resulting natural isomorphism satisfies the pentagon because length-three filtrations present every composite associator. □

If the moduli problem factorises by support, Hall multiplication is pointwise in the support variable. Disjoint-support Ext vanishing supplies the factorisation maps, so the Hall object becomes an ordered chiral algebra.

28.5 Bialgebra and double data

A quantum double requires operations beyond Hall multiplication:

$$(H^+, m, \Delta), \qquad H^0, \qquad H^-, \qquad \langle -, - \rangle,$$

together with a radical quotient and a topology in which the cross-relations and universal R-matrix converge. A geometric coproduct may require localization or stable envelopes; the Hopf pairing requires a proper duality construction.

Recognition Theorem 28.5.1 (Hall quantum-group criterion)**.** *Let an ordered chiral Hall algebra have a complete coproduct, Cartan part, negative half, nondegenerate Hopf pairing after radical quotient, and PBW triangular decomposition. Then its completed Drinfeld double is a quasi-triangular Hopf object. A presentation isomorphism of the positive half and Cartan data that preserves the pairing extends uniquely to the double.*

Proof. The Hopf pairing defines the double cross-relations. The PBW theorem makes

$$H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+ \longrightarrow D(H^+)$$

an isomorphism of complete modules. The canonical element of dual bases is the universal R-matrix. A presentation map preserving the pairing and Cartan action preserves every cross-relation and hence extends uniquely. □

28.6 Local and wrapped sectors

Let $p : Y \to X$ be a proper fibration. A support whose image in X is finite belongs to ordinary Ran locality. A support whose image contains a positive-dimensional component belongs to a wrapped sector.

Definition 28.6.1 (Hybrid local–wrapped algebra)**.** A hybrid local–wrapped algebra on X consists of two colours L, W and the following data:

1. a factorisation algebra $\mathcal{F}_\mathsf{L}$ on $\mathrm{Ran}(X)$;
2. a complete complex or category $\mathcal{H}_\mathsf{W}$ graded by wrapped charge;
3. ordered operations
$$\mu_{n,1} : \mathcal{F}_\mathsf{L}(U_1) \otimes \cdots \otimes \mathcal{F}_\mathsf{L}(U_n) \otimes \mathcal{H}_\mathsf{W} \longrightarrow \mathcal{H}_\mathsf{W}$$
for pairwise disjoint opens U_i, compatible with refinement and ordered fusion;
4. wrapped extension products
$$m_\mathsf{W} : \mathcal{H}_{\mathsf{W},\gamma} \otimes \mathcal{H}_{\mathsf{W},\delta} \longrightarrow \mathcal{H}_{\mathsf{W},\gamma+\delta};$$

5. the mixed associativity identity

$$m_{\mathsf{W}}(\mu(a, w_1), w_2) = \mu(a, m_{\mathsf{W}}(w_1, w_2)) = m_{\mathsf{W}}(w_1, \mu(a, w_2))$$

   with the factorisation and Koszul signs determined by the ordered inputs.

Morphisms preserve both colours and all displayed operations.

The two-colour operad generated by local factorisation, local action, and wrapped multiplication presents this definition. Its arity-three relations are ordinary associativity in each colour and the mixed identity above. Consequently the direct sum

$$\mathcal{F}_{\mathsf{L}} \ltimes \mathcal{H}_{\mathsf{W}}$$

is an ordered algebra object in the corresponding two-colour coefficient category. The local and wrapped sectors remain coupled through the action.

Lemma 28.6.2 (Positive elliptic degree is wrapped)**.** *Let S be a proper complex variety, let E be an elliptic curve, and let $C \subset S \times E$ be a proper one-dimensional closed subscheme. If*

$$(p_E)_*[C] = d[E], \qquad d > 0,$$

then $p_E(\operatorname{Supp} C) = E$.

Proof. The positive pushforward degree is the sum of the degrees of the irreducible components that dominate E. Hence at least one component of $\operatorname{Supp} C$ has one-dimensional image. Properness makes that image closed, and the only one-dimensional closed subset of the irreducible curve E is E itself. Thus the image of the full support is E. □

For $K3 \times E \to E$, Theorem 28.6.2 places every positive elliptic-degree curve sector in $\mathcal{H}_{\mathsf{W}}$. Projection-finite zero-degree sectors form $\mathcal{F}_{\mathsf{L}}$. Hall extension correspondences provide m_{W} and the mixed local action whenever their pull–push maps and coefficient systems satisfy Theorem 28.4.1. This is the support geometry required by the positive p-direction of the Igusa product.

Chapter 29

Bar Euler products and Borcherds denominators

29.1 Graded Lie superalgebras

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Gamma \setminus \{0\}} \mathfrak{g}_\alpha$$

be a root-graded Lie superalgebra, choose a positive monoid $\Gamma_+ \subset \Gamma$, and put

$$\mathfrak{n}_+ = \bigoplus_{\alpha \in \Gamma_+} \mathfrak{g}_\alpha.$$

Assume that every root space is finite dimensional and that every degree has finitely many decompositions in the chosen Γ_+-adic completion. PBW and Theorem 13.3.1 give

$$\overline{B}^{\mathrm{ord}}\big(U(\mathfrak{n}_+)\big) \simeq \mathrm{CE}_*(\mathfrak{n}_+) = \mathrm{Sym}(\mathfrak{n}_+[1]).$$

Theorem 29.1.1 (Bar denominator product)**.** *The completed graded Euler supercharacter of the ordered bar is*

$$\chi_\Gamma \overline{B}^{\mathrm{ord}}\big(U(\mathfrak{n}_+)\big) = \prod_{\alpha \in \Gamma_+} (1 - e^{-\alpha})^{\mathrm{sdim}\, \mathfrak{g}_\alpha}.$$

If $\mathfrak{g}$ is a Borcherds–Kac–Moody superalgebra with Weyl vector ρ, then

$$e^{-\rho} \chi_\Gamma \overline{B}^{\mathrm{ord}}\big(U(\mathfrak{n}_+)\big)$$

is the product side of its Weyl–Kac–Borcherds denominator identity.

Proof. Choose a homogeneous basis of each root space. An even root vector becomes odd after the suspension in $\mathrm{CE}_*(\mathfrak{n}_+)$ and contributes $1 - e^{-\alpha}$ to the Euler supercharacter. An odd root vector becomes even and contributes $(1 - e^{-\alpha})^{-1}$. Multiplying over the basis gives the exponent $\dim(\mathfrak{g}_{\alpha,\bar{0}}) - \dim(\mathfrak{g}_{\alpha,\bar{1}})$. The finite-decomposition hypothesis makes every coefficient of the completed product finite. The factor $e^{-\rho}$ is the leading Weyl monomial in the denominator identity. □

The theorem is an object-to-character bridge. The Lie bracket controls the Chevalley differential. The Euler product remembers the signed root spaces.

29.2 The Igusa algebra

Let

$$\phi_{0,1}(\tau,z)=\sum_{n\geq 0,\ell\in\mathbb{Z}} f(n,\ell)q^n y^\ell$$

be the weight-zero, index-one weak Jacobi form normalized by $f(0,0)=10$ and $f(0,\pm 1)=1$. In the standard cusp chamber its monic Borcherds product is

$$D_5(Z)=q^{1/2}y^{1/2}p^{1/2}\prod_{(n,\ell,m)>0}(1-q^n y^\ell p^m)^{f(nm,\ell)}.$$

The theta-constant normalization is

$$\Delta_5(Z)=64D_5(Z).$$

The Gritsenko–Nikulin Borcherds algebra $\mathfrak{g}_{\Delta_5}$ has signed root multiplicities

$$\operatorname{sdim}(\mathfrak{g}_{\Delta_5})_{\alpha(n,\ell,m)}=f(nm,\ell)$$

and denominator

$$\operatorname{den}(\mathfrak{g}_{\Delta_5})=D_5(2Z)=64^{-1}\Delta_5(2Z)$$

[GN97].

Corollary 29.2.1 (Igusa bar identity)**.** *For the positive nilpotent algebra* $\mathfrak{n}_{\Delta_5,+}$*,*

$$\boxed{e^{-\rho}\chi_\Gamma\overline{B}^{\mathrm{ord}}\big(U(\mathfrak{n}_{\Delta_5,+})\big)=D_5(2Z)=64^{-1}\Delta_5(2Z).}$$

Proof. Insert the Gritsenko–Nikulin root supermultiplicities into Theorem 29.1.1. The Weyl monomial and the doubled period variable are those of the type-II denominator chamber. The theta-product leading coefficient converts the monic product to $64^{-1}\Delta_5(2Z)$. □

29.3 Finite positive windows and the current envelope

Assume that the positive monoid carries a proper additive height

$$h:\Gamma_+\longrightarrow\mathbb{N}_{>0},\qquad h(\alpha+\beta)=h(\alpha)+h(\beta),$$

whose fibres are finite. For $H\geq 1$ put

$$F^{>H}\mathfrak{n}_+=\bigoplus_{h(\alpha)>H}\mathfrak{g}_\alpha,\qquad \mathfrak{n}_+^{(H)}=\mathfrak{n}_+/F^{>H}\mathfrak{n}_+.$$

Additivity of h makes $F^{>H}\mathfrak{n}_+$ a Lie ideal. Finite fibres make $\mathfrak{n}_+^{(H)}$ finite dimensional, and every iterated bracket of more than H positive-height generators vanishes in the quotient. Thus $\mathfrak{n}_+^{(H)}$ is nilpotent.

Let

$$V_0^{(H)}=U_{\mathrm{VA}}\big(\operatorname{Cur}\mathfrak{n}_+^{(H)}\big)$$

be its universal level-zero current vertex superalgebra. The quotient $\mathfrak{n}_+^{(H+1)}\twoheadrightarrow\mathfrak{n}_+^{(H)}$ induces a surjection $V_0^{(H+1)}\twoheadrightarrow V_0^{(H)}$.

Definition 29.3.1 (Completed positive current envelope)**.**

$$\widehat{V}_0(\mathfrak{n}_+)=\varprojlim_H V_0^{(H)}.$$

Theorem 29.3.2 (PBW character of the completed current envelope)**.** *The inverse limit is separated and complete for the height filtration, and its completed PBW supercharacter is*

$$\operatorname{sch} \widehat{V}_0(\mathfrak{n}_+) = \prod_{r\geq 1} \prod_{\alpha\in\Gamma_+} (1 - t^r e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_\alpha}.$$

Proof. PBW for $V_0^{(H)}$ identifies its state space with the supersymmetric algebra on

$$t^{-1}\mathfrak{n}_+^{(H)}[t^{-1}].$$

A mode of frequency r and root α contributes $(1 - t^r e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_\alpha}$. For a fixed coefficient of $t^N e^{-\beta}$, only roots with $h(\alpha) \leq h(\beta)$ and frequencies $r \leq N$ occur; properness of h makes this set finite. The coefficient therefore stabilizes for $H \geq h(\beta)$, which proves the product formula and coefficientwise convergence. The kernels of the projections to $V_0^{(H)}$ have zero intersection, giving separatedness; the inverse-limit definition gives completeness. ◻

The completion is formed on the positive algebra, where the height tails are genuine ideals. The full Borcherds algebra has its algebraic affine-current vertex superalgebra as a separate object.

29.4 Quadratic charge and the polarized Gram transform

Let Γ_{phys} and Λ be abelian groups. A quadratic map

$$\Pi : \Gamma_{\text{phys}} \longrightarrow \Lambda$$

has symmetric biadditive polarization

$$\mathsf{B}(\gamma,\delta) = \Pi(\gamma+\delta) - \Pi(\gamma) - \Pi(\delta).$$

Let $A = \bigoplus_\gamma A_\gamma$ be a charge-graded algebra, and write u^λ for the basis of the group algebra $\Bbbk[\Lambda]$. The Gram embedding is

$$\iota_\Pi(a_\gamma) = a_\gamma u^{\Pi(\gamma)}.$$

Under the ordinary product in $A \otimes \Bbbk[\Lambda]$,

$$\iota_\Pi(a_\gamma)\iota_\Pi(b_\delta) = u^{-\mathsf{B}(\gamma,\delta)}\iota_\Pi(a_\gamma b_\delta).$$

Thus B is the exact defect of additivity of the automorphic degree.

Define a second product on homogeneous tensors by

$$(a_\gamma u^\lambda) \star_{\mathsf{B}} (b_\delta u^\mu) = a_\gamma b_\delta\, u^{\lambda+\mu+\mathsf{B}(\gamma,\delta)}.$$

Theorem 29.4.1 (Polarized Gram transform)**.** *The product $\star_{\mathsf{B}}$ is associative and the Gram embedding is an algebra homomorphism:*

$$\iota_\Pi(a_\gamma) \star_{\mathsf{B}} \iota_\Pi(b_\delta) = \iota_\Pi(a_\gamma b_\delta).$$

If A is a Lie superalgebra, the formula

$$[a_\gamma u^\lambda, b_\delta u^\mu]_{\mathsf{B}} = [a_\gamma, b_\delta] u^{\lambda+\mu+\mathsf{B}(\gamma,\delta)}$$

defines a Lie superalgebra and makes ι_Π a Lie homomorphism.

Proof. Biadditivity gives the cocycle identity

$$\mathsf{B}(\gamma,\delta)+\mathsf{B}(\gamma+\delta,\epsilon)=\mathsf{B}(\delta,\epsilon)+\mathsf{B}(\gamma,\delta+\epsilon).$$

The two parenthesizations of a triple $\star_{\mathsf{B}}$-product therefore have the same exponent. The homomorphism identity follows from

$$\Pi(\gamma+\delta)=\Pi(\gamma)+\Pi(\delta)+\mathsf{B}(\gamma,\delta).$$

For the Lie statement, symmetry of B preserves graded antisymmetry. Every term of the graded Jacobi identity acquires the common factor

$$u^{\lambda+\mu+\nu+\mathsf{B}(\gamma,\delta)+\mathsf{B}(\gamma,\epsilon)+\mathsf{B}(\delta,\epsilon)},$$

so Jacobi follows from Jacobi in A. □

For $K3\times E$, write a physical charge as $c=(P,Q)$ with P,Q in the Mukai lattice. Then

$$\Pi(P,Q)=\left(\frac{P^2}{2},(P,Q),\frac{Q^2}{2}\right)$$

and

$$\mathsf{B}\big((P,Q),(P',Q')\big)=\big((P,P'),(P,Q')+(P',Q),(Q,Q')\big).$$

The polarized Gram transform therefore preserves additive physical charge and turns the three quadratic Fourier indices into an honest additive grading for the twisted product.

29.5 Character and geometric realization

The automorphic denominator constructs the target root algebra and its character. A compact Hall realization is a further geometric theorem: it must construct the source moduli, orientations, extension bracket, PBW comparison, and the map to $\mathfrak{g}_{\Delta_5}$. Once such a map is supplied, Theorem 29.2.1 becomes the internal bar character of that geometric algebra. The hybrid local–wrapped datum of Chapter 28 provides the support geometry required by the positive elliptic-degree sector.

Chapter 30

The structure theorem

30.1 The invariant object

The local object of the theory is

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}\big(\mathrm{FactD}(X), \otimes^{!}\big).$$

Its primitive operations are the chiral coefficient on the curve and the ordered associative multiplication in the transverse line. Every further operation enters through a specified geometric or categorical construction.

Theorem 30.1.1 (Noncommutative chiral Koszul geometry)**.** *Let X be a smooth complex algebraic curve and let A be an augmented ordered chiral algebra. The following statements hold on the natural complete subcategories specified in the preceding chapters.*

1. *The block-Ran section of A is Segal in the ordered blocks and factorising in the chiral labels.*
2. *Linear species carry the aligned-decomposition inclusion, projection, and idempotent of Theorem 3.1.1. For factorisation D-modules, locality is the family of disjoint-locus squares of Theorem 3.2.1.*
3. *The ordered bar is the interval amplitude*

$$\overline{B}^{\mathrm{ord}}_X(A) \simeq \Bbbk \otimes^{\mathbb{L}}_A \Bbbk,$$

and is a conilpotent coassociative coalgebra in factorisation coefficients.

4. *Bar and cobar are inverse on nilcomplete algebras and conilcomplete coalgebras. Positive weight gives an explicit convergent subcategory.*
5. *The augmentation-boundary algebra is $A^{!} = \mathrm{RHom}_A(\Bbbk, \Bbbk)$, and the double centralizer is the derived augmentation completion.*
6. *The two-sided Koszul transform carries the diagonal bimodule to the diagonal bicomodule, relative tensor to derived cotensor, and Hochschild to coHochschild theory.*
7. *The endomorphisms of the identity A-module functor are*

$$\mathrm{End}(\mathrm{id}_{\mathrm{Mod}_A}) \simeq \mathrm{RHom}_{A^e}(A, A),$$

and carry the universal internal E_2 structure acting centrally on A.

8. *The filtered principal-part object, its normal symbols, extension classes, and regular quotient reconstruct the complete diagonal operation.*
9. *The two-coloured forest resolution has transverse and chiral edge contractions whose differentials anticommute. Its codimension-two boundary contains associativity, chiral Jacobi, and locality interchange.*

10. *A second transverse direction or flat meromorphic connection supplies exchange. A rigid line category with a fibre functor reconstructs a coquasitriangular Hopf algebra.*
11. *A compatible BRST derivation produces the bar–BRST bicomplex and its reduction spectral sequence.*
12. *A cyclic Calabi–Yau structure and a chiral modular action produce the two-edge master equation with independent ribbon and curve genera.*
13. *A factorising perturbative BV quantization satisfying QME produces A and the mixed-forest coefficient morphism.*
14. *Factorisation pushforward, boundary endomorphisms, and Hall extension correspondences provide higher-dimensional geometric sources.*
15. *For a positive graded Lie superalgebra, the Euler character of the ordered bar of its enveloping algebra is its denominator product.*

Proof. Items 1–2 are Theorems 3.1.1, 3.2.1 and 4.3.1. Items 3–6 are Theorems 6.2.1, 6.3.1, 7.2.1, 7.4.2, 8.3.1, 8.4.1 and 8.5.1. Item 7 is Theorems 9.1.1 and 9.2.1. Item 8 is Theorem 10.2.1. Item 9 is Theorems 5.2.2 and 5.2.3. Item 10 is Theorems 14.1.1 and 15.3.1. Item 11 is Theorems 17.1.1 and 17.2.1. Item 12 is Theorems 18.2.1, 19.2.1 and 20.3.1. Item 13 is Theorems 5.4.1 and 21.2.2. Item 14 is Theorems 28.1.1, 28.2.1, 28.4.1 and 28.6.2. Item 15 is Theorem 29.1.1. □

30.2 The functorial atlas

The structure theorem is summarized by

$$\begin{tikzcd}
\text{boundary category} \arrow[r, "\mathrm{End}(b)"] \arrow[d, "\text{bulk action}"'] & A \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \arrow[r, "\overline{B}^{\mathrm{ord}}"] \arrow[d, "Z^{\mathrm{der}}"'] & C \in \mathrm{Coalg}^{\mathrm{conil}}(\mathrm{FactD}(X)) \arrow[d, "\mathrm{coHoch}"] \\
\mathrm{Obs}_{\mathrm{bulk}} \arrow[r] & Z^{\mathrm{der}}(A) \arrow[r] & \mathrm{coHoch}(C).
\end{tikzcd}$$

Exchange enriches the middle column by a braided line category. A cyclic trace extends it to ribbon surfaces. Modular sewing extends it over stable curves. Hall geometry supplies noncommutative multiplication from extensions. Borcherds products supply arithmetic characters of graded Lie sectors.

30.3 Hierarchy of scalarisation

The object-level theory precedes its scalar traces. A sequence of functors may pass through

$$\text{factorisation category} \to A \to \overline{B}^{\mathrm{ord}}(A) \to Z^{\mathrm{der}}(A) \to \mathrm{Rep}(H) \to K_0 \to \text{character}.$$

A central charge, Hall polynomial, knot invariant, determinant, or automorphic form occurs at a specified functor. An equality of scalars is explained by a comparison morphism between the structured objects that precede those scalars.

30.4 The exact examples

The examples occupy increasing levels of structure:

model	ordered product	chiral coefficient	higher structure
$T(M)$	free concatenation	arbitrary	primitive bar
$\Bbbk \oplus M$	square zero	arbitrary	completed dual
A_q	$yx = qxy$	constant or factorising	Koszul PBW
$A_\hbar$	$[y, x] = \hbar$	factorising pairing	curved dual
ZF	R-exchange	spectral meromorphy	YBE
$Y_\hbar(\mathfrak{gl}_n)$	RTT	rational spectral curve	Hopf and PBW
$\mathcal{H}(A_2)$	extension count	support factorisation	$U_v^+(\mathfrak{sl}_3)$
matrix branes	necklace composition	$\mathbb{C}^2$ formal disc	stable trace Poisson

Each row is computed from generators, relations, structure maps, and a PBW, convergence, or descent theorem.

30.5 Coda: the geometry remembers

The diagonal remembers singularity. The interval remembers order. The plane remembers exchange. The circle remembers trace. The stable curve remembers sewing. The bar remembers every subdivision of the interval. The Fulton–MacPherson screen remembers every relative collision direction. The derived centre remembers coherent bulk action. The Hall correspondence remembers extensions. The automorphic denominator remembers the signed root spectrum.

The language of the subject is fixed by these motions. Every operation is assigned to the space that moves it, every equation to the boundary that proves it, and every scalar to the trace that produces it.

Appendix A

Suspensions, orientations, and signs

A.1 Cohomological suspension

For a graded object V, the suspension $sV = V[1]$ has $|sv| = |v| + 1$. The tensor permutation is governed by the Koszul rule

$$\tau(v \otimes w) = (-1)^{|v||w|} w \otimes v.$$

The suspension isomorphism

$$sV \otimes sW \longrightarrow s^2(V \otimes W)$$

is taken to be

$$sv \otimes sw \longmapsto (-1)^{|v|} s^2(v \otimes w).$$

This convention fixes every bar sign below.

A.2 Bar differential

For a strict dg algebra, write a reduced bar word as $[a_1|\cdots|a_r] = sa_1 \otimes \cdots \otimes sa_r$. The internal and multiplication differentials are

$$d_{\mathrm{int}}[a_1|\cdots|a_r] = \sum_{i=1}^{r} (-1)^{\sum_{j<i}(|a_j|+1)} [a_1|\cdots|d_A a_i|\cdots|a_r],$$
$$b_m[a_1|\cdots|a_r] = \sum_{i=1}^{r-1} (-1)^{\sum_{j\le i}|a_j|+i-1} [a_1|\cdots|a_i a_{i+1}|\cdots|a_r].$$

Proposition A.2.1 (Bar square)**.** *With these signs,*

$$d_{\mathrm{int}}^2 = 0, \qquad d_{\mathrm{int}} b_m + b_m d_{\mathrm{int}} = 0, \qquad b_m^2 = 0.$$

Proof. The first identity is the tensor differential. The second is the derivation rule for d_A. In b_m^2, contractions of disjoint adjacent pairs occur in opposite orders with opposite signs. Contractions of three consecutive letters leave the suspended associator, which vanishes. □

A.3 Deconcatenation

The unsuspended deconcatenation formula carries sign $+1$ because the left and right subwords retain their order. The coderivation identity is

$$\Delta d_B = (d_B \otimes 1 + 1 \otimes d_B)\Delta,$$

where the differential on the tensor product includes the standard Koszul sign.

A.4 Partition orientations

For a partition π of I, let
$$N_\pi = \Bbbk^I/\Bbbk^\pi$$
be the normal vector space of the partial diagonal at the combinatorial level. Set
$$\mathrm{or}(I/\pi) = \det N_\pi.$$

A cover relation $\pi \prec \sigma$ merging two blocks has a one-dimensional quotient N_σ/N_π. The collision differential inserts its oriented basis vector at the front of the determinant. Two independent mergers anticommute because their basis vectors are exchanged.

For three blocks A, B, C, orient the three pairwise merger paths cyclically. The induced signs are the three signs of the Jacobi identity. This convention agrees with the boundary orientation of the corresponding Fulton–MacPherson corner.

A.5 Two edge colours

For a mixed forest T, use
$$\mathrm{or}(T) = \det \Bbbk^{E_\perp(T)} \otimes \det \Bbbk^{E_{\mathrm{ch}}(T)}.$$

Contracting a transverse edge removes one basis vector from the first factor; contracting a chiral edge removes one from the second. Reversing their order exchanges two odd determinant directions, so
$$d_\perp d_{\mathrm{ch}} = -d_{\mathrm{ch}} d_\perp.$$

The same convention gives $\Delta_\perp \Delta_X = -\Delta_X \Delta_\perp$ in the two-edge graph complex.

A.6 BRST totalization

If Q has cohomological degree one and the bar differential has degree one, the total differential is
$$d_{\mathrm{tot}} = d_B + Q_B.$$

The sign in the factorwise definition of Q_B in Theorem 17.1.1 makes the two operators anticommute. A second filtration can instead use $d_B + (-1)^\ell Q_B$ on bar length ℓ; the two conventions are isomorphic by multiplying length-ℓ words by $(-1)^{\ell(\ell-1)/2}$.

Appendix B

Low-arity identities

B.1 Four-letter bar word

For even elements, the multiplication differential is

$$b[a|b|c|d] = [ab|c|d] - [a|bc|d] + [a|b|cd].$$

A second application gives

$$\begin{aligned} b^2[a|b|c|d] = {} & [(ab)c|d] - [ab|cd] \\ & - [(ab)c|d] + [a|(bc)d] \\ & + [ab|cd] - [a|b(cd)]. \end{aligned}$$

The disjoint contractions cancel, and the remaining terms are the two associators. This is the cellular boundary of the pentagon K_4.

B.2 Three-block collision

Let $[x, y]_{\text{ch}}$ be a chiral bracket. The three merger paths from $\{1\}|\{2\}|\{3\}$ to $\{1, 2, 3\}$ produce

$$[x, [y, z]_{\text{ch}}]_{\text{ch}} - (-1)^{|x||y|}[y, [x, z]_{\text{ch}}]_{\text{ch}} - [[x, y]_{\text{ch}}, z]_{\text{ch}}.$$

The partition orientation identifies this sum with the total-diagonal component of d^2_{coll}.

B.3 Arnold relation and KZ curvature

For three distinct points,

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

The triple component of the KZ curvature is

$$\begin{aligned} \hbar^2\big(& [\Omega_{12}, \Omega_{13}]\omega_{12} \wedge \omega_{13} + [\Omega_{12}, \Omega_{23}]\omega_{12} \wedge \omega_{23} \\ & + [\Omega_{13}, \Omega_{23}]\omega_{13} \wedge \omega_{23}\big). \end{aligned}$$

Invariance rewrites all three commutators as one tensor with the signs needed for the Arnold relation, so the curvature vanishes.

B.4 Rational Yang–Baxter expansion

Set $a = u - v$, $b = u - w$, $c = v - w$, so $b = a + c$. Expand

$$(1 - \hbar P_{12}/a)(1 - \hbar P_{13}/b)(1 - \hbar P_{23}/c)$$

and the reverse product. The linear terms agree. The cubic terms agree because $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$. The quadratic difference is

$$\hbar^2(P_{12}P_{13} - P_{13}P_{12})\left(\frac{1}{ab} - \frac{1}{ac} + \frac{1}{bc}\right),$$

which vanishes because $b = a + c$.

B.5 The first Hall relation

Using the notation of Chapter 24, the coefficient table is

	$X = 2S_1 \oplus S_2$	$Y = S_1 \oplus M$
$e_1^2e_2$	$q+1$	$q+1$
$e_1e_2e_1$	$q+1$	1
$e_2e_1^2$	$q+1$	0

The vector

$$(1, -(q+1), q)$$

is the unique normalized linear relation among the three rows. Twisting by the Euler form transforms it into the quantum Serre vector $(1, -(v + v^{-1}), 1)$.

B.6 Two-brane Poisson cone

With $\{x, y\} = 2$,

$$A = x^2/2, \quad B = xy/2, \quad C = y^2/2$$

give

$$B^2 - AC = 0, \quad \{A, B\} = 2A, \quad \{B, C\} = 2C, \quad \{A, C\} = 4B.$$

The ideal $(B^2 - AC)$ is Poisson because its bracket with each generator vanishes modulo the ideal. The singular point $A = B = C = 0$ is the coincident brane locus.

Appendix C

Exact computation tables

C.1 Quantum-plane normal forms

For a word w in x, y, let $\mathrm{inv}(w)$ be the number of pairs in which a y occurs to the left of an x. Rewriting $yx = qxy$ gives

$$w = q^{\mathrm{inv}(w)} x^{\#x(w)} y^{\#y(w)}.$$

For example,

$$yxyx = q^3 x^2 y^2, \qquad yyxxx = q^6 x^3 y^2.$$

The exponent is independent of the reduction sequence because each rewrite removes exactly one inversion.

C.2 Weyl normal ordering

The Weyl relation $yx = xy + \hbar$ implies

$$yx^m = x^m y + m\hbar x^{m-1}$$

by induction. Iterating gives

$$y^n x^m = \sum_{r=0}^{\min(m,n)} \binom{n}{r} m(m-1)\cdots(m-r+1)\hbar^r x^{m-r} y^{n-r}.$$

Equivalently,

$$y^n x^m = \sum_{r=0}^{\min(m,n)} r! \binom{n}{r}\binom{m}{r} \hbar^r x^{m-r} y^{n-r}.$$

This is the complete normal-ordering formula.

C.3 Stable trace brackets

The first power-sum brackets are

$$\{p_{10}, p_{01}\} = p_{00} = N,$$

$$\{p_{20}, p_{01}\} = 2p_{10}, \qquad \{p_{10}, p_{02}\} = 2p_{01},$$

$$\{p_{20}, p_{02}\} = 4p_{11}, \qquad \{p_{11}, p_{20}\} = -2p_{20}, \qquad \{p_{11}, p_{02}\} = 2p_{02}.$$

The last three generators form the $\mathfrak{sl}_2$ Poisson triple in every rank.

C.4 First Jacobi coefficients of the Igusa product

For the normalized weak Jacobi form

$$\phi_{0,1} = y^{-1} + 10 + y + O(q),$$

the first rows are

	$\ell = -3$	-2	-1	0	1	2	3
$n = 0$	0	0	1	10	1	0	0
$n = 1$	0	10	-64	108	-64	10	0
$n = 2$	1	108	-513	808	-513	108	1

The constant coefficient gives the Borcherds weight $10/2 = 5$. The coefficients at $(0, \pm 1)$ give the simple reflective roots. The remaining coefficients give signed imaginary-root supermultiplicities.

C.5 Euler product from a super root space

Let a root space have even dimension p and odd dimension q. After Chevalley suspension, its chain contribution is

$$\Lambda^*(\mathbb{C}^p e^{-\alpha}) \otimes \mathrm{Sym}^*(\mathbb{C}^q e^{-\alpha}).$$

Its Euler supercharacter is

$$(1 - e^{-\alpha})^p (1 - e^{-\alpha})^{-q} = (1 - e^{-\alpha})^{p-q}.$$

This proves the local factor in Theorem 29.1.1.

Appendix D

Typed dictionary

Term	Invariant meaning
factorisation coefficient	Ran D-module with multiplicative equivalences on disjoint supports and a filtered extension across collision diagonals
ordered chiral algebra	E_1 algebra in $\mathrm{FactD}(X)$; holomorphic collision and transverse ordered fusion in one object
chiral operation	diagonal-supported D-module morphism $j_* j^!(M \boxtimes N) \to \Delta_! P$
transverse commutator	graded commutator of the internal E_1 multiplication
exchange	braid transport supplied by a second transverse direction or a meromorphic flat connection
ordered bar	interval factorisation homology with augmentation modules at the endpoints
chiral Chevalley chains	cofree commutative coalgebra for chiral convolution with differential induced by the chiral bracket
Koszul dual $A^!$	derived endomorphism algebra $\mathrm{RHom}_A(\Bbbk, \Bbbk)$ of the augmentation boundary
derived centre	$\mathrm{RHom}_{A^e}(A, A)$ with its internal E_2 brace structure
physical bulk	supplied bulk factorisation algebra together with its open–closed map to the derived centre
Morita chart	endomorphism algebra of a compact generator in a factorisation category
principal part	filtered normal coefficient of the diagonal extension; pole order $r + 1$ gives the $(r + 1)$st normal symbol
screen	relative configuration of a collision cluster modulo translation and dilation
aligned decomposition	one disjoint-support decomposition used simultaneously in two convolution factors
Ran bilax idempotent	projection onto and inclusion of the aligned-decomposition direct summand
cyclic sewing	ribbon-surface operations generated by a perfect trace on the ordered algebra
chiral modular sewing	plumbing operations on conformal blocks over moduli of stable curves
two-edge master equation	graph Maurer–Cartan equation with independent transverse and chiral brackets and loop contractions
Hall product	extension correspondence pull–push with a multiplicative coefficient system
quantum double	Hopf construction from positive and negative halves, Cartan, pairing, radical quotient, and completion
Borcherds denominator	Weyl monomial times the Euler product of signed root spaces
polarized Gram transform	quadratic Rees construction retaining additive physical charge and quadratic automorphic degree

Five objects kept distinct

$$\begin{aligned}
C &= \text{factorisation category of boundary conditions,} \\
A_b &= \mathrm{RHom}_C(b, b), \\
\overline{B}^{\mathrm{ord}}(A_b) &= \text{interval coalgebra resolving the augmentation,} \\
A_b^! &= \mathrm{RHom}_{A_b}(\Bbbk, \Bbbk), \\
Z^{\mathrm{der}}(A_b) &= \mathrm{RHom}_{A_b^e}(A_b, A_b).
\end{aligned}$$

The first is Morita invariant. The second is a chart. The third is a coalgebraic resolution. The fourth is the augmentation-boundary dual. The fifth is the universal algebraic bulk.

Three forms of noncommutativity

1. Transverse noncommutativity: ab and ba are distinct ordered fusion products.
2. Chiral noncommutativity: analytic continuation and diagonal singularity produce nontrivial OPE commutators.
3. Braided noncommutativity: exchange is governed by an R-matrix rather than by the ordinary flip.

An ordered chiral quantum group may carry all three simultaneously.

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